

HOMOLOGICAL ALGEBRA

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1. Introduction

Basic homological algebra will be explained in this document. We add as needed in the other parts, since there is clearly an infinite amount of this stuff around. A reference is [ML63].

2. Basic notions

The following notions are considered basic and will not be defined, and or proved. This does not mean they are all necessarily easy or well known.

- (1) Nothing yet.

3. Preadditive and additive categories

Here is the definition of a preadditive category.

Definition 3.1. A category $\mathcal{A}$ is called *preadditive* if each morphism set $\text{Mor}_{\mathcal{A}}(x, y)$ is endowed with the structure of an abelian group such that the compositions

$$\text{Mor}(x, y) \times \text{Mor}(y, z) \longrightarrow \text{Mor}(x, z)$$

are bilinear. A functor $F : \mathcal{A} \rightarrow \mathcal{B}$ of preadditive categories is called *additive* if and only if $F : \text{Mor}(x, y) \rightarrow \text{Mor}(F(x), F(y))$ is a homomorphism of abelian groups for all $x, y \in \text{Ob}(\mathcal{A})$.

In particular for every x, y there exists at least one morphism $x \rightarrow y$, namely the zero map.

Lemma 3.2. *Let $\mathcal{A}$ be a preadditive category. Let x be an object of $\mathcal{A}$. The following are equivalent*

- (1) x is an initial object,
- (2) x is a final object, and
- (3) $\text{id}_x = 0$ in $\text{Mor}_{\mathcal{A}}(x, x)$.

Furthermore, if such an object 0 exists, then a morphism $\alpha : y \rightarrow z$ factors through 0 if and only if $\alpha = 0$.

Proof. First assume that x is either (1) initial or (2) final. In both cases, it follows that $\text{Mor}(x, x)$ is a trivial abelian group containing id_x , thus $\text{id}_x = 0$ in $\text{Mor}(x, x)$, which shows that each of (1) and (2) implies (3).

Now assume that $\text{id}_x = 0$ in $\text{Mor}(x, x)$. Let y be an arbitrary object of $\mathcal{A}$ and let $f \in \text{Mor}(x, y)$. Denote $C : \text{Mor}(x, x) \times \text{Mor}(x, y) \rightarrow \text{Mor}(x, y)$ the composition map. Then $f = C(0, f)$ and since C is bilinear we have $C(0, f) = 0$. Thus $f = 0$. Hence x is initial in $\mathcal{A}$. A similar argument for $f \in \text{Mor}(y, x)$ can be used to show that x is also final. Thus (3) implies both (1) and (2). $\square$

Definition 3.3. In a preadditive category $\mathcal{A}$, we call an object that is both final and initial (as in Lemma 3.2) a *zero object* and denote it 0 .

Lemma 3.4. *Let $\mathcal{A}$ be a preadditive category. Let $x, y \in \text{Ob}(\mathcal{A})$. If the product $x \times y$ exists, then so does the coproduct $x \amalg y$. If the coproduct $x \amalg y$ exists, then so does the product $x \times y$. In this case also $x \amalg y \cong x \times y$.*

Proof. Suppose that $z = x \times y$ with projections $p : z \rightarrow x$ and $q : z \rightarrow y$. Denote $i : x \rightarrow z$ the morphism corresponding to $(1, 0)$. Denote $j : y \rightarrow z$ the morphism corresponding to $(0, 1)$. Thus we have the commutative diagram

$$\begin{array}{ccc}
 x & \xrightarrow{1} & x \\
 & \searrow i & \nearrow p \\
 & & z \\
 & \nearrow j & \searrow q \\
 y & \xrightarrow{1} & y
 \end{array}$$

where the diagonal compositions are zero. It follows that $i \circ p + j \circ q : z \rightarrow z$ is the identity since it is a morphism which upon composing with p gives p and upon composing with q gives q . Suppose given morphisms $a : x \rightarrow w$ and $b : y \rightarrow w$. Then we can form the map $a \circ p + b \circ q : z \rightarrow w$. In this way we get a bijection $\text{Mor}(z, w) = \text{Mor}(x, w) \times \text{Mor}(y, w)$ which shows that $z = x \amalg y$.

We leave it to the reader to construct the morphisms p, q given a coproduct $x \amalg y$ instead of a product. $\square$

Definition 3.5. Given a pair of objects x, y in a preadditive category $\mathcal{A}$, the *direct sum* $x \oplus y$ of x and y is the direct product $x \times y$ endowed with the morphisms i, j, p, q as in Lemma 3.4 above.

Remark 3.6. Note that the proof of Lemma 3.4 shows that given p and q the morphisms i, j are uniquely determined by the rules $p \circ i = \text{id}_x$, $q \circ j = \text{id}_y$, $p \circ j = 0$, $q \circ i = 0$. Moreover, we automatically have $i \circ p + j \circ q = \text{id}_{x \oplus y}$. Similarly, given i, j the morphisms p and q are uniquely determined. Finally, given objects x, y, z and morphisms $i : x \rightarrow z$, $j : y \rightarrow z$, $p : z \rightarrow x$ and $q : z \rightarrow y$ such that $p \circ i = \text{id}_x$, $q \circ j = \text{id}_y$, $p \circ j = 0$, $q \circ i = 0$ and $i \circ p + j \circ q = \text{id}_z$, then z is the direct sum of x and y with the four morphisms equal to i, j, p, q .

Lemma 3.7. Let $\mathcal{A}, \mathcal{B}$ be preadditive categories. Let $F : \mathcal{A} \rightarrow \mathcal{B}$ be an additive functor. Then F transforms direct sums to direct sums and zero to zero.

Proof. Suppose F is additive. A direct sum z of x and y is characterized by having morphisms $i : x \rightarrow z$, $j : y \rightarrow z$, $p : z \rightarrow x$ and $q : z \rightarrow y$ such that $p \circ i = \text{id}_x$, $q \circ j = \text{id}_y$, $p \circ j = 0$, $q \circ i = 0$ and $i \circ p + j \circ q = \text{id}_z$, according to Remark 3.6. Clearly $F(x), F(y), F(z)$ and the morphisms $F(i), F(j), F(p), F(q)$ satisfy exactly the same relations (by additivity) and we see that $F(z)$ is a direct sum of $F(x)$ and $F(y)$. Hence, F transforms direct sums to direct sums.

To see that F transforms zero to zero, use the characterization (3) of the zero object in Lemma 3.2. $\square$

Definition 3.8. A category $\mathcal{A}$ is called *additive* if it is preadditive and finite products exist, in other words it has a zero object and direct sums.

Namely the empty product is a finite product and if it exists, then it is a final object.

Definition 3.9. Let $\mathcal{A}$ be a preadditive category. Let $f : x \rightarrow y$ be a morphism.

- (1) A *kernel* of f is a morphism $i : z \rightarrow x$ such that (a) $f \circ i = 0$ and (b) for any $i' : z' \rightarrow x$ such that $f \circ i' = 0$ there exists a unique morphism $g : z' \rightarrow z$ such that $i' = i \circ g$.
- (2) If the kernel of f exists, then we denote this $\text{Ker}(f) \rightarrow x$.
- (3) A *cokernel* of f is a morphism $p : y \rightarrow z$ such that (a) $p \circ f = 0$ and (b) for any $p' : y \rightarrow z'$ such that $p' \circ f = 0$ there exists a unique morphism $g : z \rightarrow z'$ such that $p' = g \circ p$.
- (4) If a cokernel of f exists we denote this $y \rightarrow \text{Coker}(f)$.
- (5) If a kernel of f exists, then a *coimage* of f is a cokernel for the morphism $\text{Ker}(f) \rightarrow x$.
- (6) If a kernel and coimage exist then we denote this $x \rightarrow \text{Coim}(f)$.
- (7) If a cokernel of f exists, then the *image* of f is a kernel of the morphism $y \rightarrow \text{Coker}(f)$.
- (8) If a cokernel and image of f exist then we denote this $\text{Im}(f) \rightarrow y$.

In the above definition, we have spoken of “the kernel” and “the cokernel”, tacitly using their uniqueness up to unique isomorphism. This follows from the Yoneda lemma (Categories, Section ??) because the kernel of $f : x \rightarrow y$ represents the functor sending an object z to the set $\text{Ker}(\text{Mor}_{\mathcal{A}}(z, x) \rightarrow \text{Mor}_{\mathcal{A}}(z, y))$. The case of cokernels is dual.

We first relate the direct sum to kernels as follows.

Lemma 3.10. *Let $\mathcal{C}$ be a preadditive category. Let $x \oplus y$ with morphisms i, j, p, q as in Lemma 3.4 be a direct sum in $\mathcal{C}$. Then $i : x \rightarrow x \oplus y$ is a kernel of $q : x \oplus y \rightarrow y$. Dually, p is a cokernel for j .*

Proof. Let $f : z' \rightarrow x \oplus y$ be a morphism such that $q \circ f = 0$. We have to show that there exists a unique morphism $g : z' \rightarrow x$ such that $f = i \circ g$. Since $i \circ p + j \circ q$ is the identity on $x \oplus y$ we see that

$$f = (i \circ p + j \circ q) \circ f = i \circ p \circ f$$

and hence $g = p \circ f$ works. Uniqueness holds because $p \circ i$ is the identity on x . The proof of the second statement is dual. $\square$

Lemma 3.11. *Let $\mathcal{C}$ be a preadditive category. Let $f : x \rightarrow y$ be a morphism in $\mathcal{C}$.*

- (1) *If a kernel of f exists, then this kernel is a monomorphism.*
- (2) *If a cokernel of f exists, then this cokernel is an epimorphism.*
- (3) *If a kernel and coimage of f exist, then the coimage is an epimorphism.*
- (4) *If a cokernel and image of f exist, then the image is a monomorphism.*

Proof. Part (1) follows easily from the uniqueness required in the definition of a kernel. The proof of (2) is dual. Part (3) follows from (2), since the coimage is a cokernel. Similarly, (4) follows from (1). $\square$

Lemma 3.12. *Let $f : x \rightarrow y$ be a morphism in a preadditive category such that the kernel, cokernel, image and coimage all exist. Then f can be factored uniquely as $x \rightarrow \text{Coim}(f) \rightarrow \text{Im}(f) \rightarrow y$.*

Proof. There is a canonical morphism $\text{Coim}(f) \rightarrow y$ because $\text{Ker}(f) \rightarrow x \rightarrow y$ is zero. The composition $\text{Coim}(f) \rightarrow y \rightarrow \text{Coker}(f)$ is zero, because it is the unique morphism which gives rise to the morphism $x \rightarrow y \rightarrow \text{Coker}(f)$ which is zero (the

uniqueness follows from Lemma 3.11 (3)). Hence $\text{Coim}(f) \rightarrow y$ factors uniquely through $\text{Im}(f) \rightarrow y$, which gives us the desired map. $\square$

Example 3.13. Let k be a field. Consider the category of filtered vector spaces over k . (See Definition 19.1.) Consider the filtered vector spaces (V, F) and (W, F) with $V = W = k$ and

$$F^i V = \begin{cases} V & \text{if } i < 0 \\ 0 & \text{if } i \geq 0 \end{cases} \text{ and } F^i W = \begin{cases} W & \text{if } i \leq 0 \\ 0 & \text{if } i > 0 \end{cases}$$

The map $f : V \rightarrow W$ corresponding to id_k on the underlying vector spaces has trivial kernel and cokernel but is not an isomorphism. Note also that $\text{Coim}(f) = V$ and $\text{Im}(f) = W$. This means that the category of filtered vector spaces over k is not abelian.

We finish this section by some lemmas on splitting images of idempotent endomorphisms. Recall that an endomorphism f is idempotent if $f \circ f = f$.

Remark 3.14. Let $\mathcal{A}$ be a preadditive category, x an object of $\mathcal{A}$, and $f : x \rightarrow x$ idempotent. Writing $g = \text{id}_x - f$, one has $f \circ g = 0$, $g \circ f = 0$, and $g \circ g = g$.

Lemma 3.15. *Let $\mathcal{A}$ be a preadditive category, x be an object of $\mathcal{A}$, and $f : x \rightarrow x$ be idempotent. Write $g = \text{id}_x - f$ and recall by Remark 3.14 that $f \circ g = g \circ f = 0$.*

- (1) *If f has kernel $i : \text{Ker}(f) \rightarrow x$ and p is the unique morphism such that $g = i \circ p$, then $p \circ i = \text{id}_{\text{Ker}(f)}$ and p is the cokernel of f .*
- (2) *If f has cokernel $p : x \rightarrow \text{Coker}(f)$ and i is the unique morphism such that $g = i \circ p$, then $p \circ i = \text{id}_{\text{Coker}(f)}$ and i is the kernel of f .*
- (3) *If g has kernel $j : \text{Ker}(g) \rightarrow x$ and q is the unique morphism such that $f = j \circ q$, then $q \circ j = \text{id}_{\text{Ker}(g)}$ and q is the cokernel of g .*
- (4) *If g has cokernel $q : x \rightarrow \text{Coker}(g)$ and j is the unique morphism such that $f = j \circ q$, then $q \circ j = \text{id}_{\text{Coker}(g)}$ and j is the kernel of g .*

Proof. As for (1), compute first that $i = (f + g) \circ i = g \circ i$. By the construction of p , this implies that $i \circ p \circ i = g \circ i = i \circ \text{id}_{\text{Ker}(f)}$. By Lemma 3.11 we conclude that $p \circ i = \text{id}_{\text{Ker}(f)}$.

It remains to show that $p : x \rightarrow \text{Ker}(f)$ satisfies the universal property of the cokernel of f , i.e., that given a morphism $a : x \rightarrow y$ satisfying $a \circ f = 0$ there exists a unique morphism $b : \text{Ker}(f) \rightarrow y$ such that $a = b \circ p$. Existence follows by first computing that $a = a \circ (f + g) = a \circ g$ and concluding by the construction of p that $a = a \circ g = a \circ i \circ p$ (so that we can take $b = a \circ i$). Uniqueness follows from that $b = b \circ p \circ i = a \circ i$ (so that a determines b).

By duality, symmetry (i.e., that g is itself idempotent; cf. Remark 3.14), and duality and symmetry respectively, (2), (3), and (4) follow as did (1). $\square$

Lemma 3.16. *Let $\mathcal{A}$ be a preadditive category, x be an object of $\mathcal{A}$, and $f : x \rightarrow x$ be idempotent. Write $g = \text{id}_x - f$. If each of f and g have at least one of a kernel or cokernel and in turn morphisms i, j, p, q are constructed as in the statement of*

Lemma 3.15, then all four (co)kernels exist and

$$\begin{aligned} x &\simeq \text{Ker}(f) \oplus \text{Ker}(g) \\ &\simeq \text{Coker}(f) \oplus \text{Ker}(g) \\ &\simeq \text{Ker}(f) \oplus \text{Coker}(g) \\ &\simeq \text{Coker}(f) \oplus \text{Coker}(g) \end{aligned}$$

with the structure of each direct product given by i, j, p, q .

Proof. By Lemma 3.15 $p \circ i = \text{id}_{\text{Ker}(f)} = \text{id}_{\text{Coker}(f)}$ and $q \circ j = \text{id}_{\text{Ker}(g)} = \text{id}_{\text{Coker}(g)}$. By construction, $i \circ p + j \circ q = f + g = \text{id}_x$. By Remark 3.6 the claim follows. $\square$

Lemma 3.17. Let $\mathcal{A}$ be a preadditive category, x and y be objects of $\mathcal{A}$, and $j : y \rightarrow x$ and $q : x \rightarrow y$ be morphisms satisfying $q \circ j = \text{id}_y$. Then

- (1) $\text{id}_x - j \circ q$ has kernel j .
- (2) $\text{id}_x - j \circ q$ has cokernel q .

Proof. As for (1), compute first that $(\text{id}_x - j \circ q) \circ j = 0$. We must show that j satisfies the universal property of the kernel of $\text{id}_x - j \circ q$, i.e., that given a morphism $a : z \rightarrow x$ satisfying $(\text{id}_x - j \circ q) \circ a = 0$ there exists a unique morphism $b : z \rightarrow y$ such that $a = j \circ b$. Existence follows by computing that $a = j \circ q \circ a$ (so that we can take $b = q \circ a$). Uniqueness follows from that $b = q \circ j \circ b = q \circ a$ (so that a determines b).

By duality, (2) follows as did (1). $\square$

Lemma 3.18. Let $\mathcal{A}$ be a preadditive category, x and y be objects of $\mathcal{A}$, and $j : y \rightarrow x$ and $q : x \rightarrow y$ be morphisms satisfying $q \circ j = \text{id}_y$. If $j \circ q$ has at least one of a kernel or cokernel, then both (co)kernels exist and

$$\begin{aligned} x &\simeq \text{Ker}(j \circ q) \oplus y \\ &\simeq \text{Coker}(j \circ q) \oplus y \end{aligned}$$

with the structure of either direct product including j, q , and the canonical (co)kernel morphism, and the remaining map uniquely determined.

Proof. Write $f = j \circ q$, compute that $f \circ f = f$, and apply Lemma 3.16 and Lemma 3.17. $\square$

4. Karoubian categories

Skip this section on a first reading.

Definition 4.1. Let $\mathcal{C}$ be a preadditive category. We say $\mathcal{C}$ is *Karoubian* if every idempotent endomorphism of an object of $\mathcal{C}$ has a kernel.

The dual notion would be that every idempotent endomorphism of an object has a cokernel. However, in view of the (dual of the) following lemma that would be an equivalent notion.

Lemma 4.2. Let $\mathcal{C}$ be a preadditive category. The following are equivalent

- (1) $\mathcal{C}$ is Karoubian,
- (2) every idempotent endomorphism of an object of $\mathcal{C}$ has a cokernel, and
- (3) given an idempotent endomorphism $p : z \rightarrow z$ of $\mathcal{C}$ there exists a direct sum decomposition $z = x \oplus y$ such that p corresponds to the projection onto y .

Proof. Immediate via Lemmas 3.15, 3.16, and 3.10. $\square$

Lemma 4.3. *Let $\mathcal{D}$ be a preadditive category.*

- (1) *If $\mathcal{D}$ has countable products and kernels of maps which have a right inverse, then $\mathcal{D}$ is Karoubian.*
- (2) *If $\mathcal{D}$ has countable coproducts and cokernels of maps which have a left inverse, then $\mathcal{D}$ is Karoubian.*

Proof. Let X be an object of $\mathcal{D}$ and let $e : X \rightarrow X$ be an idempotent. The functor

$$W \longmapsto \text{Ker}(\text{Mor}_{\mathcal{D}}(W, X) \xrightarrow{e} \text{Mor}_{\mathcal{D}}(W, X))$$

is representable if and only if e has a kernel. Note that for any abelian group A and idempotent endomorphism $e : A \rightarrow A$ we have

$$\text{Ker}(e : A \rightarrow A) = \text{Ker}(\Phi : \prod_{n \in \mathbf{N}} A \rightarrow \prod_{n \in \mathbf{N}} A)$$

where

$$\Phi(a_1, a_2, a_3, \dots) = (ea_1 + (1-e)a_2, ea_2 + (1-e)a_3, \dots)$$

Moreover, Φ has the right inverse

$$\Psi(a_1, a_2, a_3, \dots) = (a_1, (1-e)a_1 + ea_2, (1-e)a_2 + ea_3, \dots).$$

Hence (1) holds. The proof of (2) is dual (using the dual definition of a Karoubian category, namely condition (2) of Lemma 4.2). $\square$

5. Abelian categories

An abelian category is a category satisfying just enough axioms so the snake lemma holds. An axiom (that is sometimes forgotten) is that the canonical map $\text{Coim}(f) \rightarrow \text{Im}(f)$ of Lemma 3.12 is always an isomorphism. Example 3.13 shows that it is necessary.

Definition 5.1. A category $\mathcal{A}$ is *abelian* if it is additive, if all kernels and cokernels exist, and if the natural map $\text{Coim}(f) \rightarrow \text{Im}(f)$ is an isomorphism for all morphisms f of $\mathcal{A}$.

Lemma 5.2. *Let $\mathcal{A}$ be a preadditive category. The additions on sets of morphisms make $\mathcal{A}^{opp}$ into a preadditive category. Furthermore, $\mathcal{A}$ is additive if and only if $\mathcal{A}^{opp}$ is additive, and $\mathcal{A}$ is abelian if and only if $\mathcal{A}^{opp}$ is abelian.*

Proof. The first statement is straightforward. To see that $\mathcal{A}$ is additive if and only if $\mathcal{A}^{opp}$ is additive, recall that additivity can be characterized by the existence of a zero object and direct sums, which are both preserved when passing to the opposite category. Finally, to see that $\mathcal{A}$ is abelian if and only if $\mathcal{A}^{opp}$ is abelian, observe that kernels, cokernels, images and coimages in $\mathcal{A}^{opp}$ correspond to cokernels, kernels, coimages and images in $\mathcal{A}$, respectively. $\square$

Definition 5.3. Let $f : x \rightarrow y$ be a morphism in an abelian category.

- (1) We say f is *injective* if $\text{Ker}(f) = 0$.
- (2) We say f is *surjective* if $\text{Coker}(f) = 0$.

If $x \rightarrow y$ is injective, then we say that x is a *subobject* of y and we use the notation $x \subset y$. If $x \rightarrow y$ is surjective, then we say that y is a *quotient* of x .

Lemma 5.4. *Let $f : x \rightarrow y$ be a morphism in an abelian category $\mathcal{A}$. Then*

- (1) f is injective if and only if f is a monomorphism,
- (2) f is surjective if and only if f is an epimorphism, and
- (3) f is an isomorphism if and only if f is injective and surjective.

Proof. Proof of (1). Recall that $\text{Ker}(f)$ is an object representing the functor sending z to $\text{Ker}(\text{Mor}_{\mathcal{A}}(z, x) \rightarrow \text{Mor}_{\mathcal{A}}(z, y))$, see Definition 3.9. Thus $\text{Ker}(f)$ is 0 if and only if $\text{Mor}_{\mathcal{A}}(z, x) \rightarrow \text{Mor}_{\mathcal{A}}(z, y)$ is injective for all z if and only if f is a monomorphism. The proof of (2) is similar. For the proof of (3) note that an isomorphism is both a monomorphism and epimorphism, which by (1), (2) proves f is injective and surjective. If f is both injective and surjective, then $x = \text{Coim}(f)$ and $y = \text{Im}(f)$ whence f is an isomorphism. $\square$

In an abelian category, if $x \subset y$ is a subobject, then we denote

$$y/x = \text{Coker}(x \rightarrow y).$$

Lemma 5.5. *Let $\mathcal{A}$ be an abelian category. All finite limits and finite colimits exist in $\mathcal{A}$.*

Proof. To show that finite limits exist it suffices to show that finite products and equalizers exist, see Categories, Lemma ???. Finite products exist by definition and the equalizer of $a, b : x \rightarrow y$ is the kernel of $a - b$. The argument for finite colimits is similar but dual to this. $\square$

Example 5.6. Let $\mathcal{A}$ be an abelian category. Pushouts and fibre products in $\mathcal{A}$ have the following simple descriptions:

- (1) If $a : x \rightarrow y, b : z \rightarrow y$ are morphisms in $\mathcal{A}$, then we have the fibre product:
 $x \times_y z = \text{Ker}((a, -b) : x \oplus z \rightarrow y).$
- (2) If $a : y \rightarrow x, b : y \rightarrow z$ are morphisms in $\mathcal{A}$, then we have the pushout:
 $x \amalg_y z = \text{Coker}((a, -b) : y \rightarrow x \oplus z).$

Definition 5.7. Let $\mathcal{A}$ be an additive category. Consider a sequence of morphisms

$$\dots \rightarrow x \rightarrow y \rightarrow z \rightarrow \dots \quad \text{or} \quad x_1 \rightarrow x_2 \rightarrow \dots \rightarrow x_n$$

in $\mathcal{A}$. We say such a sequence is a *complex* if the composition of any two consecutive (drawn) arrows is zero. If $\mathcal{A}$ is abelian then we say a complex of the first type above is *exact at y* if $\text{Im}(x \rightarrow y) = \text{Ker}(y \rightarrow z)$ and we say a complex of the second kind is *exact at x_i* where $1 < i < n$ if $\text{Im}(x_{i-1} \rightarrow x_i) = \text{Ker}(x_i \rightarrow x_{i+1})$. We say a sequence as above is *exact* or is an *exact sequence* or is an *exact complex* if it is a complex and exact at every object (in the first case) or exact at x_i for all $1 < i < n$ (in the second case). There are variants of these notions for sequences of the form

$$\dots \rightarrow x_{-3} \rightarrow x_{-2} \rightarrow x_{-1} \quad \text{and} \quad x_1 \rightarrow x_2 \rightarrow x_3 \rightarrow \dots$$

A *short exact sequence* is an exact complex of the form

$$0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0.$$

In the following lemma we assume the reader knows what it means for a sequence of abelian groups to be exact.

Lemma 5.8. *Let $\mathcal{A}$ be an abelian category. Let $0 \rightarrow M_1 \rightarrow M_2 \rightarrow M_3 \rightarrow 0$ be a complex of $\mathcal{A}$.*

- (1) $M_1 \rightarrow M_2 \rightarrow M_3 \rightarrow 0$ is exact if and only if
- $$0 \rightarrow \operatorname{Hom}_{\mathcal{A}}(M_3, N) \rightarrow \operatorname{Hom}_{\mathcal{A}}(M_2, N) \rightarrow \operatorname{Hom}_{\mathcal{A}}(M_1, N)$$
- is an exact sequence of abelian groups for all objects N of $\mathcal{A}$, and
- (2) $0 \rightarrow M_1 \rightarrow M_2 \rightarrow M_3$ is exact if and only if
- $$0 \rightarrow \operatorname{Hom}_{\mathcal{A}}(N, M_1) \rightarrow \operatorname{Hom}_{\mathcal{A}}(N, M_2) \rightarrow \operatorname{Hom}_{\mathcal{A}}(N, M_3)$$
- is an exact sequence of abelian groups for all objects N of $\mathcal{A}$.

Proof. Omitted. Hint: See Algebra, Lemma ??.

□

Definition 5.9. Let $\mathcal{A}$ be an abelian category. Let $i : A \rightarrow B$ and $q : B \rightarrow C$ be morphisms of $\mathcal{A}$ such that $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$ is a short exact sequence. We say the short exact sequence is *split* if there exist morphisms $j : C \rightarrow B$ and $p : B \rightarrow A$ such that (B, i, j, p, q) is the direct sum of A and C .

Lemma 5.10. Let $\mathcal{A}$ be an abelian category. Let $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$ be a short exact sequence.

- (1) Given a morphism $s : C \rightarrow B$ right inverse to $B \rightarrow C$, there exists a unique $\pi : B \rightarrow A$ such that (s, π) splits the short exact sequence as in Definition 5.9.
- (2) Given a morphism $\pi : B \rightarrow A$ left inverse to $A \rightarrow B$, there exists a unique $s : C \rightarrow B$ such that (s, π) splits the short exact sequence as in Definition 5.9.

Proof. Immediate corollary of Lemma 3.18.

□

Lemma 5.11. Let $\mathcal{A}$ be an abelian category. Let

$$\begin{array}{ccc} w & \xrightarrow{f} & y \\ g \downarrow & & \downarrow h \\ x & \xrightarrow{k} & z \end{array}$$

be a commutative diagram.

- (1) The diagram is cartesian if and only if

$$0 \rightarrow w \xrightarrow{(g, f)} x \oplus y \xrightarrow{(k, -h)} z$$

is exact.

- (2) The diagram is cocartesian if and only if

$$w \xrightarrow{(g, -f)} x \oplus y \xrightarrow{(k, h)} z \rightarrow 0$$

is exact.

Proof. Let $u = (g, f) : w \rightarrow x \oplus y$ and $v = (k, -h) : x \oplus y \rightarrow z$. Let $p : x \oplus y \rightarrow x$ and $q : x \oplus y \rightarrow y$ be the canonical projections. Let $i : \operatorname{Ker}(v) \rightarrow x \oplus y$ be the canonical injection. By Example 5.6, the diagram is cartesian if and only if there exists an isomorphism $r : \operatorname{Ker}(v) \rightarrow w$ with $f \circ r = q \circ i$ and $g \circ r = p \circ i$. The sequence $0 \rightarrow w \xrightarrow{u} x \oplus y \xrightarrow{v} z$ is exact if and only if there exists an isomorphism $r : \operatorname{Ker}(v) \rightarrow w$ with $u \circ r = i$. But given $r : \operatorname{Ker}(v) \rightarrow w$, we have $f \circ r = q \circ i$ and $g \circ r = p \circ i$ if and only if $q \circ u \circ r = f \circ r = q \circ i$ and $p \circ u \circ r = g \circ r = p \circ i$, hence if and only if $u \circ r = i$. This proves (1), and then (2) follows by duality. □

Lemma 5.12. *Let $\mathcal{A}$ be an abelian category. Let*

$$\begin{array}{ccc} w & \xrightarrow{f} & y \\ g \downarrow & & \downarrow h \\ x & \xrightarrow{k} & z \end{array}$$

be a commutative diagram.

- (1) *If the diagram is cartesian, then the morphism $\text{Ker}(f) \rightarrow \text{Ker}(k)$ induced by g is an isomorphism.*
- (2) *If the diagram is cocartesian, then the morphism $\text{Coker}(f) \rightarrow \text{Coker}(k)$ induced by h is an isomorphism.*

Proof. Suppose the diagram is cartesian. Let $e : \text{Ker}(f) \rightarrow \text{Ker}(k)$ be induced by g . Let $i : \text{Ker}(f) \rightarrow w$ and $j : \text{Ker}(k) \rightarrow x$ be the canonical injections. There exists $t : \text{Ker}(k) \rightarrow w$ with $f \circ t = 0$ and $g \circ t = j$. Hence, there exists $u : \text{Ker}(k) \rightarrow \text{Ker}(f)$ with $i \circ u = t$. It follows $g \circ i \circ u \circ e = g \circ t \circ e = j \circ e = g \circ i$ and $f \circ i \circ u \circ e = 0 = f \circ i$, hence $i \circ u \circ e = i$. Since i is a monomorphism this implies $u \circ e = \text{id}_{\text{Ker}(f)}$. Furthermore, we have $j \circ e \circ u = g \circ i \circ u = g \circ t = j$. Since j is a monomorphism this implies $e \circ u = \text{id}_{\text{Ker}(k)}$. This proves (1). Now, (2) follows by duality. $\square$

Lemma 5.13. *Let $\mathcal{A}$ be an abelian category. Let*

$$\begin{array}{ccc} w & \xrightarrow{f} & y \\ g \downarrow & & \downarrow h \\ x & \xrightarrow{k} & z \end{array}$$

be a commutative diagram.

- (1) *If the diagram is cartesian and k is an epimorphism, then the diagram is cocartesian and f is an epimorphism.*
- (2) *If the diagram is cocartesian and g is a monomorphism, then the diagram is cartesian and h is a monomorphism.*

Proof. Suppose the diagram is cartesian and k is an epimorphism. Let $u = (g, f) : w \rightarrow x \oplus y$ and let $v = (k, -h) : x \oplus y \rightarrow z$. As k is an epimorphism, v is an epimorphism, too. Therefore and by Lemma 5.11, the sequence $0 \rightarrow w \xrightarrow{u} x \oplus y \xrightarrow{v} z \rightarrow 0$ is exact. Thus, the diagram is cocartesian by Lemma 5.11. Finally, f is an epimorphism by Lemma 5.12 and Lemma 5.4. This proves (1), and (2) follows by duality. $\square$

Lemma 5.14. *Let $\mathcal{A}$ be an abelian category.*

- (1) *If $x \rightarrow y$ is surjective, then for every $z \rightarrow y$ the projection $x \times_y z \rightarrow z$ is surjective.*
- (2) *If $x \rightarrow y$ is injective, then for every $x \rightarrow z$ the morphism $z \rightarrow z \amalg_x y$ is injective.*

Proof. Immediately from Lemma 5.4 and Lemma 5.13. $\square$

Lemma 5.15. *Let $\mathcal{A}$ be an abelian category. Let $f : x \rightarrow y$ and $g : y \rightarrow z$ be morphisms with $g \circ f = 0$. Then, the following statements are equivalent:*

- (1) *The sequence $x \xrightarrow{f} y \xrightarrow{g} z$ is exact.*

- (2) For every $h : w \rightarrow y$ with $g \circ h = 0$ there exist an object v , an epimorphism $k : v \rightarrow w$ and a morphism $l : v \rightarrow x$ with $h \circ k = f \circ l$.

Proof. Let $i : \text{Ker}(g) \rightarrow y$ be the canonical injection. Let $p : x \rightarrow \text{Coim}(f)$ be the canonical projection. Let $j : \text{Im}(f) \rightarrow \text{Ker}(g)$ be the canonical injection.

Suppose (1) holds. Let $h : w \rightarrow y$ with $g \circ h = 0$. There exists $c : w \rightarrow \text{Ker}(g)$ with $i \circ c = h$. Let $v = x \times_{\text{Ker}(g)} w$ with canonical projections $k : v \rightarrow w$ and $l : v \rightarrow x$, so that $c \circ k = j \circ p \circ l$. Then, $h \circ k = i \circ c \circ k = i \circ j \circ p \circ l = f \circ l$. As $j \circ p$ is an epimorphism by hypothesis, k is an epimorphism by Lemma 5.13. This implies (2).

Suppose (2) holds. Then, $g \circ i = 0$. So, there are an object w , an epimorphism $k : w \rightarrow \text{Ker}(g)$ and a morphism $l : w \rightarrow x$ with $f \circ l = i \circ k$. It follows $i \circ j \circ p \circ l = f \circ l = i \circ k$. Since i is a monomorphism we see that $j \circ p \circ l = k$ is an epimorphism. So, j is an epimorphism and thus an isomorphism. This implies (1). $\square$

Lemma 5.16. Let $\mathcal{A}$ be an abelian category. Let

$$\begin{array}{ccccc} x & \xrightarrow{f} & y & \xrightarrow{g} & z \\ \downarrow \alpha & & \downarrow \beta & & \downarrow \gamma \\ u & \xrightarrow{k} & v & \xrightarrow{l} & w \end{array}$$

be a commutative diagram.

- (1) If the first row is exact and k is a monomorphism, then the induced sequence $\text{Ker}(\alpha) \rightarrow \text{Ker}(\beta) \rightarrow \text{Ker}(\gamma)$ is exact.
- (2) If the second row is exact and g is an epimorphism, then the induced sequence $\text{Coker}(\alpha) \rightarrow \text{Coker}(\beta) \rightarrow \text{Coker}(\gamma)$ is exact.

Proof. Suppose the first row is exact and k is a monomorphism. Let $a : \text{Ker}(\alpha) \rightarrow \text{Ker}(\beta)$ and $b : \text{Ker}(\beta) \rightarrow \text{Ker}(\gamma)$ be the induced morphisms. Let $h : \text{Ker}(\alpha) \rightarrow x$, $i : \text{Ker}(\beta) \rightarrow y$ and $j : \text{Ker}(\gamma) \rightarrow z$ be the canonical injections. As j is a monomorphism we have $b \circ a = 0$. Let $c : s \rightarrow \text{Ker}(\beta)$ with $b \circ c = 0$. Then, $g \circ i \circ c = j \circ b \circ c = 0$. By Lemma 5.15 there are an object t , an epimorphism $d : t \rightarrow s$ and a morphism $e : t \rightarrow x$ with $i \circ c \circ d = f \circ e$. Then, $k \circ \alpha \circ e = \beta \circ f \circ e = \beta \circ i \circ c \circ d = 0$. As k is a monomorphism we get $\alpha \circ e = 0$. So, there exists $m : t \rightarrow \text{Ker}(\alpha)$ with $h \circ m = e$. It follows $i \circ a \circ m = f \circ h \circ m = f \circ e = i \circ c \circ d$. As i is a monomorphism we get $a \circ m = c \circ d$. Thus, Lemma 5.15 implies (1), and then (2) follows by duality. $\square$

Lemma 5.17. Let $\mathcal{A}$ be an abelian category. Let

$$\begin{array}{ccccccc} x & \xrightarrow{f} & y & \xrightarrow{g} & z & \longrightarrow & 0 \\ \downarrow \alpha & & \downarrow \beta & & \downarrow \gamma & & \\ 0 \longrightarrow & u & \xrightarrow{k} & v & \xrightarrow{l} & w & \end{array}$$

be a commutative diagram with exact rows.

- (1) *There exists a unique morphism $\delta : \text{Ker}(\gamma) \rightarrow \text{Coker}(\alpha)$ such that the diagram*

$$\begin{array}{ccccc} y & \xleftarrow{\pi'} & y \times_z \text{Ker}(\gamma) & \xrightarrow{\pi} & \text{Ker}(\gamma) \\ \beta \downarrow & & & & \downarrow \delta \\ v & \xrightarrow{\iota'} & \text{Coker}(\alpha) \amalg_u v & \xleftarrow{\iota} & \text{Coker}(\alpha) \end{array}$$

commutes, where π and π' are the canonical projections and ι and ι' are the canonical coprojections.

- (2) *The induced sequence*

$$\text{Ker}(\alpha) \xrightarrow{f'} \text{Ker}(\beta) \xrightarrow{g'} \text{Ker}(\gamma) \xrightarrow{\delta} \text{Coker}(\alpha) \xrightarrow{k'} \text{Coker}(\beta) \xrightarrow{l'} \text{Coker}(\gamma)$$

is exact. If f is injective then so is f' , and if l is surjective then so is l' .

Proof. As π is an epimorphism and ι is a monomorphism by Lemma 5.13, uniqueness of δ is clear. Let $p = y \times_z \text{Ker}(\gamma)$ and $q = \text{Coker}(\alpha) \amalg_u v$. Let $h : \text{Ker}(\beta) \rightarrow y$, $i : \text{Ker}(\gamma) \rightarrow z$ and $j : \text{Ker}(\pi) \rightarrow p$ be the canonical injections. Let $\pi'' : u \rightarrow \text{Coker}(\alpha)$ be the canonical projection. Keeping in mind Lemma 5.13 we get a commutative diagram with exact rows

$$\begin{array}{ccccccc} 0 & \longrightarrow & \text{Ker}(\pi) & \xrightarrow{j} & p & \xrightarrow{\pi} & \text{Ker}(\gamma) \longrightarrow 0 \\ & & & & \downarrow \pi' & & \downarrow i \\ & & x & \xrightarrow{f} & y & \xrightarrow{g} & z \longrightarrow 0 \\ & & \downarrow \alpha & & \downarrow \beta & & \downarrow \gamma \\ 0 & \longrightarrow & u & \xrightarrow{k} & v & \xrightarrow{l} & w \\ & & \downarrow \pi'' & & \downarrow \iota' & & \\ & & 0 & \longrightarrow & \text{Coker}(\alpha) & \xrightarrow{\iota} & q \end{array}$$

As $l \circ \beta \circ \pi' = \gamma \circ i \circ \pi = 0$ and as the third row of the diagram above is exact, there is an $a : p \rightarrow u$ with $k \circ a = \beta \circ \pi'$. As the upper right quadrangle of the diagram above is cartesian, Lemma 5.12 yields an epimorphism $b : x \rightarrow \text{Ker}(\pi)$ with $\pi' \circ j \circ b = f$. It follows $k \circ a \circ j \circ b = \beta \circ \pi' \circ j \circ b = \beta \circ f = k \circ \alpha$. As k is a monomorphism this implies $a \circ j \circ b = \alpha$. It follows $\pi'' \circ a \circ j \circ b = \pi'' \circ \alpha = 0$. As b is an epimorphism this implies $\pi'' \circ a \circ j = 0$. Therefore, as the top row of the diagram above is exact, there exists $\delta : \text{Ker}(\gamma) \rightarrow \text{Coker}(\alpha)$ with $\delta \circ \pi = \pi'' \circ a$. It follows $\iota \circ \delta \circ \pi = \iota \circ \pi'' \circ a = \iota' \circ k \circ a = \iota' \circ \beta \circ \pi'$ as desired.

As the upper right quadrangle in the diagram above is cartesian there is a $c : \text{Ker}(\beta) \rightarrow p$ with $\pi' \circ c = h$ and $\pi \circ c = g'$. It follows $\iota \circ \delta \circ g' = \iota \circ \delta \circ \pi \circ c = \iota' \circ \beta \circ \pi' \circ c = \iota' \circ \beta \circ h = 0$. As ι is a monomorphism this implies $\delta \circ g' = 0$.

Next, let $d : r \rightarrow \text{Ker}(\gamma)$ with $\delta \circ d = 0$. Applying Lemma 5.15 to the exact sequence $p \xrightarrow{\pi} \text{Ker}(\gamma) \rightarrow 0$ and d yields an object s , an epimorphism $m : s \rightarrow r$ and a morphism $n : s \rightarrow p$ with $\pi \circ n = d \circ m$. As $\pi'' \circ a \circ n = \delta \circ d \circ m = 0$, applying Lemma 5.15 to the exact sequence $x \xrightarrow{\alpha} u \xrightarrow{\pi''} \text{Coker}(\alpha)$ and $a \circ n$ yields an object t , an epimorphism $\varepsilon : t \rightarrow s$ and a morphism $\zeta : t \rightarrow x$ with $a \circ n \circ \varepsilon = \alpha \circ \zeta$. It holds $\beta \circ \pi' \circ n \circ \varepsilon = k \circ \alpha \circ \zeta = \beta \circ f \circ \zeta$. Let $\eta = \pi' \circ n \circ \varepsilon - f \circ \zeta : t \rightarrow y$.

Then, $\beta \circ \eta = 0$. It follows that there is a $\vartheta : t \rightarrow \text{Ker}(\beta)$ with $\eta = h \circ \vartheta$. It holds $i \circ g' \circ \vartheta = g \circ h \circ \vartheta = g \circ \pi' \circ n \circ \varepsilon - g \circ f \circ \zeta = i \circ \pi \circ n \circ \varepsilon = i \circ d \circ m \circ \varepsilon$. As i is a monomorphism we get $g' \circ \vartheta = d \circ m \circ \varepsilon$. Thus, as $m \circ \varepsilon$ is an epimorphism, Lemma 5.15 implies that $\text{Ker}(\beta) \xrightarrow{g'} \text{Ker}(\gamma) \xrightarrow{\delta} \text{Coker}(\alpha)$ is exact. Then, the claim follows by Lemma 5.16 and duality. $\square$

Lemma 5.18. *Let $\mathcal{A}$ be an abelian category. Let*

$$\begin{array}{ccccccc}
 & & x & \xrightarrow{\quad} & y & \xrightarrow{\quad} & z \xrightarrow{\quad} 0 \\
 & \swarrow & \downarrow \alpha & \swarrow & \downarrow \beta & \swarrow & \downarrow \gamma \\
 & x' & \xrightarrow{\quad} & y' & \xrightarrow{\quad} & z' & \xrightarrow{\quad} 0 \\
 & \downarrow \alpha' & \downarrow \beta' & \downarrow \gamma' & & & \\
 0 & \xrightarrow{\quad} & u & \xrightarrow{\quad} & v & \xrightarrow{\quad} & w \\
 & \downarrow & \downarrow & \downarrow & & & \\
 0 & \xrightarrow{\quad} & u' & \xrightarrow{\quad} & v' & \xrightarrow{\quad} & w'
 \end{array}$$

be a commutative diagram with exact rows. Then, the induced diagram

$$\begin{array}{ccccccc}
 \text{Ker}(\alpha) & \longrightarrow & \text{Ker}(\beta) & \longrightarrow & \text{Ker}(\gamma) & \xrightarrow{\delta} & \text{Coker}(\alpha) \longrightarrow \text{Coker}(\beta) \longrightarrow \text{Coker}(\gamma) \\
 \downarrow & & \downarrow & & \downarrow & & \downarrow \\
 \text{Ker}(\alpha') & \longrightarrow & \text{Ker}(\beta') & \longrightarrow & \text{Ker}(\gamma') & \xrightarrow{\delta'} & \text{Coker}(\alpha') \longrightarrow \text{Coker}(\beta') \longrightarrow \text{Coker}(\gamma')
 \end{array}$$

commutes.

Proof. Omitted. $\square$

Lemma 5.19. *Let $\mathcal{A}$ be an abelian category. Let*

$$\begin{array}{ccccccc}
 w & \longrightarrow & x & \longrightarrow & y & \longrightarrow & z \\
 \downarrow \alpha & & \downarrow \beta & & \downarrow \gamma & & \downarrow \delta \\
 w' & \longrightarrow & x' & \longrightarrow & y' & \longrightarrow & z'
 \end{array}$$

be a commutative diagram with exact rows.

- (1) *If α, γ are surjective and δ is injective, then β is surjective.*
- (2) *If β, δ are injective and α is surjective, then γ is injective.*

Proof. Assume α, γ are surjective and δ is injective. We may replace w' by $\text{Im}(w' \rightarrow x')$, i.e., we may assume that $w' \rightarrow x'$ is injective. We may replace z by $\text{Im}(y \rightarrow z)$, i.e., we may assume that $y \rightarrow z$ is surjective. Then we may apply Lemma 5.17 to

$$\begin{array}{ccccccc}
 \text{Ker}(y \rightarrow z) & \longrightarrow & y & \longrightarrow & z & \longrightarrow & 0 \\
 \downarrow & & \downarrow & & \downarrow & & \\
 0 & \longrightarrow & \text{Ker}(y' \rightarrow z') & \longrightarrow & y' & \longrightarrow & z'
 \end{array}$$

to conclude that $\text{Ker}(y \rightarrow z) \rightarrow \text{Ker}(y' \rightarrow z')$ is surjective. Finally, we apply Lemma 5.17 to

$$\begin{array}{ccccccc} w & \longrightarrow & x & \longrightarrow & \text{Ker}(y \rightarrow z) & \longrightarrow & 0 \\ \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & w' & \longrightarrow & x' & \longrightarrow & \text{Ker}(y' \rightarrow z') \end{array}$$

to conclude that $x \rightarrow x'$ is surjective. This proves (1). The proof of (2) is dual to this. $\square$

Lemma 5.20. *Let $\mathcal{A}$ be an abelian category. Let*

$$\begin{array}{ccccccccc} v & \longrightarrow & w & \longrightarrow & x & \longrightarrow & y & \longrightarrow & z \\ \downarrow \alpha & & \downarrow \beta & & \downarrow \gamma & & \downarrow \delta & & \downarrow \epsilon \\ v' & \longrightarrow & w' & \longrightarrow & x' & \longrightarrow & y' & \longrightarrow & z' \end{array}$$

be a commutative diagram with exact rows. If β, δ are isomorphisms, ϵ is injective, and α is surjective then γ is an isomorphism.

Proof. Immediate consequence of Lemma 5.19. $\square$

6. Extensions

Definition 6.1. Let $\mathcal{A}$ be an abelian category. Let $A, B \in \text{Ob}(\mathcal{A})$. An *extension* E of B by A is a short exact sequence

$$0 \rightarrow A \rightarrow E \rightarrow B \rightarrow 0.$$

A *morphism of extensions* between two extensions $0 \rightarrow A \rightarrow E \rightarrow B \rightarrow 0$ and $0 \rightarrow A \rightarrow F \rightarrow B \rightarrow 0$ means a morphism $f : E \rightarrow F$ in $\mathcal{A}$ making the diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & A & \longrightarrow & E & \longrightarrow & B \longrightarrow 0 \\ & & \downarrow \text{id} & & \downarrow f & & \downarrow \text{id} \\ 0 & \longrightarrow & A & \longrightarrow & F & \longrightarrow & B \longrightarrow 0 \end{array}$$

commutative. Thus, the extensions of B by A form a category.

By abuse of language we often omit mention of the morphisms $A \rightarrow E$ and $E \rightarrow B$, although they are definitively part of the structure of an extension.

Definition 6.2. Let $\mathcal{A}$ be an abelian category. Let $A, B \in \text{Ob}(\mathcal{A})$. The set of isomorphism classes of extensions of B by A is denoted

$$\text{Ext}_{\mathcal{A}}(B, A).$$

This is called the *Ext-group*.

This definition works, because by our conventions $\text{Ob}(\mathcal{A})$ is a set, and hence $\text{Ext}_{\mathcal{A}}(B, A)$ is a set. In any of the cases of “big” abelian categories listed in Categories, Remark ?? one can check by hand that $\text{Ext}_{\mathcal{A}}(B, A)$ is a set as well. Also, we will see later that this is always the case when $\mathcal{A}$ has either enough projectives or enough injectives. Insert future reference here.

Actually we can turn $\text{Ext}_{\mathcal{A}}(-, -)$ into a functor

$$\mathcal{A} \times \mathcal{A}^{\text{opp}} \longrightarrow \text{Sets}, \quad (A, B) \longmapsto \text{Ext}_{\mathcal{A}}(B, A)$$

as follows:

- (1) Given a morphism $B' \rightarrow B$ and an extension E of B by A we define $E' = E \times_B B'$. By Lemma 5.12 we have the following commutative diagram of short exact sequences

$$\begin{array}{ccccccccc} 0 & \longrightarrow & A & \longrightarrow & E' & \longrightarrow & B' & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & A & \longrightarrow & E & \longrightarrow & B & \longrightarrow & 0 \end{array}$$

The extension E' is called the *pullback of E via $B' \rightarrow B$* .

- (2) Given a morphism $A \rightarrow A'$ and an extension E of B by A we define $E' = A' \amalg_A E$. By Lemma 5.13 we have the following commutative diagram of short exact sequences

$$\begin{array}{ccccccccc} 0 & \longrightarrow & A & \longrightarrow & E & \longrightarrow & B & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & A' & \longrightarrow & E' & \longrightarrow & B & \longrightarrow & 0 \end{array}$$

The extension E' is called the *pushout of E via $A \rightarrow A'$* .

To see that this defines a functor as indicated above there are several things to verify. First of all functoriality in the variable B requires that $(E \times_B B') \times_{B'} B'' = E \times_B B''$ which is a general property of fibre products. Dually one deals with functoriality in the variable A . Finally, given $A \rightarrow A'$ and $B' \rightarrow B$ we have to show that

$$A' \amalg_A (E \times_B B') \cong (A' \amalg_A E) \times_B B'$$

as extensions of B' by A' . Recall that $A' \amalg_A E$ is a quotient of $A' \oplus E$. Thus the right hand side is a quotient of $A' \oplus E \times_B B'$, and it is straightforward to see that the kernel is exactly what you need in order to get the left hand side.

Note that if E_1 and E_2 are extensions of B by A , then $E_1 \oplus E_2$ is an extension of $B \oplus B$ by $A \oplus A$. We push out by the sum map $A \oplus A \rightarrow A$ and we pull back by the diagonal map $B \rightarrow B \oplus B$ to get an extension $E_1 + E_2$ of B by A .

$$\begin{array}{ccccccccc} 0 & \longrightarrow & A \oplus A & \longrightarrow & E_1 \oplus E_2 & \longrightarrow & B \oplus B & \longrightarrow & 0 \\ & & \downarrow \Sigma & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & A & \longrightarrow & E' & \longrightarrow & B \oplus B & \longrightarrow & 0 \\ & & \uparrow & & \uparrow & & \uparrow \Delta & & \\ 0 & \longrightarrow & A & \longrightarrow & E_1 + E_2 & \longrightarrow & B & \longrightarrow & 0 \end{array}$$

The extension $E_1 + E_2$ is called the *Baer sum* of the given extensions.

Lemma 6.3. *The construction $(E_1, E_2) \mapsto E_1 + E_2$ above defines a commutative group law on $\text{Ext}_{\mathcal{A}}(B, A)$ which is functorial in both variables.*

Proof. Omitted. □

Lemma 6.4. *Let $\mathcal{A}$ be an abelian category. Let $0 \rightarrow M_1 \rightarrow M_2 \rightarrow M_3 \rightarrow 0$ be a short exact sequence in $\mathcal{A}$.*

- (1) *There is a canonical six term exact sequence of abelian groups*

$$\begin{array}{ccccccc}
 0 & \longrightarrow & \mathrm{Hom}_{\mathcal{A}}(M_3, N) & \longrightarrow & \mathrm{Hom}_{\mathcal{A}}(M_2, N) & \longrightarrow & \mathrm{Hom}_{\mathcal{A}}(M_1, N) \\
 & & & & & \swarrow & \\
 & & \mathrm{Ext}_{\mathcal{A}}(M_3, N) & \longrightarrow & \mathrm{Ext}_{\mathcal{A}}(M_2, N) & \longrightarrow & \mathrm{Ext}_{\mathcal{A}}(M_1, N)
 \end{array}$$

for all objects N of $\mathcal{A}$, and

- (2) *there is a canonical six term exact sequence of abelian groups*

$$\begin{array}{ccccccc}
 0 & \longrightarrow & \mathrm{Hom}_{\mathcal{A}}(N, M_1) & \longrightarrow & \mathrm{Hom}_{\mathcal{A}}(N, M_2) & \longrightarrow & \mathrm{Hom}_{\mathcal{A}}(N, M_3) \\
 & & & & & \swarrow & \\
 & & \mathrm{Ext}_{\mathcal{A}}(N, M_1) & \longrightarrow & \mathrm{Ext}_{\mathcal{A}}(N, M_2) & \longrightarrow & \mathrm{Ext}_{\mathcal{A}}(N, M_3)
 \end{array}$$

for all objects N of $\mathcal{A}$.

Proof. Omitted. Hint: The boundary maps are defined using either the pushout or pullback of the given short exact sequence. $\square$

7. Additive functors

First a completely silly lemma characterizing additive functors between additive categories.

Lemma 7.1. *Let $\mathcal{A}$ and $\mathcal{B}$ be additive categories. Let $F : \mathcal{A} \rightarrow \mathcal{B}$ be a functor. The following are equivalent*

- (1) *F is additive,*
- (2) *$F(A) \oplus F(B) \rightarrow F(A \oplus B)$ is an isomorphism for all $A, B \in \mathcal{A}$, and*
- (3) *$F(A \oplus B) \rightarrow F(A) \oplus F(B)$ is an isomorphism for all $A, B \in \mathcal{A}$.*

Proof. Additive functors commute with direct sums by Lemma 3.7 hence (1) implies (2) and (3). If (2) holds and 0 denotes the zero object of $\mathcal{A}$, then the map $F(0) \oplus F(0) \rightarrow F(0 \oplus 0) \cong F(0)$ being an isomorphism implies that the diagonal map $\mathrm{Hom}(F(0), C) \rightarrow \mathrm{Hom}(F(0), C) \oplus \mathrm{Hom}(F(0), C)$ is bijective for all C in $\mathcal{B}$. Thus $F(0)$ is the zero object of $\mathcal{B}$ and F maps the zero morphism between any two objects to the zero morphism. The same holds if (3) holds. It follows that (2) and (3) are equivalent because the composition $F(A) \oplus F(B) \rightarrow F(A \oplus B) \rightarrow F(A) \oplus F(B)$ is given by the matrix

$$\begin{pmatrix} F(1) & F(0) \\ F(0) & F(1) \end{pmatrix}$$

and therefore is the identity map. Assume (2) and (3) hold. Let $f, g : A \rightarrow B$ be maps. Then $f + g$ is equal to the composition

$$A \rightarrow A \oplus A \xrightarrow{\mathrm{diag}(f, g)} B \oplus B \rightarrow B$$

Apply the functor F and consider the following diagram

$$\begin{array}{ccccccc}
 F(A) & \longrightarrow & F(A \oplus A) & \xrightarrow{F(\mathrm{diag}(f, g))} & F(B \oplus B) & \longrightarrow & F(B) \\
 & \searrow & \uparrow & & \downarrow & \nearrow & \\
 & & F(A) \oplus F(A) & \xrightarrow{\mathrm{diag}(F(f), F(g))} & F(B) \oplus F(B) & &
 \end{array}$$

We claim this is commutative. For the middle square we can verify it separately for each of the four induced maps $F(A) \rightarrow F(B)$ where it follows from the fact that F is a functor and transforms zero morphisms into zero morphisms. For the triangle on the left, we use that $F(A \oplus A) \rightarrow F(A) \oplus F(A)$ is an isomorphism to see that it suffices to check after composition with this map and this check is trivial. Dually for the other triangle. Thus going around the bottom is equal to $F(f + g)$ and we conclude. $\square$

Recall that we defined, in Categories, Definition ?? the notion of a “right exact”, “left exact” and “exact” functor in the setting of a functor between categories that have finite (co)limits. Thus this applies in particular to functors between abelian categories.

Lemma 7.2. *Let $\mathcal{A}$ and $\mathcal{B}$ be abelian categories. Let $F : \mathcal{A} \rightarrow \mathcal{B}$ be a functor.*

- (1) *If F is either left or right exact, then it is additive.*
- (2) *F is left exact if and only if for every short exact sequence $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$ the sequence $0 \rightarrow F(A) \rightarrow F(B) \rightarrow F(C)$ is exact.*
- (3) *F is right exact if and only if for every short exact sequence $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$ the sequence $F(A) \rightarrow F(B) \rightarrow F(C) \rightarrow 0$ is exact.*
- (4) *F is exact if and only if for every short exact sequence $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$ the sequence $0 \rightarrow F(A) \rightarrow F(B) \rightarrow F(C) \rightarrow 0$ is exact.*

Proof. If F is left exact, i.e., F commutes with finite limits, then F sends products to products, hence F preserves direct sums, hence F is additive by Lemma 7.1. On the other hand, suppose that for every short exact sequence $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$ the sequence $0 \rightarrow F(A) \rightarrow F(B) \rightarrow F(C)$ is exact. Let A, B be two objects. Then we have a short exact sequence

$$0 \rightarrow A \rightarrow A \oplus B \rightarrow B \rightarrow 0$$

see for example Lemma 3.10. By assumption, the lower row in the commutative diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & F(A) & \longrightarrow & F(A) \oplus F(B) & \longrightarrow & F(B) \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & F(A) & \longrightarrow & F(A \oplus B) & \longrightarrow & F(B) \end{array}$$

is exact. Hence by the snake lemma (Lemma 5.17) we conclude that $F(A) \oplus F(B) \rightarrow F(A \oplus B)$ is an isomorphism. Hence F is additive in this case as well. Thus for the rest of the proof we may assume F is additive.

Denote $f : B \rightarrow C$ a map from B to C . Exactness of $0 \rightarrow A \rightarrow B \rightarrow C$ just means that $A = \text{Ker}(f)$. Clearly the kernel of f is the equalizer of the two maps f and 0 from B to C . Hence if F commutes with limits, then $F(\text{Ker}(f)) = \text{Ker}(F(f))$ which exactly means that $0 \rightarrow F(A) \rightarrow F(B) \rightarrow F(C)$ is exact.

Conversely, suppose that F is additive and transforms any short exact sequence $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$ into an exact sequence $0 \rightarrow F(A) \rightarrow F(B) \rightarrow F(C)$. Because it is additive it commutes with direct sums and hence finite products in $\mathcal{A}$. To show it commutes with finite limits it therefore suffices to show that it commutes with equalizers. But equalizers in an abelian category are the same as the kernel of the difference map, hence it suffices to show that F commutes with taking kernels. Let $f : A \rightarrow B$ be a morphism. Factor f as $A \rightarrow I \rightarrow B$ with $f' : A \rightarrow I$ surjective

and $i : I \rightarrow B$ injective. (This is possible by the definition of an abelian category.) Then it is clear that $\text{Ker}(f) = \text{Ker}(f')$. Also $0 \rightarrow \text{Ker}(f') \rightarrow A \rightarrow I \rightarrow 0$ and $0 \rightarrow I \rightarrow B \rightarrow B/I \rightarrow 0$ are short exact. By the condition imposed on F we see that $0 \rightarrow F(\text{Ker}(f')) \rightarrow F(A) \rightarrow F(I)$ and $0 \rightarrow F(I) \rightarrow F(B) \rightarrow F(B/I)$ are exact. Hence it is also the case that $F(\text{Ker}(f'))$ is the kernel of the map $F(A) \rightarrow F(B)$, and we win.

The proof of (3) is similar to the proof of (2). Statement (4) is a combination of (2) and (3). $\square$

Lemma 7.3. *Let $\mathcal{A}$ and $\mathcal{B}$ be abelian categories. Let $F : \mathcal{A} \rightarrow \mathcal{B}$ be an exact functor. For every pair of objects A, B of $\mathcal{A}$ the functor F induces an abelian group homomorphism*

$$\text{Ext}_{\mathcal{A}}(B, A) \longrightarrow \text{Ext}_{\mathcal{B}}(F(B), F(A))$$

which maps the extension E to $F(E)$.

Proof. Omitted. $\square$

The following lemma is used in the proof that the category of abelian sheaves on a site is abelian, where the functor b is sheafification.

Lemma 7.4. *Let $a : \mathcal{A} \rightarrow \mathcal{B}$ and $b : \mathcal{B} \rightarrow \mathcal{A}$ be functors. Assume that*

- (1) $\mathcal{A}, \mathcal{B}$ are additive categories, a, b are additive functors, and a is right adjoint to b ,
- (2) $\mathcal{B}$ is abelian and b is left exact, and
- (3) $ba \cong \text{id}_{\mathcal{A}}$.

Then $\mathcal{A}$ is abelian.

Proof. As $\mathcal{B}$ is abelian we see that all finite limits and colimits exist in $\mathcal{B}$ by Lemma 5.5. Since b is a left adjoint we see that b is also right exact and hence exact, see Categories, Lemma ???. Let $\varphi : B_1 \rightarrow B_2$ be a morphism of $\mathcal{B}$. In particular, if $K = \text{Ker}(B_1 \rightarrow B_2)$, then K is the equalizer of 0 and φ and hence bK is the equalizer of 0 and $b\varphi$, hence bK is the kernel of $b\varphi$. Similarly, if $Q = \text{Coker}(B_1 \rightarrow B_2)$, then Q is the coequalizer of 0 and φ and hence bQ is the coequalizer of 0 and $b\varphi$, hence bQ is the cokernel of $b\varphi$. Thus we see that every morphism of the form $b\varphi$ in $\mathcal{A}$ has a kernel and a cokernel. However, since $ba \cong \text{id}$ we see that every morphism of $\mathcal{A}$ is of this form, and we conclude that kernels and cokernels exist in $\mathcal{A}$. In fact, the argument shows that if $\psi : A_1 \rightarrow A_2$ is a morphism then

$$\text{Ker}(\psi) = b\text{Ker}(a\psi), \quad \text{and} \quad \text{Coker}(\psi) = b\text{Coker}(a\psi).$$

Now we still have to show that $\text{Coim}(\psi) = \text{Im}(\psi)$. We do this as follows. First note that since $\mathcal{A}$ has kernels and cokernels it has all finite limits and colimits (see proof of Lemma 5.5). Hence we see by Categories, Lemma ??? that a is left exact

and hence transforms kernels (=equalizers) into kernels.

$$\begin{aligned}
\text{Coim}(\psi) &= \text{Coker}(\text{Ker}(\psi) \rightarrow A_1) && \text{by definition} \\
&= b \text{Coker}(a(\text{Ker}(\psi) \rightarrow A_1)) && \text{by formula above} \\
&= b \text{Coker}(\text{Ker}(a\psi) \rightarrow aA_1) && a \text{ preserves kernels} \\
&= b \text{Coim}(a\psi) && \text{by definition} \\
&= b \text{Im}(a\psi) && \mathcal{B} \text{ is abelian} \\
&= b \text{Ker}(aA_2 \rightarrow \text{Coker}(a\psi)) && \text{by definition} \\
&= \text{Ker}(baA_2 \rightarrow b \text{Coker}(a\psi)) && b \text{ preserves kernels} \\
&= \text{Ker}(A_2 \rightarrow b \text{Coker}(a\psi)) && ba = \text{id}_A \\
&= \text{Ker}(A_2 \rightarrow \text{Coker}(\psi)) && \text{by formula above} \\
&= \text{Im}(\psi) && \text{by definition}
\end{aligned}$$

Thus the lemma holds. $\square$

8. Localization

In this section we note how Gabriel-Zisman localization interacts with the additive structure on a category.

Lemma 8.1. *Let $\mathcal{C}$ be a preadditive category. Let S be a left or right multiplicative system. There exists a unique preadditive structure on $S^{-1}\mathcal{C}$ such that the localization functor $Q : \mathcal{C} \rightarrow S^{-1}\mathcal{C}$ is additive.*

Proof. We will prove this in the case S is a left multiplicative system. The case where S is a right multiplicative system is dual. Suppose that X, Y are objects of $\mathcal{C}$ and that $\alpha, \beta : X \rightarrow Y$ are morphisms in $S^{-1}\mathcal{C}$. According to Categories, Lemma ?? we may represent these by pairs $s^{-1}f, s^{-1}g$ with common denominator s . Suppose that we have a pre-additive structure on $S^{-1}\mathcal{C}$ and that Q is additive. Then

$$\begin{aligned}
s^{-1}f + s^{-1}g &= Q(s)^{-1} \circ Q(f) + Q(s)^{-1} \circ Q(g) \\
&= Q(s)^{-1} \circ (Q(f) + Q(g)) \\
&= Q(s)^{-1} \circ Q(f + g) \\
&= s^{-1}(f + g).
\end{aligned}$$

Hence, the pre-additive structure on $S^{-1}\mathcal{C}$ is unique if it exists. Correspondingly, we define $\alpha + \beta$ to be the equivalence class of $s^{-1}(f + g)$ and in the rest of the proof we show that this is well defined and that composition is bilinear. Once this is done it is clear that Q is an additive functor.

Let us show the construction above is well defined. An abstract way of saying this is that filtered colimits of abelian groups agree with filtered colimits of sets and to use Categories, Equation (??). We can work this out in a bit more detail as follows. Say $s : Y \rightarrow Y_1$ and $f, g : X \rightarrow Y_1$. Suppose we have a second representation of α, β as $(s')^{-1}f', (s')^{-1}g'$ with $s' : Y \rightarrow Y_2$ and $f', g' : X \rightarrow Y_2$. By Categories, Remark ?? we can find a morphism $s_3 : Y \rightarrow Y_3$ and morphisms $a_1 : Y_1 \rightarrow Y_3$, $a_2 : Y_2 \rightarrow Y_3$ such that $a_1 \circ s = s_3 = a_2 \circ s'$ and also $a_1 \circ f = a_2 \circ f'$ and $a_1 \circ g = a_2 \circ g'$. Hence

we see that $s^{-1}(f + g)$ is equivalent to

$$\begin{aligned} s_3^{-1}(a_1 \circ (f + g)) &= s_3^{-1}(a_1 \circ f + a_1 \circ g) \\ &= s_3^{-1}(a_2 \circ f' + a_2 \circ g') \\ &= s_3^{-1}(a_2 \circ (f' + g')) \end{aligned}$$

which is equivalent to $(s')^{-1}(f' + g')$.

Fix $s : Y \rightarrow Y'$ and $f, g : X \rightarrow Y'$ with $\alpha = s^{-1}f$ and $\beta = s^{-1}g$ as morphisms $X \rightarrow Y$ in $S^{-1}\mathcal{C}$. To show that composition is bilinear first consider the case of a morphism $\gamma : Y \rightarrow Z$ in $S^{-1}\mathcal{C}$. Say $\gamma = t^{-1}h$ for some $h : Y \rightarrow Z'$ and $t : Z \rightarrow Z'$ in S . Using LMS2 we choose morphisms $a : Y' \rightarrow Z''$ and $t' : Z' \rightarrow Z''$ in S such that $a \circ s = t' \circ h$. Picture

$$\begin{array}{ccccc} & & & Z & \\ & & & \downarrow t & \\ & Y & \xrightarrow{h} & Z' & \\ & \downarrow s & & \downarrow t' & \\ X & \xrightarrow{f,g} & Y' & \xrightarrow{a} & Z'' \end{array}$$

Then $\gamma \circ \alpha = (t' \circ t)^{-1}(a \circ f)$ and $\gamma \circ \beta = (t' \circ t)^{-1}(a \circ g)$. Hence we see that $\gamma \circ (\alpha + \beta)$ is represented by $(t' \circ t)^{-1}(a \circ (f + g)) = (t' \circ t)^{-1}(a \circ f + a \circ g)$ which represents $\gamma \circ \alpha + \gamma \circ \beta$.

Finally, assume that $\delta : W \rightarrow X$ is another morphism of $S^{-1}\mathcal{C}$. Say $\delta = r^{-1}i$ for some $i : W \rightarrow X'$ and $r : X \rightarrow X'$ in S . We claim that we can find a morphism $s' : Y' \rightarrow Y''$ in S and morphisms $a'', b'' : X' \rightarrow Y''$ such that the following diagram commutes

$$\begin{array}{ccccc} & & & Y & \\ & & & \downarrow s & \\ & X & \xrightarrow{f,g,f+g} & Y' & \\ & \downarrow r & & \downarrow s' & \\ W & \xrightarrow{i} & X' & \xrightarrow{a'',b'',a''+b''} & Y'' \end{array}$$

Namely, using LMS2 we can first choose $s_1 : Y' \rightarrow Y_1$, $s_2 : Y' \rightarrow Y_2$ in S and $a : X' \rightarrow Y_1$, $b : X' \rightarrow Y_2$ such that $a \circ r = s_1 \circ f$ and $b \circ r = s_2 \circ g$. Then using that the category Y'/S is filtered (see Categories, Remark ??), we can find a $s' : Y' \rightarrow Y''$ and morphisms $a' : Y_1 \rightarrow Y''$, $b' : Y_2 \rightarrow Y''$ such that $s' = a' \circ s_1$ and $s' = b' \circ s_2$. Setting $a'' = a' \circ a$ and $b'' = b' \circ b$ works. At this point we see that the compositions $\alpha \circ \delta$ and $\beta \circ \delta$ are represented by $(s' \circ s)^{-1}(a'' \circ i)$ and $(s' \circ s)^{-1}(b'' \circ i)$. Hence $\alpha \circ \delta + \beta \circ \delta$ is represented by $(s' \circ s)^{-1}(a'' \circ i + b'' \circ i) = (s' \circ s)^{-1}((a'' + b'') \circ i)$ which by the diagram again is a representative of $(\alpha + \beta) \circ \delta$. $\square$

Lemma 8.2. *Let $\mathcal{C}$ be an additive category. Let S be a left or right multiplicative system. Then $S^{-1}\mathcal{C}$ is an additive category and the localization functor $Q : \mathcal{C} \rightarrow S^{-1}\mathcal{C}$ is additive.*

Proof. By Lemma 8.1 we see that $S^{-1}\mathcal{C}$ is preadditive and that Q is additive. By Lemma 3.7 we conclude that $S^{-1}\mathcal{C}$ has a zero object and direct sums. $\square$

The following lemma describes the “kernel” of the localization functor in case we invert a multiplicative system.

Lemma 8.3. *Let $\mathcal{C}$ be an additive category. Let S be a multiplicative system. Let X be an object of $\mathcal{C}$. The following are equivalent*

- (1) $Q(X) = 0$ in $S^{-1}\mathcal{C}$,
- (2) *there exists $Y \in \text{Ob}(\mathcal{C})$ such that $0 : X \rightarrow Y$ is an element of S , and*
- (3) *there exists $Z \in \text{Ob}(\mathcal{C})$ such that $0 : Z \rightarrow X$ is an element of S .*

Proof. If (2) holds we see that $0 = Q(0) : Q(X) \rightarrow Q(Y)$ is an isomorphism. In the additive category $S^{-1}\mathcal{C}$ this implies that $Q(X) = 0$. Hence (2) $\Rightarrow$ (1). Similarly, (3) $\Rightarrow$ (1). Suppose that $Q(X) = 0$. This implies that the morphism $f : 0 \rightarrow X$ is transformed into an isomorphism in $S^{-1}\mathcal{C}$. Hence by Categories, Lemma ?? there exists a morphism $g : Z \rightarrow 0$ such that $fg \in S$. This proves (1) $\Rightarrow$ (3). Similarly, (1) $\Rightarrow$ (2). $\square$

Lemma 8.4. *Let $\mathcal{A}$ be an abelian category.*

- (1) *If S is a left multiplicative system, then the category $S^{-1}\mathcal{A}$ has cokernels and the functor $Q : \mathcal{A} \rightarrow S^{-1}\mathcal{A}$ commutes with them.*
- (2) *If S is a right multiplicative system, then the category $S^{-1}\mathcal{A}$ has kernels and the functor $Q : \mathcal{A} \rightarrow S^{-1}\mathcal{A}$ commutes with them.*
- (3) *If S is a multiplicative system, then the category $S^{-1}\mathcal{A}$ is abelian and the functor $Q : \mathcal{A} \rightarrow S^{-1}\mathcal{A}$ is exact.*

Proof. Assume S is a left multiplicative system. Let $a : X \rightarrow Y$ be a morphism of $S^{-1}\mathcal{A}$. Then $a = s^{-1}f$ for some $s : Y \rightarrow Y'$ in S and $f : X \rightarrow Y'$. Since $Q(s)$ is an isomorphism we see that the existence of $\text{Coker}(a : X \rightarrow Y)$ is equivalent to the existence of $\text{Coker}(Q(f) : X \rightarrow Y')$. Since $\text{Coker}(Q(f))$ is the coequalizer of 0 and $Q(f)$ we see that $\text{Coker}(Q(f))$ is represented by $Q(\text{Coker}(f))$ by Categories, Lemma ?. This proves (1).

Part (2) is dual to part (1).

If S is a multiplicative system, then S is both a left and a right multiplicative system. Thus we see that $S^{-1}\mathcal{A}$ has kernels and cokernels and Q commutes with kernels and cokernels. To finish the proof of (3) we have to show that $\text{Coim} = \text{Im}$ in $S^{-1}\mathcal{A}$. Again using that any arrow in $S^{-1}\mathcal{A}$ is isomorphic to an arrow $Q(f)$ we see that the result follows from the result for $\mathcal{A}$. $\square$

9. Jordan-Hölder

The Jordan-Hölder lemma is Lemma 9.7. First we state some definitions.

Definition 9.1. Let $\mathcal{A}$ be an abelian category. An object A of $\mathcal{A}$ is said to be *simple* if it is nonzero and the only subobjects of A are 0 and A .

Definition 9.2. Let $\mathcal{A}$ be an abelian category.

- (1) We say an object A of $\mathcal{A}$ is *Artinian* if and only if it satisfies the descending chain condition for subobjects.
- (2) We say $\mathcal{A}$ is *Artinian* if every object of $\mathcal{A}$ is Artinian.

Definition 9.3. Let $\mathcal{A}$ be an abelian category.

- (1) We say an object A of $\mathcal{A}$ is *Noetherian* if and only if it satisfies the ascending chain condition for subobjects.

(2) We say $\mathcal{A}$ is *Noetherian* if every object of $\mathcal{A}$ is Noetherian.

Lemma 9.4. *Let $\mathcal{A}$ be an abelian category. Let $0 \rightarrow A_1 \rightarrow A_2 \rightarrow A_3 \rightarrow 0$ be a short exact sequence of $\mathcal{A}$. Then A_2 is Artinian if and only if A_1 and A_3 are Artinian.*

Proof. Omitted. $\square$

Lemma 9.5. *Let $\mathcal{A}$ be an abelian category. Let $0 \rightarrow A_1 \rightarrow A_2 \rightarrow A_3 \rightarrow 0$ be a short exact sequence of $\mathcal{A}$. Then A_2 is Noetherian if and only if A_1 and A_3 are Noetherian.*

Proof. Omitted. $\square$

Lemma 9.6. *Let $\mathcal{A}$ be an abelian category. Let A be an object of $\mathcal{A}$. The following are equivalent*

- (1) *A is Artinian and Noetherian, and*
- (2) *there exists a filtration $0 \subset A_1 \subset A_2 \subset \dots \subset A_n = A$ by subobjects such that A_i/A_{i-1} is simple for $i = 1, \dots, n$.*

Proof. Assume (1). If A is zero, then (2) holds. If A is not zero, then there exists a smallest nonzero object $A_1 \subset A$ by the Artinian property. Of course A_1 is simple. If $A_1 = A$, then we are done. If not, then we can find $A_1 \subset A_2 \subset A$ minimal with $A_2 \neq A_1$. Then A_2/A_1 is simple. Continuing in this way, we can find a sequence $0 \subset A_1 \subset A_2 \subset \dots$ of subobjects of A such that A_i/A_{i-1} is simple. Since A is Noetherian, we conclude that the process stops. Hence (2) follows.

Assume (2). We will prove (1) by induction on n . If $n = 1$, then A is simple and clearly Noetherian and Artinian. If the result holds for $n - 1$, then we use the short exact sequence $0 \rightarrow A_{n-1} \rightarrow A_n \rightarrow A_n/A_{n-1} \rightarrow 0$ and Lemmas 9.4 and 9.5 to conclude for n . $\square$

Lemma 9.7 (Jordan-Hölder). *Let $\mathcal{A}$ be an abelian category. Let A be an object of $\mathcal{A}$ satisfying the equivalent conditions of Lemma 9.6. Given two filtrations*

$$0 \subset A_1 \subset A_2 \subset \dots \subset A_n = A \quad \text{and} \quad 0 \subset B_1 \subset B_2 \subset \dots \subset B_m = A$$

with $S_i = A_i/A_{i-1}$ and $T_j = B_j/B_{j-1}$ simple objects we have $n = m$ and there exists a permutation σ of $\{1, \dots, n\}$ such that $S_i \cong T_{\sigma(i)}$ for all $i \in \{1, \dots, n\}$.

Proof. Let j be the smallest index such that $A_1 \subset B_j$. Then the map $S_1 = A_1 \rightarrow B_j/B_{j-1} = T_j$ is an isomorphism. Moreover, the object $A/A_1 = A_n/A_1 = B_m/A_1$ has the two filtrations

$$0 \subset A_2/A_1 \subset A_3/A_1 \subset \dots \subset A_n/A_1$$

and

$$0 \subset (B_1 + A_1)/A_1 \subset \dots \subset (B_{j-1} + A_1)/A_1 = B_j/A_1 \subset B_{j+1}/A_1 \subset \dots \subset B_m/A_1$$

We conclude by induction. $\square$

10. Serre subcategories

In [Ser53, Chapter I, Section 1] a notion of a “class” of abelian groups is defined. This notion has been extended to abelian categories by many authors (in slightly different ways). We will use the following variant which is virtually identical to Serre’s original definition.

Definition 10.1. Let $\mathcal{A}$ be an abelian category.

- (1) A *Serre subcategory* of $\mathcal{A}$ is a nonempty full subcategory $\mathcal{C}$ of $\mathcal{A}$ such that given an exact sequence¹

$$A \rightarrow B \rightarrow C$$

with $A, C \in \text{Ob}(\mathcal{C})$, then also $B \in \text{Ob}(\mathcal{C})$.

- (2) A *weak Serre subcategory* of $\mathcal{A}$ is a nonempty full subcategory $\mathcal{C}$ of $\mathcal{A}$ such that given an exact sequence

$$A_0 \rightarrow A_1 \rightarrow A_2 \rightarrow A_3 \rightarrow A_4$$

with A_0, A_1, A_3, A_4 in $\mathcal{C}$, then also A_2 in $\mathcal{C}$.

In some references the second notion is called a “thick” subcategory and in other references the first notion is called a “thick” subcategory. However, it seems that the notion of a Serre subcategory is universally accepted to be the one defined above. Note that in both cases the category $\mathcal{C}$ is abelian and that the inclusion functor $\mathcal{C} \rightarrow \mathcal{A}$ is a fully faithful exact functor. Let’s characterize these types of subcategories in more detail.

Lemma 10.2. *Let $\mathcal{A}$ be an abelian category. Let $\mathcal{C}$ be a subcategory of $\mathcal{A}$. Then $\mathcal{C}$ is a Serre subcategory if and only if the following conditions are satisfied:*

- (1) $0 \in \text{Ob}(\mathcal{C})$,
- (2) $\mathcal{C}$ is a strictly full subcategory of $\mathcal{A}$,
- (3) any subobject or quotient of an object of $\mathcal{C}$ is an object of $\mathcal{C}$,
- (4) if $A \in \text{Ob}(\mathcal{A})$ is an extension of objects of $\mathcal{C}$ then also $A \in \text{Ob}(\mathcal{C})$.

Moreover, a Serre subcategory is an abelian category and the inclusion functor is exact.

Proof. Omitted. □

Lemma 10.3. *Let $\mathcal{A}$ be an abelian category. Let $\mathcal{C}$ be a subcategory of $\mathcal{A}$. Then $\mathcal{C}$ is a weak Serre subcategory if and only if the following conditions are satisfied:*

- (1) $0 \in \text{Ob}(\mathcal{C})$,
- (2) $\mathcal{C}$ is a strictly full subcategory of $\mathcal{A}$,
- (3) kernels and cokernels in $\mathcal{A}$ of morphisms between objects of $\mathcal{C}$ are in $\mathcal{C}$,
- (4) if $A \in \text{Ob}(\mathcal{A})$ is an extension of objects of $\mathcal{C}$ then also $A \in \text{Ob}(\mathcal{C})$.

Moreover, a weak Serre subcategory is an abelian category and the inclusion functor is exact.

Proof. Omitted. □

Lemma 10.4. *Let $\mathcal{A}, \mathcal{B}$ be abelian categories. Let $F : \mathcal{A} \rightarrow \mathcal{B}$ be an exact functor. Then the full subcategory of objects C of $\mathcal{A}$ such that $F(C) = 0$ forms a Serre subcategory of $\mathcal{A}$.*

Proof. Omitted. □

Definition 10.5. Let $\mathcal{A}, \mathcal{B}$ be abelian categories. Let $F : \mathcal{A} \rightarrow \mathcal{B}$ be an exact functor. Then the full subcategory of objects C of $\mathcal{A}$ such that $F(C) = 0$ is called the *kernel of the functor F* , and is sometimes denoted $\text{Ker}(F)$.

¹By Definition 5.7 this means $\text{Im}(A \rightarrow B) = \text{Ker}(B \rightarrow C)$.

Any Serre subcategory of an abelian category is the kernel of an exact functor. In Examples, Section ?? we discuss this for Serre's original example of torsion groups.

Lemma 10.6. *Let $\mathcal{A}$ be an abelian category. Let $\mathcal{C} \subset \mathcal{A}$ be a Serre subcategory. There exists an abelian category $\mathcal{A}/\mathcal{C}$ and an exact functor*

$$F : \mathcal{A} \longrightarrow \mathcal{A}/\mathcal{C}$$

which is essentially surjective and whose kernel is $\mathcal{C}$. The category $\mathcal{A}/\mathcal{C}$ and the functor F are characterized by the following universal property: For any exact functor $G : \mathcal{A} \rightarrow \mathcal{B}$ such that $\mathcal{C} \subset \text{Ker}(G)$ there exists a factorization $G = H \circ F$ for a unique exact functor $H : \mathcal{A}/\mathcal{C} \rightarrow \mathcal{B}$.

Proof. Consider the set of arrows of $\mathcal{A}$ defined by the following formula

$$S = \{f \in \text{Arrows}(\mathcal{A}) \mid \text{Ker}(f), \text{Coker}(f) \in \text{Ob}(\mathcal{C})\}.$$

We claim that S is a multiplicative system. To prove this we have to check MS1, MS2, MS3, see Categories, Definition ??.

It is clear that identities are elements of S . Suppose that $f : A \rightarrow B$ and $g : B \rightarrow C$ are elements of S . There are exact sequences

$$\begin{aligned} 0 \rightarrow \text{Ker}(f) \rightarrow \text{Ker}(gf) \rightarrow \text{Ker}(g) \\ \text{Coker}(f) \rightarrow \text{Coker}(gf) \rightarrow \text{Coker}(g) \rightarrow 0 \end{aligned}$$

Hence it follows that $gf \in S$. This proves MS1. (In fact, a similar argument will show that S is a saturated multiplicative system, see Categories, Definition ??.)

Consider a solid diagram

$$\begin{array}{ccc} A & \xrightarrow{g} & B \\ \downarrow t & & \downarrow s \\ C & \xrightarrow{f} & C \amalg_A B \end{array}$$

with $t \in S$. Set $W = C \amalg_A B = \text{Coker}((t, -g) : A \rightarrow C \oplus B)$. Then $\text{Ker}(t) \rightarrow \text{Ker}(s)$ is surjective and $\text{Coker}(t) \rightarrow \text{Coker}(s)$ is an isomorphism. Hence s is an element of S . This proves LMS2 and the proof of RMS2 is dual.

Finally, consider morphisms $f, g : B \rightarrow C$ and a morphism $s : A \rightarrow B$ in S such that $f \circ s = g \circ s$. This means that $(f - g) \circ s = 0$. In turn this means that $I = \text{Im}(f - g) \subset C$ is a quotient of $\text{Coker}(s)$ hence an object of $\mathcal{C}$. Thus $t : C \rightarrow C' = C/I$ is an element of S such that $t \circ (f - g) = 0$, i.e., such that $t \circ f = t \circ g$. This proves LMS3 and the proof of RMS3 is dual.

Having proved that S is a multiplicative system we set $\mathcal{A}/\mathcal{C} = S^{-1}\mathcal{A}$, and we set F equal to the localization functor Q . By Lemma 8.4 the category $\mathcal{A}/\mathcal{C}$ is abelian and F is exact. If X is in the kernel of $F = Q$, then by Lemma 8.3 we see that $0 : X \rightarrow Z$ is an element of S and hence X is an object of $\mathcal{C}$, i.e., the kernel of F is $\mathcal{C}$. Finally, if G is as in the statement of the lemma, then G turns every element of S into an isomorphism. Hence we obtain the functor $H : \mathcal{A}/\mathcal{C} \rightarrow \mathcal{B}$ from the universal property of localization, see Categories, Lemma ??. We still have to show the functor H is exact. To do this it suffices to show that H commutes with taking kernels and cokernels, see Lemma 7.2. Let $A \rightarrow B$ be a morphism in $\mathcal{A}/\mathcal{C}$. We may represent $A \rightarrow B$ as fs^{-1} where $s : A' \rightarrow A$ is in S and $f : A' \rightarrow B$ an arbitrary morphism of $\mathcal{A}$. Since $F = Q$ maps s to an isomorphism in the quotient category $\mathcal{A}/\mathcal{C}$, it suffices to show that H commutes with taking kernels and cokernels of

morphisms $f : A \rightarrow B$ of $\mathcal{A}$. But here we have $H(f) = G(f)$ and the result follows from the fact that G is exact. $\square$

Lemma 10.7. *Let $\mathcal{A}, \mathcal{B}$ be abelian categories. Let $F : \mathcal{A} \rightarrow \mathcal{B}$ be an exact functor. Let $\mathcal{C} \subset \mathcal{A}$ be a Serre subcategory contained in the kernel of F . Then $\mathcal{C} = \text{Ker}(F)$ if and only if the induced functor $\overline{F} : \mathcal{A}/\mathcal{C} \rightarrow \mathcal{B}$ (Lemma 10.6) is faithful.*

Proof. We will use the results of Lemma 10.6 without further mention. The “only if” direction is true because the kernel of $\overline{F}$ is zero by construction. Namely, if $f : X \rightarrow Y$ is a morphism in $\mathcal{A}/\mathcal{C}$ such that $\overline{F}(f) = 0$, then $\overline{F}(\text{Im}(f)) = \text{Im}(\overline{F}(f)) = 0$, hence $\text{Im}(f) = 0$ by the assumption on the kernel of F . Thus $f = 0$.

For the “if” direction, let X be an object of $\mathcal{A}$ such that $F(X) = 0$. Then $\overline{F}(\text{id}_X) = \text{id}_{\overline{F}(X)} = 0$, thus $\text{id}_X = 0$ in $\mathcal{A}/\mathcal{C}$ by faithfulness of $\overline{F}$. Hence $X = 0$ in $\mathcal{A}/\mathcal{C}$, that is $X \in \text{Ob}(\mathcal{C})$. $\square$

11. K-groups

A tiny bit about K_0 of an abelian category.

Definition 11.1. Let $\mathcal{A}$ be an abelian category. We denote $K_0(\mathcal{A})$ the *zeroth K-group of $\mathcal{A}$* . It is the abelian group constructed as follows. Take the free abelian group on the objects of $\mathcal{A}$ and for every short exact sequence $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$ impose the relation $[B] - [A] - [C] = 0$.

Another way to say this is that there is a presentation

$$\bigoplus_{A \rightarrow B \rightarrow C \text{ ses}} \mathbf{Z}[A \rightarrow B \rightarrow C] \longrightarrow \bigoplus_{A \in \text{Ob}(\mathcal{A})} \mathbf{Z}[A] \longrightarrow K_0(\mathcal{A}) \longrightarrow 0$$

where the first map sends $[A \rightarrow B \rightarrow C]$ to $[B] - [A] - [C]$. The short exact sequence $0 \rightarrow 0 \rightarrow 0 \rightarrow 0 \rightarrow 0$ leads to the relation $[0] = 0$ in $K_0(\mathcal{A})$. There are no set-theoretical issues as all of our categories are “small” if not mentioned otherwise. Some examples of K -groups for categories of modules over rings were computed in Algebra, Section ??.

Lemma 11.2. *Let $F : \mathcal{A} \rightarrow \mathcal{B}$ be an exact functor between abelian categories. Then F induces a homomorphism of K -groups $K_0(F) : K_0(\mathcal{A}) \rightarrow K_0(\mathcal{B})$ by simply setting $K_0(F)([A]) = [F(A)]$.*

Proof. Proves itself. $\square$

Suppose we are given an object M of an abelian category $\mathcal{A}$ and a complex of the form

$$(11.2.1) \quad \dots \longrightarrow M \xrightarrow{\varphi} M \xrightarrow{\psi} M \xrightarrow{\varphi} M \longrightarrow \dots$$

In this situation we define

$$H^0(M, \varphi, \psi) = \text{Ker}(\psi) / \text{Im}(\varphi), \quad \text{and} \quad H^1(M, \varphi, \psi) = \text{Ker}(\varphi) / \text{Im}(\psi).$$

Lemma 11.3. *Let $\mathcal{A}$ be an abelian category. Let $\mathcal{C} \subset \mathcal{A}$ be a Serre subcategory and set $\mathcal{B} = \mathcal{A}/\mathcal{C}$.*

- (1) *The exact functors $\mathcal{C} \rightarrow \mathcal{A}$ and $\mathcal{A} \rightarrow \mathcal{B}$ induce an exact sequence*

$$K_0(\mathcal{C}) \rightarrow K_0(\mathcal{A}) \rightarrow K_0(\mathcal{B}) \rightarrow 0$$

of K -groups, and

- (2) the kernel of $K_0(\mathcal{C}) \rightarrow K_0(\mathcal{A})$ is equal to the collection of elements of the form

$$[H^0(M, \varphi, \psi)] - [H^1(M, \varphi, \psi)]$$

where (M, φ, ψ) is a complex as in (11.2.1) with the property that it becomes exact in $\mathcal{B}$; in other words that $H^0(M, \varphi, \psi)$ and $H^1(M, \varphi, \psi)$ are objects of $\mathcal{C}$.

Proof. Proof of (1). It is clear that $K_0(\mathcal{A}) \rightarrow K_0(\mathcal{B})$ is surjective and that the composition $K_0(\mathcal{C}) \rightarrow K_0(\mathcal{A}) \rightarrow K_0(\mathcal{B})$ is zero. Let $x \in K_0(\mathcal{A})$ be an element mapping to zero in $K_0(\mathcal{B})$. We can write $x = [A] - [A']$ with A, A' in $\mathcal{A}$ (fun exercise). Denote B, B' the corresponding objects of $\mathcal{B}$. The fact that x maps to zero in $K_0(\mathcal{B})$ means that there exists a finite set $I = I^+ \amalg I^-$, for each $i \in I$ a short exact sequence

$$0 \rightarrow B_i \rightarrow B'_i \rightarrow B''_i \rightarrow 0$$

in $\mathcal{B}$ such that we have

$$[B] - [B'] = \sum_{i \in I^+} ([B'_i] - [B_i] - [B''_i]) - \sum_{i \in I^-} ([B'_i] - [B_i] - [B''_i])$$

in the free abelian group on isomorphism classes of objects of $\mathcal{B}$. We can rewrite this as

$$[B] + \sum_{i \in I^+} ([B_i] + [B''_i]) + \sum_{i \in I^-} [B'_i] = [B'] + \sum_{i \in I^-} ([B_i] + [B''_i]) + \sum_{i \in I^+} [B'_i].$$

Since the right and left hand side should contain the same isomorphism classes of objects of $\mathcal{B}$ counted with multiplicity, this means there should be a bijection

$$\tau : \{B\} \amalg \{B_i, B''_i; i \in I^+\} \amalg \{B'_i; i \in I^-\} \longrightarrow \{B'\} \amalg \{B_i, B''_i; i \in I^-\} \amalg \{B'_i; i \in I^+\}$$

such that N and $\tau(N)$ are isomorphic in $\mathcal{B}$. The proofs of Lemmas 10.6 and 8.4 show that we can choose for $i \in I$ a short exact sequence

$$0 \rightarrow A_i \rightarrow A'_i \rightarrow A''_i \rightarrow 0$$

in $\mathcal{A}$ such that B_i, B'_i, B''_i are isomorphic to the images of A_i, A'_i, A''_i in $\mathcal{B}$. This implies that the corresponding bijection

$$\tau : \{A\} \amalg \{A_i, A''_i; i \in I^+\} \amalg \{A'_i; i \in I^-\} \longrightarrow \{A'\} \amalg \{A_i, A''_i; i \in I^-\} \amalg \{A'_i; i \in I^+\}$$

satisfies the property that M and $\tau(M)$ are objects of $\mathcal{A}$ which become isomorphic in $\mathcal{B}$. This means $[M] - [\tau(M)]$ is in the image of $K_0(\mathcal{C}) \rightarrow K_0(\mathcal{A})$. Namely, the isomorphism in $\mathcal{B}$ is given by a diagram $M \leftarrow M' \rightarrow \tau(M)$ in $\mathcal{A}$ where both $M' \rightarrow M$ and $M' \rightarrow \tau(M)$ have kernel and cokernel in $\mathcal{C}$. Working backwards we conclude that $x = [A] - [A']$ is in the image of $K_0(\mathcal{C}) \rightarrow K_0(\mathcal{A})$ and the proof of part (1) is complete.

Proof of (2). The proof is similar to the proof of (1) but slightly more bookkeeping is involved. First we remark that any class of the type $[H^0(M, \varphi, \psi)] - [H^1(M, \varphi, \psi)]$ is zero in $K_0(\mathcal{A})$ by the following calculation

$$\begin{aligned} 0 &= [M] - [M] \\ &= [\text{Ker}(\varphi)] + [\text{Im}(\varphi)] - [\text{Ker}(\psi)] - [\text{Im}(\psi)] \\ &= [\text{Ker}(\varphi)/\text{Im}(\psi)] - [\text{Ker}(\psi)/\text{Im}(\varphi)] \\ &= [H^1(M, \varphi, \psi)] - [H^0(M, \varphi, \psi)] \end{aligned}$$

as desired. Hence it suffices to show that any element in the kernel of $K_0(\mathcal{C}) \rightarrow K_0(\mathcal{A})$ is of this form.

Any element x in $K_0(\mathcal{C})$ can be represented as the difference $x = [P] - [Q]$ of two objects of $\mathcal{C}$ (fun exercise). Suppose that this element maps to zero in $K_0(\mathcal{A})$. This means that there exist

- (1) a finite set $I = I^+ \amalg I^-$,
- (2) for $i \in I$ a short exact sequence $0 \rightarrow A_i \rightarrow B_i \rightarrow C_i \rightarrow 0$ in $\mathcal{A}$

such that

$$[P] - [Q] = \sum_{i \in I^+} ([B_i] - [A_i] - [C_i]) - \sum_{i \in I^-} ([B_i] - [A_i] - [C_i])$$

in the free abelian group on the objects of $\mathcal{A}$. We can rewrite this as

$$[P] + \sum_{i \in I^+} ([A_i] + [C_i]) + \sum_{i \in I^-} [B_i] = [Q] + \sum_{i \in I^-} ([A_i] + [C_i]) + \sum_{i \in I^+} [B_i].$$

Since the right and left hand side should contain the same objects of $\mathcal{A}$ counted with multiplicity, this means there should be a bijection τ between the terms which occur above. Set

$$T^+ = \{p\} \amalg \{a, c\} \times I^+ \amalg \{b\} \times I^-$$

and

$$T^- = \{q\} \amalg \{a, c\} \times I^- \amalg \{b\} \times I^+.$$

Set $T = T^+ \amalg T^- = \{p, q\} \amalg \{a, b, c\} \times I$. For $t \in T$ define

$$O(t) = \begin{cases} P & \text{if } t = p \\ Q & \text{if } t = q \\ A_i & \text{if } t = (a, i) \\ B_i & \text{if } t = (b, i) \\ C_i & \text{if } t = (c, i) \end{cases}$$

Hence we can view $\tau : T^+ \rightarrow T^-$ as a bijection such that $O(t) = O(\tau(t))$ for all $t \in T^+$. Let $t_0^- = \tau(p)$ and let $t_0^+ \in T^+$ be the unique element such that $\tau(t_0^+) = q$. Consider the object

$$M^+ = \bigoplus_{t \in T^+} O(t)$$

By using τ we see that it is equal to the object

$$M^- = \bigoplus_{t \in T^-} O(t)$$

Consider the map

$$\varphi : M^+ \longrightarrow M^-$$

which on the summand $O(t) = A_i$ corresponding to $t = (a, i)$, $i \in I^+$ uses the map $A_i \rightarrow B_i$ into the summand $O((b, i)) = B_i$ of M^- and on the summand $O(t) = B_i$ corresponding to (b, i) , $i \in I^-$ uses the map $B_i \rightarrow C_i$ into the summand $O((c, i)) = C_i$ of M^- . The map is zero on the summands corresponding to p and (c, i) , $i \in I^+$. Similarly, consider the map

$$\psi : M^- \longrightarrow M^+$$

which on the summand $O(t) = A_i$ corresponding to $t = (a, i)$, $i \in I^-$ uses the map $A_i \rightarrow B_i$ into the summand $O((b, i)) = B_i$ of M^+ and on the summand $O(t) = B_i$ corresponding to (b, i) , $i \in I^+$ uses the map $B_i \rightarrow C_i$ into the summand $O((c, i)) = C_i$ of M^+ . The map is zero on the summands corresponding to q and (c, i) , $i \in I^-$.

Note that the kernel of φ is equal to the direct sum of the summand P and the summands $O((c, i)) = C_i$, $i \in I^+$ and the subobjects A_i inside the summands

$O((b, i)) = B_i$, $i \in I^-$. The image of ψ is equal to the direct sum of the summands $O((c, i)) = C_i$, $i \in I^+$ and the subobjects A_i inside the summands $O((b, i)) = B_i$, $i \in I^-$. In other words we see that

$$P \cong \text{Ker}(\varphi) / \text{Im}(\psi).$$

In exactly the same way we see that

$$Q \cong \text{Ker}(\psi) / \text{Im}(\varphi).$$

Since as we remarked above the existence of the bijection τ shows that $M^+ = M^-$ we see that the lemma follows. $\square$

12. Cohomological delta-functors

Definition 12.1. Let $\mathcal{A}, \mathcal{B}$ be abelian categories. A *cohomological δ -functor* or simply a *δ -functor* from $\mathcal{A}$ to $\mathcal{B}$ is given by the following data:

- (1) a collection $F^n : \mathcal{A} \rightarrow \mathcal{B}$, $n \geq 0$ of additive functors, and
- (2) for every short exact sequence $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$ of $\mathcal{A}$ a collection $\delta_{A \rightarrow B \rightarrow C} : F^n(C) \rightarrow F^{n+1}(A)$, $n \geq 0$ of morphisms of $\mathcal{B}$.

These data are assumed to satisfy the following axioms

- (1) for every short exact sequence as above the sequence

$$\begin{array}{ccccccc} 0 & \longrightarrow & F^0(A) & \longrightarrow & F^0(B) & \longrightarrow & F^0(C) \\ & & & & \searrow \delta_{A \rightarrow B \rightarrow C} & & \\ & & F^1(A) & \longrightarrow & F^1(B) & \longrightarrow & F^1(C) \\ & & & & \searrow \delta_{A \rightarrow B \rightarrow C} & & \\ & & F^2(A) & \longrightarrow & F^2(B) & \longrightarrow & \dots \end{array}$$

is exact, and

- (2) for every morphism $(A \rightarrow B \rightarrow C) \rightarrow (A' \rightarrow B' \rightarrow C')$ of short exact sequences of $\mathcal{A}$ the diagrams

$$\begin{array}{ccc} F^n(C) & \xrightarrow{\delta_{A \rightarrow B \rightarrow C}} & F^{n+1}(A) \\ \downarrow & & \downarrow \\ F^n(C') & \xrightarrow{\delta_{A' \rightarrow B' \rightarrow C'}} & F^{n+1}(A') \end{array}$$

are commutative.

Note that this in particular implies that F^0 is left exact.

Definition 12.2. Let $\mathcal{A}, \mathcal{B}$ be abelian categories. Let (F^n, δ_F) and (G^n, δ_G) be δ -functors from $\mathcal{A}$ to $\mathcal{B}$. A *morphism of δ -functors from F to G* is a collection of transformations of functors $t^n : F^n \rightarrow G^n$, $n \geq 0$ such that for every short exact sequence $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$ of $\mathcal{A}$ the diagrams

$$\begin{array}{ccc} F^n(C) & \xrightarrow{\delta_{F, A \rightarrow B \rightarrow C}} & F^{n+1}(A) \\ t^n \downarrow & & \downarrow t^{n+1} \\ G^n(C) & \xrightarrow{\delta_{G, A \rightarrow B \rightarrow C}} & G^{n+1}(A) \end{array}$$

are commutative.

Definition 12.3. Let $\mathcal{A}, \mathcal{B}$ be abelian categories. Let $F = (F^n, \delta_F)$ be a δ -functor from $\mathcal{A}$ to $\mathcal{B}$. We say F is a *universal δ -functor* if and only if for every δ -functor $G = (G^n, \delta_G)$ and any morphism of functors $t : F^0 \rightarrow G^0$ there exists a unique morphism of δ -functors $\{t^n\}_{n \geq 0} : F \rightarrow G$ such that $t = t^0$.

Lemma 12.4. Let $\mathcal{A}, \mathcal{B}$ be abelian categories. Let $F = (F^n, \delta_F)$ be a δ -functor from $\mathcal{A}$ to $\mathcal{B}$. Suppose that for every $n > 0$ and any $A \in \text{Ob}(\mathcal{A})$ there exists an injective morphism $u : A \rightarrow B$ (depending on A and n) such that $F^n(u) : F^n(A) \rightarrow F^n(B)$ is zero. Then F is a universal δ -functor.

Proof. Let $G = (G^n, \delta_G)$ be a δ -functor from $\mathcal{A}$ to $\mathcal{B}$ and let $t : F^0 \rightarrow G^0$ be a morphism of functors. We have to show there exists a unique morphism of δ -functors $\{t^n\}_{n \geq 0} : F \rightarrow G$ such that $t = t^0$. We construct t^n by induction on n . For $n = 0$ we set $t^0 = t$. Suppose we have already constructed a unique sequence of transformations of functors t^i for $i \leq n$ compatible with the maps δ in degrees $\leq n$.

Let $A \in \text{Ob}(\mathcal{A})$. By assumption we may choose an embedding $u : A \rightarrow B$ such that $F^{n+1}(u) = 0$. Let $C = B/u(A)$. The long exact cohomology sequence for the short exact sequence $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$ and the δ -functor F gives that $F^{n+1}(A) = \text{Coker}(F^n(B) \rightarrow F^n(C))$ by our choice of u . Since we have already defined t^n we can set

$$t_A^{n+1} : F^{n+1}(A) \rightarrow G^{n+1}(A)$$

equal to the unique map such that

$$\begin{array}{ccc} \text{Coker}(F^n(B) \rightarrow F^n(C)) & \xrightarrow{t^n} & \text{Coker}(G^n(B) \rightarrow G^n(C)) \\ \delta_{F, A \rightarrow B \rightarrow C} \downarrow & & \downarrow \delta_{G, A \rightarrow B \rightarrow C} \\ F^{n+1}(A) & \xrightarrow{t_A^{n+1}} & G^{n+1}(A) \end{array}$$

commutes. This is clearly uniquely determined by the requirements imposed. We omit the verification that this defines a transformation of functors. $\square$

Lemma 12.5. Let $\mathcal{A}, \mathcal{B}$ be abelian categories. Let $F : \mathcal{A} \rightarrow \mathcal{B}$ be a functor. If there exists a universal δ -functor (F^n, δ_F) from $\mathcal{A}$ to $\mathcal{B}$ with $F^0 = F$, then it is determined up to unique isomorphism of δ -functors.

Proof. Immediate from the definitions. $\square$

13. Complexes

Of course the notions of a chain complex and a cochain complex are dual and you only have to read one of the two parts of this section. So pick the one you like. (Actually, this doesn't quite work right since the conventions on numbering things are not adapted to an easy transition between chain and cochain complexes.)

A *chain complex* $A_\bullet$ in an additive category $\mathcal{A}$ is a complex

$$\dots \rightarrow A_{n+1} \xrightarrow{d_{n+1}} A_n \xrightarrow{d_n} A_{n-1} \rightarrow \dots$$

of $\mathcal{A}$. In other words, we are given an object A_i of $\mathcal{A}$ for all $i \in \mathbf{Z}$ and for all $i \in \mathbf{Z}$ a morphism $d_i : A_i \rightarrow A_{i-1}$ such that $d_{i-1} \circ d_i = 0$ for all i . A *morphism of chain*

complexes $f : A_\bullet \rightarrow B_\bullet$ is given by a family of morphisms $f_i : A_i \rightarrow B_i$ such that all the diagrams

$$\begin{array}{ccc} A_i & \xrightarrow{d_i} & A_{i-1} \\ f_i \downarrow & & \downarrow f_{i-1} \\ B_i & \xrightarrow{d_i} & B_{i-1} \end{array}$$

commute. The *category of chain complexes of $\mathcal{A}$* is denoted $\text{Ch}(\mathcal{A})$. The full subcategory consisting of objects of the form

$$\dots \rightarrow A_2 \rightarrow A_1 \rightarrow A_0 \rightarrow 0 \rightarrow 0 \rightarrow \dots$$

is denoted $\text{Ch}_{\geq 0}(\mathcal{A})$. In other words, a chain complex $A_\bullet$ belongs to $\text{Ch}_{\geq 0}(\mathcal{A})$ if and only if $A_i = 0$ for all $i < 0$.

Given an additive category $\mathcal{A}$ we identify $\mathcal{A}$ with the full subcategory of $\text{Ch}(\mathcal{A})$ consisting of chain complexes zero except in degree 0 by the functor

$$\mathcal{A} \longrightarrow \text{Ch}(\mathcal{A}), \quad A \longmapsto (\dots \rightarrow 0 \rightarrow A \rightarrow 0 \rightarrow \dots)$$

By abuse of notation we often denote the object on the right hand side simply A . If we want to stress that we are viewing A as a chain complex we may sometimes use the notation $A[0]$, see Section 14.

A *homotopy* h between a pair of morphisms of chain complexes $f, g : A_\bullet \rightarrow B_\bullet$ is a collection of morphisms $h_i : A_i \rightarrow B_{i+1}$ such that we have

$$f_i - g_i = d_{i+1} \circ h_i + h_{i-1} \circ d_i$$

for all i . Two morphisms $f, g : A_\bullet \rightarrow B_\bullet$ are said to be *homotopic* if a homotopy between f and g exists. Clearly, the notions of chain complex, morphism of chain complexes, and homotopies between morphisms of chain complexes make sense even in a preadditive category.

Lemma 13.1. *Let $\mathcal{A}$ be an additive category. Let $f, g : B_\bullet \rightarrow C_\bullet$ be morphisms of chain complexes. Suppose given morphisms of chain complexes $a : A_\bullet \rightarrow B_\bullet$, and $c : C_\bullet \rightarrow D_\bullet$. If $\{h_i : B_i \rightarrow C_{i+1}\}$ defines a homotopy between f and g , then $\{c_{i+1} \circ h_i \circ a_i\}$ defines a homotopy between $c \circ f \circ a$ and $c \circ g \circ a$.*

Proof. Omitted. □

In particular this means that it makes sense to define the category of chain complexes with maps up to homotopy. We'll return to this later.

Definition 13.2. Let $\mathcal{A}$ be an additive category. We say a morphism $a : A_\bullet \rightarrow B_\bullet$ is a *homotopy equivalence* if there exists a morphism $b : B_\bullet \rightarrow A_\bullet$ such that there exists a homotopy between $b \circ a$ and id_A and there exists a homotopy between $a \circ b$ and id_B . If there exists such a morphism between $A_\bullet$ and $B_\bullet$, then we say that $A_\bullet$ and $B_\bullet$ are *homotopy equivalent*.

In other words, two complexes are homotopy equivalent if they become isomorphic in the category of complexes up to homotopy.

Lemma 13.3. *Let $\mathcal{A}$ be an abelian category.*

- (1) *The category of chain complexes in $\mathcal{A}$ is abelian.*
- (2) *A morphism of complexes $f : A_\bullet \rightarrow B_\bullet$ is injective if and only if each $f_n : A_n \rightarrow B_n$ is injective.*

- (3) A morphism of complexes $f : A_\bullet \rightarrow B_\bullet$ is surjective if and only if each $f_n : A_n \rightarrow B_n$ is surjective.
- (4) A sequence of chain complexes

$$A_\bullet \xrightarrow{f} B_\bullet \xrightarrow{g} C_\bullet$$

is exact at $B_\bullet$ if and only if each sequence

$$A_i \xrightarrow{f_i} B_i \xrightarrow{g_i} C_i$$

is exact at B_i .

Proof. Omitted. □

For any $i \in \mathbf{Z}$ the i th homology group of a chain complex $A_\bullet$ in an abelian category is defined by the following formula

$$H_i(A_\bullet) = \text{Ker}(d_i) / \text{Im}(d_{i+1}).$$

If $f : A_\bullet \rightarrow B_\bullet$ is a morphism of chain complexes of $\mathcal{A}$ then we get an induced morphism $H_i(f) : H_i(A_\bullet) \rightarrow H_i(B_\bullet)$ because clearly $f_i(\text{Ker}(d_i : A_i \rightarrow A_{i-1})) \subset \text{Ker}(d_i : B_i \rightarrow B_{i-1})$, and similarly for $\text{Im}(d_{i+1})$. Thus we obtain a functor

$$H_i : \text{Ch}(\mathcal{A}) \longrightarrow \mathcal{A}.$$

Definition 13.4. Let $\mathcal{A}$ be an abelian category.

- (1) A morphism of chain complexes $f : A_\bullet \rightarrow B_\bullet$ is called a *quasi-isomorphism* if the induced map $H_i(f) : H_i(A_\bullet) \rightarrow H_i(B_\bullet)$ is an isomorphism for all $i \in \mathbf{Z}$.
- (2) A chain complex $A_\bullet$ is called *acyclic* if all of its homology objects $H_i(A_\bullet)$ are zero.

Lemma 13.5. Let $\mathcal{A}$ be an abelian category.

- (1) If the maps $f, g : A_\bullet \rightarrow B_\bullet$ are homotopic, then the induced maps $H_i(f)$ and $H_i(g)$ are equal.
- (2) If the map $f : A_\bullet \rightarrow B_\bullet$ is a homotopy equivalence, then f is a quasi-isomorphism.

Proof. Omitted. □

Lemma 13.6. Let $\mathcal{A}$ be an abelian category. Suppose that

$$0 \rightarrow A_\bullet \rightarrow B_\bullet \rightarrow C_\bullet \rightarrow 0$$

is a short exact sequence of chain complexes of $\mathcal{A}$. Then there is a canonical long exact homology sequence

$$\begin{array}{ccccccc}
 & & \dots & & \dots & & \dots \\
 & & & \swarrow & & \searrow & \\
 & & H_i(A_\bullet) & \longrightarrow & H_i(B_\bullet) & \longrightarrow & H_i(C_\bullet) \\
 & & & \swarrow & & \searrow & \\
 & & H_{i-1}(A_\bullet) & \longrightarrow & H_{i-1}(B_\bullet) & \longrightarrow & H_{i-1}(C_\bullet) \\
 & & \dots & & \dots & & \dots
 \end{array}$$

Proof. Omitted. The maps come from the Snake Lemma 5.17 applied to the diagrams

$$\begin{array}{ccccccc}
 A_i / \text{Im}(d_{A,i+1}) & \longrightarrow & B_i / \text{Im}(d_{B,i+1}) & \longrightarrow & C_i / \text{Im}(d_{C,i+1}) & \longrightarrow & 0 \\
 \downarrow d_{A,i} & & \downarrow d_{B,i} & & \downarrow d_{C,i} & & \\
 0 \longrightarrow & \text{Ker}(d_{A,i-1}) & \longrightarrow & \text{Ker}(d_{B,i-1}) & \longrightarrow & \text{Ker}(d_{C,i-1}) &
 \end{array}$$

□

A *cochain complex* $A^\bullet$ in an additive category $\mathcal{A}$ is a complex

$$\dots \rightarrow A^{n-1} \xrightarrow{d^{n-1}} A^n \xrightarrow{d^n} A^{n+1} \rightarrow \dots$$

of $\mathcal{A}$. In other words, we are given an object A^i of $\mathcal{A}$ for all $i \in \mathbf{Z}$ and for all $i \in \mathbf{Z}$ a morphism $d^i : A^i \rightarrow A^{i+1}$ such that $d^{i+1} \circ d^i = 0$ for all i . A *morphism of cochain complexes* $f : A^\bullet \rightarrow B^\bullet$ is given by a family of morphisms $f^i : A^i \rightarrow B^i$ such that all the diagrams

$$\begin{array}{ccc}
 A^i & \xrightarrow{d^i} & A^{i+1} \\
 f^i \downarrow & & \downarrow f^{i+1} \\
 B^i & \xrightarrow{d^i} & B^{i+1}
 \end{array}$$

commute. The *category of cochain complexes of $\mathcal{A}$* is denoted $\text{CoCh}(\mathcal{A})$. The full subcategory consisting of objects of the form

$$\dots \rightarrow 0 \rightarrow 0 \rightarrow A^0 \rightarrow A^1 \rightarrow A^2 \rightarrow \dots$$

is denoted $\text{CoCh}_{\geq 0}(\mathcal{A})$. In other words, a cochain complex $A^\bullet$ belongs to the subcategory $\text{CoCh}_{\geq 0}(\mathcal{A})$ if and only if $A^i = 0$ for all $i < 0$.

Given an additive category $\mathcal{A}$ we identify $\mathcal{A}$ with the full subcategory of $\text{CoCh}(\mathcal{A})$ consisting of cochain complexes zero except in degree 0 by the functor

$$\mathcal{A} \longrightarrow \text{CoCh}(\mathcal{A}), \quad A \longmapsto (\dots \rightarrow 0 \rightarrow A \rightarrow 0 \rightarrow \dots)$$

By abuse of notation we often denote the object on the right hand side simply A . If we want to stress that we are viewing A as a cochain complex we may sometimes use the notation $A[0]$, see Section 14.

A *homotopy* h between a pair of morphisms of cochain complexes $f, g : A^\bullet \rightarrow B^\bullet$ is a collection of morphisms $h^i : A^i \rightarrow B^{i-1}$ such that we have

$$f^i - g^i = d^{i-1} \circ h^i + h^{i+1} \circ d^i$$

for all i . Two morphisms $f, g : A^\bullet \rightarrow B^\bullet$ are said to be *homotopic* if a homotopy between f and g exists. Clearly, the notions of cochain complex, morphism of cochain complexes, and homotopies between morphisms of cochain complexes make sense even in a preadditive category.

Lemma 13.7. *Let $\mathcal{A}$ be an additive category. Let $f, g : B^\bullet \rightarrow C^\bullet$ be morphisms of cochain complexes. Suppose given morphisms of cochain complexes $a : A^\bullet \rightarrow B^\bullet$, and $c : C^\bullet \rightarrow D^\bullet$. If $\{h^i : B^i \rightarrow C^{i-1}\}$ defines a homotopy between f and g , then $\{c^{i-1} \circ h^i \circ a^i\}$ defines a homotopy between $c \circ f \circ a$ and $c \circ g \circ a$.*

Proof. Omitted. □

In particular this means that it makes sense to define the category of cochain complexes with maps up to homotopy. We'll return to this later.

Definition 13.8. Let $\mathcal{A}$ be an additive category. We say a morphism $a : A^\bullet \rightarrow B^\bullet$ is a *homotopy equivalence* if there exists a morphism $b : B^\bullet \rightarrow A^\bullet$ such that there exists a homotopy between $b \circ a$ and id_A and there exists a homotopy between $a \circ b$ and id_B . If there exists such a morphism between $A^\bullet$ and $B^\bullet$, then we say that $A^\bullet$ and $B^\bullet$ are *homotopy equivalent*.

In other words, two complexes are homotopy equivalent if they become isomorphic in the category of complexes up to homotopy.

Lemma 13.9. *Let $\mathcal{A}$ be an abelian category.*

- (1) *The category of cochain complexes in $\mathcal{A}$ is abelian.*
- (2) *A morphism of cochain complexes $f : A^\bullet \rightarrow B^\bullet$ is injective if and only if each $f^n : A^n \rightarrow B^n$ is injective.*
- (3) *A morphism of cochain complexes $f : A^\bullet \rightarrow B^\bullet$ is surjective if and only if each $f^n : A^n \rightarrow B^n$ is surjective.*
- (4) *A sequence of cochain complexes*

$$A^\bullet \xrightarrow{f} B^\bullet \xrightarrow{g} C^\bullet$$

is exact at $B^\bullet$ if and only if each sequence

$$A^i \xrightarrow{f^i} B^i \xrightarrow{g^i} C^i$$

is exact at B^i .

Proof. Omitted. □

For any $i \in \mathbf{Z}$ the i th *cohomology group* of a cochain complex $A^\bullet$ is defined by the following formula

$$H^i(A^\bullet) = \text{Ker}(d^i) / \text{Im}(d^{i-1}).$$

If $f : A^\bullet \rightarrow B^\bullet$ is a morphism of cochain complexes of $\mathcal{A}$ then we get an induced morphism $H^i(f) : H^i(A^\bullet) \rightarrow H^i(B^\bullet)$ because clearly $f^i(\text{Ker}(d^i : A^i \rightarrow A^{i+1})) \subset \text{Ker}(d^i : B^i \rightarrow B^{i+1})$, and similarly for $\text{Im}(d^{i-1})$. Thus we obtain a functor

$$H^i : \text{CoCh}(\mathcal{A}) \longrightarrow \mathcal{A}.$$

Definition 13.10. Let $\mathcal{A}$ be an abelian category.

- (1) A morphism of cochain complexes $f : A^\bullet \rightarrow B^\bullet$ of $\mathcal{A}$ is called a *quasi-isomorphism* if the induced maps $H^i(f) : H^i(A^\bullet) \rightarrow H^i(B^\bullet)$ is an isomorphism for all $i \in \mathbf{Z}$.
- (2) A cochain complex $A^\bullet$ is called *acyclic* if all of its cohomology objects $H^i(A^\bullet)$ are zero.

Lemma 13.11. *Let $\mathcal{A}$ be an abelian category.*

- (1) *If the maps $f, g : A^\bullet \rightarrow B^\bullet$ are homotopic, then the induced maps $H^i(f)$ and $H^i(g)$ are equal.*
- (2) *If $f : A^\bullet \rightarrow B^\bullet$ is a homotopy equivalence, then f is a quasi-isomorphism.*

Proof. Omitted. □

Lemma 13.12. *Let $\mathcal{A}$ be an abelian category. Suppose that*

$$0 \rightarrow A^\bullet \rightarrow B^\bullet \rightarrow C^\bullet \rightarrow 0$$

is a short exact sequence of cochain complexes of $\mathcal{A}$. Then there is a long exact cohomology sequence

$$\begin{array}{ccccccc} & & \dots & & \dots & & \dots \\ & & & \swarrow & & \searrow & \\ & H^i(A^\bullet) & \xleftarrow{\quad} & H^i(B^\bullet) & \xrightarrow{\quad} & H^i(C^\bullet) & \\ & & & \swarrow & & \searrow & \\ & H^{i+1}(A^\bullet) & \xleftarrow{\quad} & H^{i+1}(B^\bullet) & \xrightarrow{\quad} & H^{i+1}(C^\bullet) & \\ & & & \swarrow & & \searrow & \\ & \dots & & \dots & & \dots & \end{array}$$

The construction produces long exact cohomology sequences which are functorial in the short exact sequence and compatible with shifts as in Definition 14.8.

Proof. For the horizontal maps $H^i(A^\bullet) \rightarrow H^i(B^\bullet)$ and $H^i(B^\bullet) \rightarrow H^i(C^\bullet)$ we use the fact that H^i is a functor, see above. For the “boundary map” $H^i(C^\bullet) \rightarrow H^{i+1}(A^\bullet)$ we use the map δ of the Snake Lemma 5.17 applied to the diagram

$$\begin{array}{ccccccc} A^i / \text{Im}(d_A^{i-1}) & \longrightarrow & B^i / \text{Im}(d_B^{i-1}) & \longrightarrow & C^i / \text{Im}(d_C^{i-1}) & \longrightarrow & 0 \\ \downarrow d_A^i & & \downarrow d_B^i & & \downarrow d_C^i & & \\ 0 & \longrightarrow & \text{Ker}(d_A^{i+1}) & \longrightarrow & \text{Ker}(d_B^{i+1}) & \longrightarrow & \text{Ker}(d_C^{i+1}) \end{array}$$

This works as the kernel of the right vertical map is equal to $H^i(C^\bullet)$ and the cokernel of the left vertical map is $H^{i+1}(A^\bullet)$. The exactness of the long sequence is the exactness in part (2) of Lemma 5.17. The functoriality is Lemma 5.18. Compatibility with shifts is immediate from the definitions. $\square$

14. Homotopy and the shift functor

It is an annoying feature that signs and indices have to be part of any discussion of homological algebra².

Definition 14.1. Let $\mathcal{A}$ be an additive category. Let $A_\bullet$ be a chain complex with boundary maps $d_{A,n} : A_n \rightarrow A_{n-1}$. For any $k \in \mathbf{Z}$ we define the k -shifted chain complex $A[k]_\bullet$ as follows:

- (1) we set $A[k]_n = A_{n+k}$, and
- (2) we set $d_{A[k],n} : A[k]_n \rightarrow A[k]_{n-1}$ equal to $d_{A[k],n} = (-1)^k d_{A,n+k}$.

If $f : A_\bullet \rightarrow B_\bullet$ is a morphism of chain complexes, then we let $f[k] : A[k]_\bullet \rightarrow B[k]_\bullet$ be the morphism of chain complexes with $f[k]_n = f_{k+n}$.

Of course this means we have functors $[k] : \text{Ch}(\mathcal{A}) \rightarrow \text{Ch}(\mathcal{A})$ which mutually commute (on the nose, without any intervening isomorphisms of functors), such that $A[k][l]_\bullet = A[k+l]_\bullet$ and with $[0] = \text{id}_{\text{Ch}(\mathcal{A})}$.

²Please let us know if you notice sign errors or if you have improvements to our conventions.

Recall that we view $\mathcal{A}$ as a full subcategory of $\text{Ch}(\mathcal{A})$, see Section 13. Thus for any object A of $\mathcal{A}$ the notation $A[k]$ refers to the unique chain complex zero in all degrees except having A in degree $-k$.

Definition 14.2. Let $\mathcal{A}$ be an abelian category. Let $A_\bullet$ be a chain complex with boundary maps $d_{A,n} : A_n \rightarrow A_{n-1}$. For any $k \in \mathbf{Z}$ we identify $H_{i+k}(A_\bullet) \rightarrow H_i(A[k]_\bullet)$ via the identification $A_{i+k} = A[k]_i$.

This identification is functorial in $A_\bullet$. Note that since no signs are involved in this definition we actually get a compatible system of identifications of all the homology objects $H_{i-k}(A[k]_\bullet)$, which are further compatible with the identifications $A[k][l]_\bullet = A[k+l]_\bullet$ and with $[0] = \text{id}_{\text{Ch}(\mathcal{A})}$.

Let $\mathcal{A}$ be an additive category. Suppose that $A_\bullet$ and $B_\bullet$ are chain complexes, $a, b : A_\bullet \rightarrow B_\bullet$ are morphisms of chain complexes, and $\{h_i : A_i \rightarrow B_{i+1}\}$ is a homotopy between a and b . Recall that this means that $a_i - b_i = d_{i+1} \circ h_i + h_{i-1} \circ d_i$. What if $a = b$? Then we obtain the formula $0 = d_{i+1} \circ h_i + h_{i-1} \circ d_i$, in other words, $-d_{i+1} \circ h_i = h_{i-1} \circ d_i$. By definition above this means the collection $\{h_i\}$ above defines a morphism of chain complexes

$$A_\bullet \rightarrow B[1]_\bullet.$$

Such a thing is the same as a morphism $A[-1]_\bullet \rightarrow B_\bullet$ by our remarks above. This proves the following lemma.

Lemma 14.3. *Let $\mathcal{A}$ be an additive category. Suppose that $A_\bullet$ and $B_\bullet$ are chain complexes. Given any morphism of chain complexes $a : A_\bullet \rightarrow B_\bullet$ there is a bijection between the set of homotopies from a to a and $\text{Mor}_{\text{Ch}(\mathcal{A})}(A_\bullet, B[1]_\bullet)$. More generally, the set of homotopies between a and b is either empty or a principal homogeneous space under the group $\text{Mor}_{\text{Ch}(\mathcal{A})}(A_\bullet, B[1]_\bullet)$.*

Proof. See above. □

Lemma 14.4. *Let $\mathcal{A}$ be an abelian category. Let*

$$0 \rightarrow A_\bullet \rightarrow B_\bullet \rightarrow C_\bullet \rightarrow 0$$

be a short exact sequence of complexes. Suppose that $\{s_n : C_n \rightarrow B_n\}$ is a family of morphisms which split the short exact sequences $0 \rightarrow A_n \rightarrow B_n \rightarrow C_n \rightarrow 0$. Let $\pi_n : B_n \rightarrow A_n$ be the associated projections, see Lemma 5.10. Then the family of morphisms

$$\pi_{n-1} \circ d_{B,n} \circ s_n : C_n \rightarrow A_{n-1}$$

define a morphism of complexes $\delta(s) : C_\bullet \rightarrow A[-1]_\bullet$.

Proof. Denote $i : A_\bullet \rightarrow B_\bullet$ and $q : B_\bullet \rightarrow C_\bullet$ the maps of complexes in the short exact sequence. Then $i_{n-1} \circ \pi_{n-1} \circ d_{B,n} \circ s_n = d_{B,n} \circ s_n - s_{n-1} \circ d_{C,n}$. Hence $i_{n-2} \circ d_{A,n-1} \circ \pi_{n-1} \circ d_{B,n} \circ s_n = d_{B,n-1} \circ (d_{B,n} \circ s_n - s_{n-1} \circ d_{C,n}) = -d_{B,n-1} \circ s_{n-1} \circ d_{C,n}$ as desired. □

Lemma 14.5. *Notation and assumptions as in Lemma 14.4 above. The morphism of complexes $\delta(s) : C_\bullet \rightarrow A[-1]_\bullet$ induces the maps*

$$H_i(\delta(s)) : H_i(C_\bullet) \rightarrow H_i(A[-1]_\bullet) = H_{i-1}(A_\bullet)$$

which occur in the long exact homology sequence associated to the short exact sequence of chain complexes by Lemma 13.6.

Proof. Omitted. □

Lemma 14.6. *Notation and assumptions as in Lemma 14.4 above. Suppose $\{s'_n : C_n \rightarrow B_n\}$ is a second choice of splittings. Write $s'_n = s_n + i_n \circ h_n$ for some unique morphisms $h_n : C_n \rightarrow A_n$. The family of maps $\{h_n : C_n \rightarrow A[-1]_{n+1}\}$ is a homotopy between the associated morphisms $\delta(s), \delta(s') : C_\bullet \rightarrow A[-1]_\bullet$.*

Proof. Omitted. □

Definition 14.7. Let $\mathcal{A}$ be an additive category. Let $A^\bullet$ be a cochain complex with boundary maps $d_A^n : A^n \rightarrow A^{n+1}$. For any $k \in \mathbf{Z}$ we define the k -shifted cochain complex $A[k]^\bullet$ as follows:

- (1) we set $A[k]^n = A^{n+k}$, and
- (2) we set $d_{A[k]}^n : A[k]^n \rightarrow A[k]^{n+1}$ equal to $d_{A[k]}^n = (-1)^k d_A^{n+k}$.

If $f : A^\bullet \rightarrow B^\bullet$ is a morphism of cochain complexes, then we let $f[k] : A[k]^\bullet \rightarrow B[k]^\bullet$ be the morphism of cochain complexes with $f[k]^n = f^{k+n}$.

Of course this means we have functors $[k] : \text{CoCh}(\mathcal{A}) \rightarrow \text{CoCh}(\mathcal{A})$ which mutually commute (on the nose, without any intervening isomorphisms of functors) and such that $A[k][l]^\bullet = A[k+l]^\bullet$ and with $[0] = \text{id}_{\text{CoCh}(\mathcal{A})}$.

Recall that we view $\mathcal{A}$ as a full subcategory of $\text{CoCh}(\mathcal{A})$, see Section 13. Thus for any object A of $\mathcal{A}$ the notation $A[k]$ refers to the unique cochain complex zero in all degrees except having A in degree $-k$.

Definition 14.8. Let $\mathcal{A}$ be an abelian category. Let $A^\bullet$ be a cochain complex with boundary maps $d_A^n : A^n \rightarrow A^{n+1}$. For any $k \in \mathbf{Z}$ we identify $H^{i+k}(A^\bullet) \rightarrow H^i(A[k]^\bullet)$ via the identification $A^{i+k} = A[k]^i$.

This identification is functorial in $A^\bullet$. Note that since no signs are involved in this definition we actually get a compatible system of identifications of all the homology objects $H^{i-k}(A[k]^\bullet)$, which are further compatible with the identifications $A[k][l]^\bullet = A[k+l]^\bullet$ and with $[0] = \text{id}_{\text{CoCh}(\mathcal{A})}$.

Let $\mathcal{A}$ be an additive category. Suppose that $A^\bullet$ and $B^\bullet$ are cochain complexes, $a, b : A^\bullet \rightarrow B^\bullet$ are morphisms of cochain complexes, and $\{h^i : A^i \rightarrow B^{i-1}\}$ is a homotopy between a and b . Recall that this means that $a^i - b^i = d^{i-1} \circ h^i + h^{i+1} \circ d^i$. What if $a = b$? Then we obtain the formula $0 = d^{i-1} \circ h^i + h^{i+1} \circ d^i$, in other words, $-d^{i-1} \circ h^i = h^{i+1} \circ d^i$. By definition above this means the collection $\{h^i\}$ above defines a morphism of cochain complexes

$$A^\bullet \longrightarrow B[-1]^\bullet.$$

Such a thing is the same as a morphism $A[1]^\bullet \rightarrow B^\bullet$ by our remarks above. This proves the following lemma.

Lemma 14.9. *Let $\mathcal{A}$ be an additive category. Suppose that $A^\bullet$ and $B^\bullet$ are cochain complexes. Given any morphism of cochain complexes $a : A^\bullet \rightarrow B^\bullet$ there is a bijection between the set of homotopies from a to a and $\text{Mor}_{\text{CoCh}(\mathcal{A})}(A^\bullet, B[-1]^\bullet)$. More generally, the set of homotopies between a and b is either empty or a principal homogeneous space under the group $\text{Mor}_{\text{CoCh}(\mathcal{A})}(A^\bullet, B[-1]^\bullet)$.*

Proof. See above. □

Lemma 14.10. *Let $\mathcal{A}$ be an additive category. Let*

$$0 \rightarrow A^\bullet \rightarrow B^\bullet \rightarrow C^\bullet \rightarrow 0$$

be a complex (!) of complexes. Suppose that we are given splittings $B^n = A^n \oplus C^n$ compatible with the maps in the displayed sequence. Let $s^n : C^n \rightarrow B^n$ and $\pi^n : B^n \rightarrow A^n$ be the corresponding maps. Then the family of morphisms

$$\pi^{n+1} \circ d_B^n \circ s^n : C^n \rightarrow A^{n+1}$$

define a morphism of complexes $\delta : C^\bullet \rightarrow A[1]^\bullet$.

Proof. Denote $i : A^\bullet \rightarrow B^\bullet$ and $q : B^\bullet \rightarrow C^\bullet$ the maps of complexes in the short exact sequence. Then $i^{n+1} \circ \pi^{n+1} \circ d_B^n \circ s^n = d_B^n \circ s^n - s^{n+1} \circ d_C^n$. Hence $i^{n+2} \circ d_A^{n+1} \circ \pi^{n+1} \circ d_B^n \circ s^n = d_B^{n+1} \circ (d_B^n \circ s^n - s^{n+1} \circ d_C^n) = -d_B^{n+1} \circ s^{n+1} \circ d_C^n$ as desired. $\square$

Lemma 14.11. *Notation and assumptions as in Lemma 14.10 above. Assume in addition that $\mathcal{A}$ is abelian. The morphism of complexes $\delta : C^\bullet \rightarrow A[1]^\bullet$ induces the maps*

$$H^i(\delta) : H^i(C^\bullet) \longrightarrow H^i(A[1]^\bullet) = H^{i+1}(A^\bullet)$$

which occur in the long exact cohomology sequence associated to the short exact sequence of cochain complexes by Lemma 13.12.

Proof. Omitted. $\square$

Lemma 14.12. *Notation and assumptions as in Lemma 14.10. Let $\alpha : A^\bullet \rightarrow B^\bullet$, $\beta : B^\bullet \rightarrow C^\bullet$ be the given morphisms of complexes. Suppose $(s')^n : C^n \rightarrow B^n$ and $(\pi')^n : B^n \rightarrow A^n$ is a second choice of splittings. Write $(s')^n = s^n + \alpha^n \circ h^n$ and $(\pi')^n = \pi^n + g^n \circ \beta^n$ for some unique morphisms $h^n : C^n \rightarrow A^n$ and $g^n : C^n \rightarrow A^n$. Then*

- (1) $g^n = -h^n$, and
- (2) *the family of maps $\{g^n : C^n \rightarrow A[1]^{n-1}\}$ is a homotopy between $\delta, \delta' : C^\bullet \rightarrow A[1]^\bullet$, more precisely $(\delta')^n = \delta^n + g^{n+1} \circ d_C^n + d_{A[1]}^{n-1} \circ g^n$.*

Proof. As $(s')^n$ and $(\pi')^n$ are splittings we have $(\pi')^n \circ (s')^n = 0$. Hence

$$0 = (\pi^n + g^n \circ \beta^n) \circ (s^n + \alpha^n \circ h^n) = g^n \circ \beta^n \circ s^n + \pi^n \circ \alpha^n \circ h^n = g^n + h^n$$

which proves (1). We compute $(\delta')^n$ as follows

$$(\pi^{n+1} + g^{n+1} \circ \beta^{n+1}) \circ d_B^n \circ (s^n + \alpha^n \circ h^n) = \delta^n + g^{n+1} \circ d_C^n + d_{A[1]}^n \circ h^n$$

Since $h^n = -g^n$ and since $d_{A[1]}^{n-1} = -d_A^n$ we conclude that (2) holds. $\square$

15. Truncation of complexes

Let $\mathcal{A}$ be an abelian category. Let $A_\bullet$ be a chain complex. There are several ways to *truncate* the complex $A_\bullet$.

- (1) The “stupid” truncation $\sigma_{\leq n}$ is the subcomplex $\sigma_{\leq n} A_\bullet$ defined by the rule $(\sigma_{\leq n} A_\bullet)_i = 0$ if $i > n$ and $(\sigma_{\leq n} A_\bullet)_i = A_i$ if $i \leq n$. In a picture

$$\begin{array}{ccccccc} \sigma_{\leq n} A_\bullet & & \cdots & \longrightarrow & 0 & \longrightarrow & A_n & \longrightarrow & A_{n-1} & \longrightarrow & \cdots \\ \downarrow & & & & \downarrow & & \downarrow & & \downarrow & & \\ A_\bullet & & \cdots & \longrightarrow & A_{n+1} & \longrightarrow & A_n & \longrightarrow & A_{n-1} & \longrightarrow & \cdots \end{array}$$

Note the property $\sigma_{\leq n} A_{\bullet} / \sigma_{\leq n-1} A_{\bullet} = A_n[-n]$.

- (2) The “*stupid*” truncation $\sigma_{\geq n}$ is the quotient complex $\sigma_{\geq n} A_{\bullet}$ defined by the rule $(\sigma_{\geq n} A_{\bullet})_i = A_i$ if $i \geq n$ and $(\sigma_{\geq n} A_{\bullet})_i = 0$ if $i < n$. In a picture

$$\begin{array}{ccccccc} A_{\bullet} & & \cdots \longrightarrow & A_{n+1} & \longrightarrow & A_n & \longrightarrow & A_{n-1} & \longrightarrow & \cdots \\ \downarrow & & & \downarrow & & \downarrow & & \downarrow & & \\ \sigma_{\geq n} A_{\bullet} & & \cdots \longrightarrow & A_{n+1} & \longrightarrow & A_n & \longrightarrow & 0 & \longrightarrow & \cdots \end{array}$$

The map of complexes $\sigma_{\geq n} A_{\bullet} \rightarrow \sigma_{\geq n+1} A_{\bullet}$ is surjective with kernel $A_n[-n]$.

- (3) The *canonical truncation* $\tau_{\geq n} A_{\bullet}$ is defined by the picture

$$\begin{array}{ccccccc} \tau_{\geq n} A_{\bullet} & & \cdots \longrightarrow & A_{n+1} & \longrightarrow & \text{Ker}(d_n) & \longrightarrow & 0 & \longrightarrow & \cdots \\ \downarrow & & & \downarrow & & \downarrow & & \downarrow & & \\ A_{\bullet} & & \cdots \longrightarrow & A_{n+1} & \longrightarrow & A_n & \longrightarrow & A_{n-1} & \longrightarrow & \cdots \end{array}$$

Note that these complexes have the property that

$$H_i(\tau_{\geq n} A_{\bullet}) = \begin{cases} H_i(A_{\bullet}) & \text{if } i \geq n \\ 0 & \text{if } i < n \end{cases}$$

- (4) The *canonical truncation* $\tau_{\leq n} A_{\bullet}$ is defined by the picture

$$\begin{array}{ccccccc} A_{\bullet} & & \cdots \longrightarrow & A_{n+1} & \longrightarrow & A_n & \longrightarrow & A_{n-1} & \longrightarrow & \cdots \\ \downarrow & & & \downarrow & & \downarrow & & \downarrow & & \\ \tau_{\leq n} A_{\bullet} & & \cdots \longrightarrow & 0 & \longrightarrow & \text{Coker}(d_{n+1}) & \longrightarrow & A_{n-1} & \longrightarrow & \cdots \end{array}$$

Note that these complexes have the property that

$$H_i(\tau_{\leq n} A_{\bullet}) = \begin{cases} H_i(A_{\bullet}) & \text{if } i \leq n \\ 0 & \text{if } i > n \end{cases}$$

Let $\mathcal{A}$ be an abelian category. Let $A^{\bullet}$ be a cochain complex. There are four ways to truncate the complex $A^{\bullet}$.

- (1) The “*stupid*” truncation $\sigma_{\geq n}$ is the subcomplex $\sigma_{\geq n} A^{\bullet}$ defined by the rule $(\sigma_{\geq n} A^{\bullet})^i = 0$ if $i < n$ and $(\sigma_{\geq n} A^{\bullet})^i = A^i$ if $i \geq n$. In a picture

$$\begin{array}{ccccccc} \sigma_{\geq n} A^{\bullet} & & \cdots \longrightarrow & 0 & \longrightarrow & A^n & \longrightarrow & A^{n+1} & \longrightarrow & \cdots \\ \downarrow & & & \downarrow & & \downarrow & & \downarrow & & \\ A^{\bullet} & & \cdots \longrightarrow & A^{n-1} & \longrightarrow & A^n & \longrightarrow & A^{n+1} & \longrightarrow & \cdots \end{array}$$

Note the property $\sigma_{\geq n} A^{\bullet} / \sigma_{\geq n+1} A^{\bullet} = A^n[-n]$.

- (2) The “*stupid*” truncation $\sigma_{\leq n}$ is the quotient complex $\sigma_{\leq n} A^{\bullet}$ defined by the rule $(\sigma_{\leq n} A^{\bullet})^i = 0$ if $i > n$ and $(\sigma_{\leq n} A^{\bullet})^i = A^i$ if $i \leq n$. In a picture

$$\begin{array}{ccccccc} A^{\bullet} & & \cdots \longrightarrow & A^{n-1} & \longrightarrow & A^n & \longrightarrow & A^{n+1} & \longrightarrow & \cdots \\ \downarrow & & & \downarrow & & \downarrow & & \downarrow & & \\ \sigma_{\leq n} A^{\bullet} & & \cdots \longrightarrow & A^{n-1} & \longrightarrow & A^n & \longrightarrow & 0 & \longrightarrow & \cdots \end{array}$$

The map of complexes $\sigma_{\leq n} A^{\bullet} \rightarrow \sigma_{\leq n-1} A^{\bullet}$ is surjective with kernel $A^n[-n]$.

(3) The *canonical truncation* $\tau_{\leq n}A^\bullet$ is defined by the picture

$$\begin{array}{ccccccc} \tau_{\leq n}A^\bullet & & \dots \longrightarrow & A^{n-1} & \longrightarrow & \text{Ker}(d^n) & \longrightarrow & 0 & \longrightarrow & \dots \\ \downarrow & & & \downarrow & & \downarrow & & \downarrow & & \\ A^\bullet & & \dots \longrightarrow & A^{n-1} & \longrightarrow & A^n & \longrightarrow & A^{n+1} & \longrightarrow & \dots \end{array}$$

Note that these complexes have the property that

$$H^i(\tau_{\leq n}A^\bullet) = \begin{cases} H^i(A^\bullet) & \text{if } i \leq n \\ 0 & \text{if } i > n \end{cases}$$

(4) The *canonical truncation* $\tau_{\geq n}A^\bullet$ is defined by the picture

$$\begin{array}{ccccccc} A^\bullet & & \dots \longrightarrow & A^{n-1} & \longrightarrow & A^n & \longrightarrow & A^{n+1} & \longrightarrow & \dots \\ \downarrow & & & \downarrow & & \downarrow & & \downarrow & & \\ \tau_{\geq n}A^\bullet & & \dots \longrightarrow & 0 & \longrightarrow & \text{Coker}(d^{n-1}) & \longrightarrow & A^{n+1} & \longrightarrow & \dots \end{array}$$

Note that these complexes have the property that

$$H^i(\tau_{\geq n}A^\bullet) = \begin{cases} 0 & \text{if } i < n \\ H^i(A^\bullet) & \text{if } i \geq n \end{cases}$$

16. Graded objects

We make the following definition.

Definition 16.1. Let $\mathcal{A}$ be an additive category. The *category of graded objects of $\mathcal{A}$* , denoted $\text{Gr}(\mathcal{A})$, is the category with

- (1) objects $A = (A^i)$ are families of objects A^i , $i \in \mathbf{Z}$ of objects of $\mathcal{A}$, and
- (2) morphisms $f : A = (A^i) \rightarrow B = (B^i)$ are families of morphisms $f^i : A^i \rightarrow B^i$ of $\mathcal{A}$.

If $\mathcal{A}$ has countable direct sums, then we can associate to an object $A = (A^i)$ of $\text{Gr}(\mathcal{A})$ the object

$$A = \bigoplus_{i \in \mathbf{Z}} A^i$$

and set $k^i A = A^i$. In this case $\text{Gr}(\mathcal{A})$ is equivalent to the category of pairs (A, k) consisting of an object A of $\mathcal{A}$ and a direct sum decomposition

$$A = \bigoplus_{i \in \mathbf{Z}} k^i A$$

by direct summands indexed by $\mathbf{Z}$ and a morphism $(A, k) \rightarrow (B, k)$ of such objects is given by a morphism $\varphi : A \rightarrow B$ of $\mathcal{A}$ such that $\varphi(k^i A) \subset k^i B$ for all $i \in \mathbf{Z}$. Whenever our additive category $\mathcal{A}$ has countable direct sums we will use this equivalence without further mention.

However, with our definitions an additive or abelian category does not necessarily have all (countable) direct sums. In this case our definition still makes sense. For example, if $\mathcal{A} = \text{Vect}_k$ is the category of finite dimensional vector spaces over a field k , then $\text{Gr}(\text{Vect}_k)$ is the category of vector spaces with a given gradation all of whose graded pieces are finite dimensional, and not the category of finite dimensional vector spaces with a given gradation.

Lemma 16.2. *Let $\mathcal{A}$ be an abelian category. The category of graded objects $\text{Gr}(\mathcal{A})$ is abelian.*

Proof. Let $f : A = (A^i) \rightarrow B = (B^i)$ be a morphism of graded objects of $\mathcal{A}$ given by morphisms $f^i : A^i \rightarrow B^i$ of $\mathcal{A}$. Then we have $\text{Ker}(f) = (\text{Ker}(f^i))$ and $\text{Coker}(f) = (\text{Coker}(f^i))$ in the category $\text{Gr}(\mathcal{A})$. Since we have $\text{Im} = \text{Coim}$ in $\mathcal{A}$ we see the same thing holds in $\text{Gr}(\mathcal{A})$. $\square$

Remark 16.3 (Warning). There are abelian categories $\mathcal{A}$ having countable direct sums but where countable direct sums are not exact. An example is the opposite of the category of abelian sheaves on $\mathbf{R}$. Namely, the category of abelian sheaves on $\mathbf{R}$ has countable products, but countable products are not exact. For such a category the functor $\text{Gr}(\mathcal{A}) \rightarrow \mathcal{A}$, $(A^i) \mapsto \bigoplus A^i$ described above is not exact. It is still true that $\text{Gr}(\mathcal{A})$ is equivalent to the category of graded objects (A, k) of $\mathcal{A}$, but the kernel in the category of graded objects of a map $\varphi : (A, k) \rightarrow (B, k)$ is not equal to $\text{Ker}(\varphi)$ endowed with a direct sum decomposition, but rather it is the direct sum of the kernels of the maps $k^i A \rightarrow k^i B$.

Definition 16.4. Let $\mathcal{A}$ be an additive category. If $A = (A^i)$ is a graded object, then the k th *shift* $A[k]$ is the graded object with $A[k]^i = A^{k+i}$.

If A and B are graded objects of $\mathcal{A}$, then we have

$$(16.4.1) \quad \text{Hom}_{\text{Gr}(\mathcal{A})}(A, B[k]) = \text{Hom}_{\text{Gr}(\mathcal{A})}(A[-k], B)$$

and an element of this group is sometimes called a map of graded objects *homogeneous of degree k* .

Given any set G we can define G -graded objects of $\mathcal{A}$ as the category whose objects are $A = (A^g)_{g \in G}$ families of objects parametrized by elements of G . Morphisms $f : A \rightarrow B$ are defined as families of maps $f^g : A^g \rightarrow B^g$ where g runs over the elements of G . If G is an abelian group, then we can (unambiguously) define shift functors $[g]$ on the category of G -graded objects by the rule $(A[g])^{g_0} = A^{g+g_0}$. A particular case of this type of construction is when $G = \mathbf{Z} \times \mathbf{Z}$. In this case the objects of the category are called *bigraded* objects of $\mathcal{A}$. The (p, q) component of a bigraded object A is usually denoted $A^{p,q}$. For $(a, b) \in \mathbf{Z} \times \mathbf{Z}$ we write $A[a, b]$ instead of $A[(a, b)]$. A morphism $A \rightarrow A[a, b]$ is sometimes called a *map of bidegree (a, b)* .

17. Additive monoidal categories

Some material about the interaction between a monoidal structure and an additive structure on a category.

Definition 17.1. An *additive monoidal category* is an additive category $\mathcal{A}$ endowed with a monoidal structure $\otimes, \phi$ (Categories, Definition ??) such that $\otimes$ is an additive functor in each variable.

Lemma 17.2. *Let $\mathcal{A}$ be an additive monoidal category. If Y_i , $i = 1, 2$ are left duals of X_i , $i = 1, 2$, then $Y_1 \oplus Y_2$ is a left dual of $X_1 \oplus X_2$.*

Proof. Follows from uniqueness of adjoints and Categories, Remark ??. $\square$

Lemma 17.3. *In a Karoubian additive monoidal category every summand of an object which has a left dual has a left dual.*

Proof. We will use Categories, Lemma ?? without further mention. Let X be an object which has a left dual Y . We have

$$\mathrm{Hom}(X, X) = \mathrm{Hom}(\mathbf{1}, X \otimes Y) = \mathrm{Hom}(Y, Y)$$

If $a : X \rightarrow X$ corresponds to $b : Y \rightarrow Y$ then b is the unique endomorphism of Y such that precomposing by a on

$$\mathrm{Hom}(Z' \otimes X, Z) = \mathrm{Hom}(Z', Z \otimes Y)$$

is the same as postcomposing by $1 \otimes b$. Hence the bijection $\mathrm{Hom}(X, X) \rightarrow \mathrm{Hom}(Y, Y)$, $a \mapsto b$ is an isomorphism of the opposite of the algebra $\mathrm{Hom}(X, X)$ with the algebra $\mathrm{Hom}(Y, Y)$. In particular, if $X = X_1 \oplus X_2$, then the corresponding projectors e_1, e_2 are mapped to idempotents in $\mathrm{Hom}(Y, Y)$. If $Y = Y_1 \oplus Y_2$ is the corresponding direct sum decomposition of Y (Section 4) then we see that under the bijection $\mathrm{Hom}(Z' \otimes X, Z) = \mathrm{Hom}(Z', Z \otimes Y)$ we have $\mathrm{Hom}(Z' \otimes X_i, Z) = \mathrm{Hom}(Z', Z \otimes Y_i)$ functorially as subgroups for $i = 1, 2$. It follows that Y_i is the left dual of X_i by the discussion in Categories, Remark ??. $\square$

Example 17.4. Let F be a field. Let $\mathcal{C}$ be the category of graded F -vector spaces. Given graded vector spaces V and W we let $V \otimes W$ denote the graded F -vector space whose degree n part is

$$(V \otimes W)^n = \bigoplus_{n=p+q} V^p \otimes_F W^q$$

Given a third graded vector space U as associativity constraint $\phi : U \otimes (V \otimes W) \rightarrow (U \otimes V) \otimes W$ we use the “usual” isomorphisms

$$U^p \otimes_F (V^q \otimes_F W^r) \rightarrow (U^p \otimes_F V^q) \otimes_F W^r$$

of vector spaces. As unit we use the graded F -vector space $\mathbf{1}$ which has F in degree 0 and is zero in other degrees. There are two commutativity constraints on $\mathcal{C}$ which turn $\mathcal{C}$ into a symmetric monoidal category: one involves the intervention of signs and the other does not. We will usually use the one that does. To be explicit, if V and W are graded F -vector spaces we will use the isomorphism $\psi : V \otimes W \rightarrow W \otimes V$ which in degree n uses

$$V^p \otimes_F W^q \rightarrow W^q \otimes_F V^p, \quad v \otimes w \mapsto (-1)^{pq} w \otimes v$$

We omit the verification that this works.

Lemma 17.5. *Let F be a field. Let $\mathcal{C}$ be the category of graded F -vector spaces viewed as a monoidal category as in Example 17.4. If V in $\mathcal{C}$ has a left dual W , then $\sum_n \dim_F V^n < \infty$ and the map ϵ defines nondegenerate pairings $W^{-n} \times V^n \rightarrow F$.*

Proof. As unit we take the object $\mathbf{1}$ described above. By Categories, Definition ?? we have maps

$$\eta : \mathbf{1} \rightarrow V \otimes W \quad \epsilon : W \otimes V \rightarrow \mathbf{1}$$

Since $\mathbf{1} = F$ placed in degree 0, we may think of ϵ as a sequence of pairings $W^{-n} \times V^n \rightarrow F$ as in the statement of the lemma. Choose bases $\{e_{n,i}\}_{i \in I_n}$ for V^n for all n . Write

$$\eta(1) = \sum e_{n,i} \otimes w_{-n,i}$$

for some elements $w_{-n,i} \in W^{-n}$ almost all of which are zero! The condition that $(\epsilon \otimes 1) \circ (1 \otimes \eta)$ is the identity on W means that

$$\sum_{n,i} \epsilon(w, e_{n,i}) w_{-n,i} = w$$

Thus we see that W is generated as a graded vector space by the finitely many nonzero vectors $w_{-n,i}$. The condition that $(1 \otimes \epsilon) \circ (\eta \otimes 1)$ is the identity of V means that

$$\sum_{n,i} e_{n,i} \epsilon(w_{-n,i}, v) = v$$

In particular, setting $v = e_{n,i}$ we conclude that $\epsilon(w_{-n,i}, e_{n,i'}) = \delta_{ii'}$. Thus we find that the statement of the lemma holds and that $\{w_{-n,i}\}_{i \in I_n}$ is the dual basis for W^{-n} to the chosen basis for V^n . $\square$

18. Double complexes and associated total complexes

We discuss double complexes and associated total complexes.

Definition 18.1. Let $\mathcal{A}$ be an additive category. A *double complex* in $\mathcal{A}$ is given by a system $(\{A^{p,q}, d_1^{p,q}, d_2^{p,q}\}_{p,q \in \mathbf{Z}})$, where each $A^{p,q}$ is an object of $\mathcal{A}$ and $d_1^{p,q} : A^{p,q} \rightarrow A^{p+1,q}$ and $d_2^{p,q} : A^{p,q} \rightarrow A^{p,q+1}$ are morphisms of $\mathcal{A}$ such that the following rules hold:

- (1) $d_1^{p+1,q} \circ d_1^{p,q} = 0$
- (2) $d_2^{p,q+1} \circ d_2^{p,q} = 0$
- (3) $d_1^{p,q+1} \circ d_2^{p,q} = d_2^{p+1,q} \circ d_1^{p,q}$

for all $p, q \in \mathbf{Z}$.

This is just the cochain version of the definition. It says that each $A^{p,\bullet}$ is a cochain complex and that each $d_1^{p,\bullet}$ is a morphism of complexes $A^{p,\bullet} \rightarrow A^{p+1,\bullet}$ such that $d_1^{p+1,\bullet} \circ d_1^{p,\bullet} = 0$ as morphisms of complexes. In other words a double complex can be seen as a complex of complexes. So in the diagram

$$\begin{array}{ccccccc}
 & & \cdots & & \cdots & & \cdots \\
 & & \uparrow & & \uparrow & & \\
 \cdots & \longrightarrow & A^{p,q+1} & \xrightarrow{d_1^{p,q+1}} & A^{p+1,q+1} & \longrightarrow & \cdots \\
 & & \uparrow d_2^{p,q} & & \uparrow d_2^{p+1,q} & & \\
 \cdots & \longrightarrow & A^{p,q} & \xrightarrow{d_1^{p,q}} & A^{p+1,q} & \longrightarrow & \cdots \\
 & & \uparrow & & \uparrow & & \\
 & & \cdots & & \cdots & &
 \end{array}$$

any square commutes. Warning: In the literature one encounters a different definition where a “bicomplex” or a “double complex” has the property that the squares in the diagram anti-commute.

Example 18.2. Let $\mathcal{A}, \mathcal{B}, \mathcal{C}$ be additive categories. Suppose that

$$\otimes : \mathcal{A} \times \mathcal{B} \longrightarrow \mathcal{C}, \quad (X, Y) \longmapsto X \otimes Y$$

is a functor which is bilinear on morphisms, see Categories, Definition ?? for the definition of $\mathcal{A} \times \mathcal{B}$. Given complexes $X^\bullet$ of $\mathcal{A}$ and $Y^\bullet$ of $\mathcal{B}$ we obtain a double complex

$$K^{\bullet,\bullet} = X^\bullet \otimes Y^\bullet$$

in $\mathcal{C}$. Here the first differential $K^{p,q} \rightarrow K^{p+1,q}$ is the morphism $X^p \otimes Y^q \rightarrow X^{p+1} \otimes Y^q$ induced by the morphism $X^p \rightarrow X^{p+1}$ and the identity on Y^q . Similarly for the second differential.

Definition 18.3. Let $\mathcal{A}$ be an additive category. Let $A^{\bullet,\bullet}$ be a double complex. The *associated simple complex*, denoted $sA^\bullet$, also often called the *associated total complex*, denoted $\text{Tot}(A^{\bullet,\bullet})$, is given by

$$sA^n = \text{Tot}^n(A^{\bullet,\bullet}) = \bigoplus_{n=p+q} A^{p,q}$$

(if it exists) with differential

$$d_{sA^\bullet}^n = d_{\text{Tot}(A^{\bullet,\bullet})}^n = \sum_{n=p+q} (d_1^{p,q} + (-1)^p d_2^{p,q})$$

If countable direct sums exist in $\mathcal{A}$ or if for each n at most finitely many $A^{p,n-p}$ are nonzero, then $\text{Tot}(A^{\bullet,\bullet})$ exists. Note that the definition is *not* symmetric in the indices (p, q) .

Remark 18.4. Let $\mathcal{A}$ be an additive category. Let $A^{\bullet,\bullet,\bullet}$ be a triple complex. The associated total complex is the complex with terms

$$\text{Tot}^n(A^{\bullet,\bullet,\bullet}) = \bigoplus_{p+q+r=n} A^{p,q,r}$$

and differential

$$d_{\text{Tot}(A^{\bullet,\bullet,\bullet})}^n = \sum_{p+q+r=n} (d_1^{p,q,r} + (-1)^p d_2^{p,q,r} + (-1)^{p+q} d_3^{p,q,r})$$

With this definition a simple calculation shows that the associated total complex is equal to

$$\text{Tot}(A^{\bullet,\bullet,\bullet}) = \text{Tot}(\text{Tot}_{12}(A^{\bullet,\bullet,\bullet})) = \text{Tot}(\text{Tot}_{23}(A^{\bullet,\bullet,\bullet}))$$

In other words, we can either first combine the first two of the variables and then combine the sum of those with the last, or we can first combine the last two variables and then combine the first with the sum of the last two.

Remark 18.5. Let $\mathcal{A}$ be an additive category. Let $A^{\bullet,\bullet}$ be a double complex with differentials $d_1^{p,q}$ and $d_2^{p,q}$. Denote $A^{\bullet,\bullet}[a, b]$ the double complex with

$$(A^{\bullet,\bullet}[a, b])^{p,q} = A^{p+a, q+b}$$

and differentials

$$d_{A^{\bullet,\bullet}[a,b],1}^{p,q} = (-1)^a d_1^{p+a, q+b} \quad \text{and} \quad d_{A^{\bullet,\bullet}[a,b],2}^{p,q} = (-1)^b d_2^{p+a, q+b}$$

In this situation there is a well defined isomorphism

$$\gamma : \text{Tot}(A^{\bullet,\bullet})[a+b] \longrightarrow \text{Tot}(A^{\bullet,\bullet}[a, b])$$

which in degree n is given by the map

$$\begin{aligned} (\text{Tot}(A^{\bullet,\bullet})[a+b])^n &= \bigoplus_{p+q=n+a+b} A^{p,q} \\ &\downarrow \epsilon(p,q,a,b) \text{id}_{A^{p,q}} \\ \text{Tot}(A^{\bullet,\bullet}[a, b])^n &= \bigoplus_{p'+q'=n} A^{p'+a, q'+b} \end{aligned}$$

for some sign $\epsilon(p, q, a, b)$. Of course the summand $A^{p,q}$ maps to the summand $A^{p'+a, q'+b}$ when $p = p' + a$ and $q = q' + b$. To figure out the conditions on these signs observe that on the source we have

$$d|_{A^{p,q}} = (-1)^{a+b} (d_1^{p,q} + (-1)^p d_2^{p,q})$$

whereas on the target we have

$$d|_{A^{p'+a, q'+b}} = (-1)^a d_1^{p'+a, q'+b} + (-1)^{p'} (-1)^b d_2^{p'+a, q'+b}$$

Thus our constraints are that

$$(-1)^a \epsilon(p, q, a, b) = \epsilon(p+1, q, a, b) (-1)^{a+b} \Leftrightarrow \epsilon(p+1, q, a, b) = (-1)^b \epsilon(p, q, a, b)$$

and

$$(-1)^{p'+b} \epsilon(p, q, a, b) = \epsilon(p, q+1, a, b) (-1)^{a+b+p} \Leftrightarrow \epsilon(p, q, a, b) = \epsilon(p, q+1, a, b)$$

Thus we choose $\epsilon(p, q, a, b) = (-1)^{pb}$.

Remark 18.6. Let $\mathcal{A}$ be an additive category with countable direct sums. Let $\text{DoubleComp}(\mathcal{A})$ denote the category of double complexes. We can consider an object $A^{\bullet, \bullet}$ of $\text{DoubleComp}(\mathcal{A})$ as a complex of complexes as follows

$$\dots \rightarrow A^{\bullet, -1} \rightarrow A^{\bullet, 0} \rightarrow A^{\bullet, 1} \rightarrow \dots$$

For the variant where we switch the role of the indices, see Remark 18.7. In this remark we show that taking the associated total complex is compatible with all the structures on complexes we have studied in the chapter so far.

First, observe that the shift functor on double complexes viewed as complexes of complexes in the manner given above is the functor $[0, 1]$ defined in Remark 18.5. By Remark 18.5 the functor

$$\text{Tot} : \text{DoubleComp}(\mathcal{A}) \rightarrow \text{Comp}(\mathcal{A})$$

is compatible with shift functors, in the sense that we have a functorial isomorphism $\gamma : \text{Tot}(A^{\bullet, \bullet})[1] \rightarrow \text{Tot}(A^{\bullet, \bullet}[0, 1])$.

Second, if

$$f, g : A^{\bullet, \bullet} \rightarrow B^{\bullet, \bullet}$$

are homotopic when f and g are viewed as morphisms of complexes of complexes in the manner given above, then

$$\text{Tot}(f), \text{Tot}(g) : \text{Tot}(A^{\bullet, \bullet}) \rightarrow \text{Tot}(B^{\bullet, \bullet})$$

are homotopic maps of complexes. Indeed, let $h = (h^q)$ be a homotopy between f and g . If we denote $h^{p, q} : A^{p, q} \rightarrow B^{p, q-1}$ the component in degree p of h^q , then this means that

$$f^{p, q} - g^{p, q} = d_2^{p, q-1} \circ h^{p, q} + h^{p, q+1} \circ d_2^{p, q}$$

The fact that $h^q : A^{\bullet, q} \rightarrow B^{\bullet, q-1}$ is a map of complexes means that

$$d_1^{p, q-1} \circ h^{p, q} = h^{p+1, q} \circ d_1^{p, q}$$

Let us define $h' = ((h')^n)$ the homotopy given by the maps $(h')^n : \text{Tot}^n(A^{\bullet, \bullet}) \rightarrow \text{Tot}^{n-1}(B^{\bullet, \bullet})$ using $(-1)^p h^{p, q}$ on the summand $A^{p, q}$ for $p+q = n$. Then we see that

$$d_{\text{Tot}(B^{\bullet, \bullet})} \circ h' + h' \circ d_{\text{Tot}(A^{\bullet, \bullet})}$$

restricted to the summand $A^{p, q}$ is equal to

$$d_1^{p, q-1} \circ (-1)^p h^{p, q} + (-1)^p d_2^{p, q-1} \circ (-1)^p h^{p, q} + (-1)^{p+1} h^{p+1, q} \circ d_1^{p, q} + (-1)^p h^{p, q+1} \circ (-1)^p d_2^{p, q}$$

which evaluates to $f^{p, q} - g^{p, q}$ by the equations given above. This proves the second compatibility.

Third, suppose that in the paragraph above we have $f = g$. Then the assignment $h \rightsquigarrow h'$ above is compatible with the identification of Lemma 14.9. More precisely, if we view h as a morphism of complexes of complexes $A^{\bullet,\bullet} \rightarrow B^{\bullet,\bullet}[0, -1]$ via this lemma then

$$\mathrm{Tot}(A^{\bullet,\bullet}) \xrightarrow{\mathrm{Tot}(h)} \mathrm{Tot}(B^{\bullet,\bullet}[0, -1]) \xrightarrow{\gamma^{-1}} \mathrm{Tot}(B^{\bullet,\bullet})[-1]$$

is equal to h' viewed as a morphism of complexes via the lemma. Here γ is the identification of Remark 18.5. The verification of this third point is immediate.

Fourth, let

$$0 \rightarrow A^{\bullet,\bullet} \rightarrow B^{\bullet,\bullet} \rightarrow C^{\bullet,\bullet} \rightarrow 0$$

be a complex of double complexes and suppose we are given splittings $s^q : C^{\bullet,q} \rightarrow B^{\bullet,q}$ and $\pi^q : B^{\bullet,q} \rightarrow A^{\bullet,q}$ of this as in Lemma 14.10 when we view double complexes as complexes of complexes in the manner given above. This on the one hand produces a map

$$\delta : C^{\bullet,\bullet} \rightarrow A^{\bullet,\bullet}[0, 1]$$

by the procedure in Lemma 14.10. On the other hand taking Tot we obtain a complex

$$0 \rightarrow \mathrm{Tot}(A^{\bullet,\bullet}) \rightarrow \mathrm{Tot}(B^{\bullet,\bullet}) \rightarrow \mathrm{Tot}(C^{\bullet,\bullet}) \rightarrow 0$$

which is termwise split (see below) and hence comes with a morphism

$$\delta' : \mathrm{Tot}(C^{\bullet,\bullet}) \rightarrow \mathrm{Tot}(A^{\bullet,\bullet})[1]$$

well defined up to homotopy by Lemmas 14.10 and 14.12. Claim: these maps agree in the sense that

$$\mathrm{Tot}(C^{\bullet,\bullet}) \xrightarrow{\mathrm{Tot}(\delta)} \mathrm{Tot}(A^{\bullet,\bullet}[0, 1]) \xrightarrow{\gamma^{-1}} \mathrm{Tot}(A^{\bullet,\bullet})[1]$$

is equal to δ' where γ is as in Remark 18.5. To see this denote $s^{p,q} : C^{p,q} \rightarrow B^{p,q}$ and $\pi^{p,q} : B^{p,q} \rightarrow A^{p,q}$ the components of s^q and π^q . As splittings $(s')^n : \mathrm{Tot}^n(C^{\bullet,\bullet}) \rightarrow \mathrm{Tot}^n(B^{\bullet,\bullet})$ and $(\pi')^n : \mathrm{Tot}^n(B^{\bullet,\bullet}) \rightarrow \mathrm{Tot}^n(A^{\bullet,\bullet})$ we use the maps whose components are $s^{p,q}$ and $\pi^{p,q}$ for $p + q = n$. We recall that

$$(\delta')^n = (\pi')^{n+1} \circ d_{\mathrm{Tot}(B^{\bullet,\bullet})}^n \circ (s')^n : \mathrm{Tot}^n(C^{\bullet,\bullet}) \rightarrow \mathrm{Tot}^{n+1}(A^{\bullet,\bullet})$$

The restriction of this to the summand $C^{p,q}$ is equal to

$$\pi^{p+1,q} \circ d_1^{p,q} \circ s^{p,q} + \pi^{p,q+1} \circ (-1)^p d_2^{p,q} \circ s^{p,q} = \pi^{p,q+1} \circ (-1)^p d_2^{p,q} \circ s^{p,q}$$

The equality holds because s^q is a morphism of complexes (with d_1 as differential) and because $\pi^{p+1,q} \circ s^{p+1,q} = 0$ as s and π correspond to a direct sum decomposition of B in every bidegree. On the other hand, for δ we have

$$\delta^q = \pi^q \circ d_2 \circ s^q : C^{\bullet,q} \rightarrow A^{\bullet,q+1}$$

whose restriction to the summand $C^{p,q}$ is equal to $\pi^{p,q+1} \circ d_2^{p,q} \circ s^{p,q}$. The difference in signs is exactly canceled out by the sign of $(-1)^p$ in the isomorphism γ and the fourth claim is proven.

Remark 18.7. Let $\mathcal{A}$ be an additive category with countable direct sums. Let $\mathrm{DoubleComp}(\mathcal{A})$ denote the category of double complexes. We can consider an object $A^{\bullet,\bullet}$ of $\mathrm{DoubleComp}(\mathcal{A})$ as a complex of complexes as follows

$$\dots \rightarrow A^{-1,\bullet} \rightarrow A^{0,\bullet} \rightarrow A^{1,\bullet} \rightarrow \dots$$

For the variant where we switch the role of the indices, see Remark 18.6. In this remark we show that taking the associated total complex is compatible with all the structures on complexes we have studied in the chapter so far.

First, observe that the shift functor on double complexes viewed as complexes of complexes in the manner given above is the functor $[1, 0]$ defined in Remark 18.5. By Remark 18.5 the functor

$$\text{Tot} : \text{DoubleComp}(\mathcal{A}) \rightarrow \text{Comp}(\mathcal{A})$$

is compatible with shift functors, in the sense that we have a functorial isomorphism $\gamma : \text{Tot}(A^{\bullet, \bullet})[1] \rightarrow \text{Tot}(A^{\bullet, \bullet}[1, 0])$.

Second, if

$$f, g : A^{\bullet, \bullet} \rightarrow B^{\bullet, \bullet}$$

are homotopic when f and g are viewed as morphisms of complexes of complexes in the manner given above, then

$$\text{Tot}(f), \text{Tot}(g) : \text{Tot}(A^{\bullet, \bullet}) \rightarrow \text{Tot}(B^{\bullet, \bullet})$$

are homotopic maps of complexes. Indeed, let $h = (h^p)$ be a homotopy between f and g . If we denote $h^{p,q} : A^{p,q} \rightarrow B^{p-1,q}$ the component in degree q of h^p , then this means that

$$f^{p,q} - g^{p,q} = d_1^{p-1,q} \circ h^{p,q} + h^{p+1,q} \circ d_1^{p,q}$$

The fact that $h^p : A^{p, \bullet} \rightarrow B^{p-1, \bullet}$ is a map of complexes means that

$$d_2^{p-1,q} \circ h^{p,q} = h^{p,q+1} \circ d_2^{p,q}$$

Let us define $h' = ((h')^n)$ the homotopy given by the maps $(h')^n : \text{Tot}^n(A^{\bullet, \bullet}) \rightarrow \text{Tot}^{n-1}(B^{\bullet, \bullet})$ using $h^{p,q}$ on the summand $A^{p,q}$ for $p+q=n$. Then we see that

$$d_{\text{Tot}(B^{\bullet, \bullet})} \circ h' + h' \circ d_{\text{Tot}(A^{\bullet, \bullet})}$$

restricted to the summand $A^{p,q}$ is equal to

$$d_1^{p-1,q} \circ h^{p,q} + (-1)^{p-1} d_2^{p-1,q} \circ h^{p,q} + h^{p+1,q} \circ d_1^{p,q} + h^{p,q+1} \circ (-1)^p d_2^{p,q}$$

which evaluates to $f^{p,q} - g^{p,q}$ by the equations given above. This proves the second compatibility.

Third, suppose that in the paragraph above we have $f = g$. Then the assignment $h \rightsquigarrow h'$ above is compatible with the identification of Lemma 14.9. More precisely, if we view h as a morphism of complexes of complexes $A^{\bullet, \bullet} \rightarrow B^{\bullet, \bullet}[-1, 0]$ via this lemma then

$$\text{Tot}(A^{\bullet, \bullet}) \xrightarrow{\text{Tot}(h)} \text{Tot}(B^{\bullet, \bullet}[-1, 0]) \xrightarrow{\gamma^{-1}} \text{Tot}(B^{\bullet, \bullet})[-1]$$

is equal to h' viewed as a morphism of complexes via the lemma. Here γ is the identification of Remark 18.5. The verification of this third point is immediate.

Fourth, let

$$0 \rightarrow A^{\bullet, \bullet} \rightarrow B^{\bullet, \bullet} \rightarrow C^{\bullet, \bullet} \rightarrow 0$$

be a complex of double complexes and suppose we are given splittings $s^p : C^{p, \bullet} \rightarrow B^{p, \bullet}$ and $\pi^p : B^{p, \bullet} \rightarrow A^{p, \bullet}$ of this as in Lemma 14.10 when we view double complexes as complexes of complexes in the manner given above. This on the one hand produces a map

$$\delta : C^{\bullet, \bullet} \longrightarrow A^{\bullet, \bullet}[1, 0]$$

by the procedure in Lemma 14.10. On the other hand taking Tot we obtain a complex

$$0 \rightarrow \text{Tot}(A^{\bullet,\bullet}) \rightarrow \text{Tot}(B^{\bullet,\bullet}) \rightarrow \text{Tot}(C^{\bullet,\bullet}) \rightarrow 0$$

which is termwise split (see below) and hence comes with a morphism

$$\delta' : \text{Tot}(C^{\bullet,\bullet}) \longrightarrow \text{Tot}(A^{\bullet,\bullet})[1]$$

well defined up to homotopy by Lemmas 14.10 and 14.12. Claim: these maps agree in the sense that

$$\text{Tot}(C^{\bullet,\bullet}) \xrightarrow{\text{Tot}(\delta)} \text{Tot}(A^{\bullet,\bullet}[1,0]) \xrightarrow{\gamma^{-1}} \text{Tot}(A^{\bullet,\bullet})[1]$$

is equal to δ' where γ is as in Remark 18.5. To see this denote $s^{p,q} : C^{p,q} \rightarrow B^{p,q}$ and $\pi^{p,q} : B^{p,q} \rightarrow A^{p,q}$ the components of s^p and π^p . As splittings $(s')^n : \text{Tot}^n(C^{\bullet,\bullet}) \rightarrow \text{Tot}^n(B^{\bullet,\bullet})$ and $(\pi')^n : \text{Tot}^n(B^{\bullet,\bullet}) \rightarrow \text{Tot}^n(A^{\bullet,\bullet})$ we use the maps whose components are $s^{p,q}$ and $\pi^{p,q}$ for $p+q=n$. We recall that

$$(\delta')^n = (\pi')^{n+1} \circ d_{\text{Tot}(B^{\bullet,\bullet})}^n \circ (s')^n : \text{Tot}^n(C^{\bullet,\bullet}) \rightarrow \text{Tot}^{n+1}(A^{\bullet,\bullet})$$

The restriction of this to the summand $C^{p,q}$ is equal to

$$\pi^{p+1,q} \circ d_1^{p,q} \circ s^{p,q} + \pi^{p,q+1} \circ (-1)^p d_2^{p,q} \circ s^{p,q} = \pi^{p+1,q} \circ d_1^{p,q} \circ s^{p,q}$$

The equality holds because s^p is a morphism of complexes (with d_2 as differential) and because $\pi^{p,q+1} \circ s^{p,q+1} = 0$ as s and π correspond to a direct sum decomposition of B in every bidegree. On the other hand, for δ we have

$$\delta^p = \pi^p \circ d_1 \circ s^p : C^{p,\bullet} \rightarrow A^{p+1,\bullet}$$

whose restriction to the summand $C^{p,q}$ is equal to $\pi^{p+1,q} \circ d_1^{p,q} \circ s^{p,q}$. Thus we get the same as before which matches with the fact that the isomorphism $\gamma : \text{Tot}(A^{\bullet,\bullet})[1] \rightarrow \text{Tot}(A^{\bullet,\bullet}[1,0])$ is defined without the intervention of signs.

19. Filtrations

A nice reference for this material is [Del71, Section 1]. (Note that our conventions regarding abelian categories are different.)

Definition 19.1. Let $\mathcal{A}$ be an abelian category.

- (1) A *decreasing filtration* F on an object A is a family $(F^n A)_{n \in \mathbf{Z}}$ of subobjects of A such that

$$A \supset \dots \supset F^n A \supset F^{n+1} A \supset \dots \supset 0$$

- (2) A *filtered object* of $\mathcal{A}$ is pair (A, F) consisting of an object A of $\mathcal{A}$ and a decreasing filtration F on A .
- (3) A *morphism* $(A, F) \rightarrow (B, F)$ of *filtered objects* is given by a morphism $\varphi : A \rightarrow B$ of $\mathcal{A}$ such that $\varphi(F^i A) \subset F^i B$ for all $i \in \mathbf{Z}$.
- (4) The category of filtered objects is denoted $\text{Fil}(\mathcal{A})$.
- (5) Given a filtered object (A, F) and a subobject $X \subset A$ the *induced filtration* on X is the filtration with $F^n X = X \cap F^n A$.
- (6) Given a filtered object (A, F) and a surjection $\pi : A \rightarrow Y$ the *quotient filtration* is the filtration with $F^n Y = \pi(F^n A)$.
- (7) A filtration F on an object A is said to be *finite* if there exist n, m such that $F^n A = A$ and $F^m A = 0$.

- (8) Given a filtered object (A, F) we say $\bigcap F^i A$ exists if there exists a biggest subobject of A contained in all $F^i A$. We say $\bigcup F^i A$ exists if there exists a smallest subobject of A containing all $F^i A$.
- (9) The filtration on a filtered object (A, F) is said to be *separated* if $\bigcap F^i A = 0$ and *exhaustive* if $\bigcup F^i A = A$.

By abuse of notation we say that a morphism $f : (A, F) \rightarrow (B, F)$ of filtered objects is *injective* if $f : A \rightarrow B$ is injective in the abelian category $\mathcal{A}$. Similarly we say f is *surjective* if $f : A \rightarrow B$ is surjective in the category $\mathcal{A}$. Being injective (resp. surjective) is equivalent to being a monomorphism (resp. epimorphism) in $\text{Fil}(\mathcal{A})$. By Lemma 19.2 this is also equivalent to having zero kernel (resp. cokernel).

Lemma 19.2. *Let $\mathcal{A}$ be an abelian category. The category of filtered objects $\text{Fil}(\mathcal{A})$ has the following properties:*

- (1) *It is an additive category.*
- (2) *It has a zero object.*
- (3) *It has kernels and cokernels, images and coimages.*
- (4) *In general it is not an abelian category.*

Proof. It is clear that $\text{Fil}(\mathcal{A})$ is additive with direct sum given by $(A, F) \oplus (B, F) = (A \oplus B, F)$ where $F^p(A \oplus B) = F^p A \oplus F^p B$. The kernel of a morphism $f : (A, F) \rightarrow (B, F)$ of filtered objects is the injection $\text{Ker}(f) \subset A$ where $\text{Ker}(f)$ is endowed with the induced filtration. The cokernel of a morphism $f : A \rightarrow B$ of filtered objects is the surjection $B \rightarrow \text{Coker}(f)$ where $\text{Coker}(f)$ is endowed with the quotient filtration. Since all kernels and cokernels exist, so do all coimages and images. See Example 3.13 for the last statement. $\square$

Definition 19.3. Let $\mathcal{A}$ be an abelian category. A morphism $f : A \rightarrow B$ of filtered objects of $\mathcal{A}$ is said to be *strict* if $f(F^i A) = f(A) \cap F^i B$ for all $i \in \mathbf{Z}$.

This is also equivalent to requiring that $f^{-1}(F^i B) = F^i A + \text{Ker}(f)$ for all $i \in \mathbf{Z}$. We characterize strict morphisms as follows.

Lemma 19.4. *Let $\mathcal{A}$ be an abelian category. Let $f : A \rightarrow B$ be a morphism of filtered objects of $\mathcal{A}$. The following are equivalent*

- (1) *f is strict,*
- (2) *the morphism $\text{Coim}(f) \rightarrow \text{Im}(f)$ of Lemma 3.12 is an isomorphism.*

Proof. Note that $\text{Coim}(f) \rightarrow \text{Im}(f)$ is an isomorphism of objects of $\mathcal{A}$, and that part (2) signifies that it is an isomorphism of filtered objects. By the description of kernels and cokernels in the proof of Lemma 19.2 we see that the filtration on $\text{Coim}(f)$ is the quotient filtration coming from $A \rightarrow \text{Coim}(f)$. Similarly, the filtration on $\text{Im}(f)$ is the induced filtration coming from the injection $\text{Im}(f) \rightarrow B$. The definition of strict is exactly that the quotient filtration is the induced filtration. $\square$

Lemma 19.5. *Let $\mathcal{A}$ be an abelian category. Let $f : A \rightarrow B$ be a strict monomorphism of filtered objects. Let $g : A \rightarrow C$ be a morphism of filtered objects. Then $f \oplus g : A \rightarrow B \oplus C$ is a strict monomorphism.*

Proof. Clear from the definitions. $\square$

Lemma 19.6. *Let $\mathcal{A}$ be an abelian category. Let $f : B \rightarrow A$ be a strict epimorphism of filtered objects. Let $g : C \rightarrow A$ be a morphism of filtered objects. Then $f \oplus g : B \oplus C \rightarrow A$ is a strict epimorphism.*

Proof. Clear from the definitions. $\square$

Lemma 19.7. *Let $\mathcal{A}$ be an abelian category. Let $(A, F), (B, F)$ be filtered objects. Let $u : A \rightarrow B$ be a morphism of filtered objects. If u is injective then u is strict if and only if the filtration on A is the induced filtration. If u is surjective then u is strict if and only if the filtration on B is the quotient filtration.*

Proof. This is immediate from the definition. $\square$

Lemma 19.8. *Let $\mathcal{A}$ be an abelian category. Let $f : A \rightarrow B, g : B \rightarrow C$ be strict morphisms of filtered objects.*

- (1) *In general the composition $g \circ f$ is not strict.*
- (2) *If g is injective, then $g \circ f$ is strict.*
- (3) *If f is surjective, then $g \circ f$ is strict.*

Proof. Let B a vector space over a field k with basis e_1, e_2 , with the filtration $F^n B = B$ for $n < 0$, with $F^0 B = ke_1$, and $F^n B = 0$ for $n > 0$. Now take $A = k(e_1 + e_2)$ and $C = B/ke_2$ with filtrations induced by B , i.e., such that $A \rightarrow B$ and $B \rightarrow C$ are strict (Lemma 19.7). Then $F^n(A) = A$ for $n < 0$ and $F^n(A) = 0$ for $n \geq 0$. Also $F^n(C) = C$ for $n \leq 0$ and $F^n(C) = 0$ for $n > 0$. So the (nonzero) composition $A \rightarrow C$ is not strict.

Assume g is injective. Then

$$\begin{aligned} g(f(F^p A)) &= g(f(A) \cap F^p B) \\ &= g(f(A)) \cap g(F^p(B)) \\ &= (g \circ f)(A) \cap (g(B) \cap F^p C) \\ &= (g \circ f)(A) \cap F^p C. \end{aligned}$$

The first equality as f is strict, the second because g is injective, the third because g is strict, and the fourth because $(g \circ f)(A) \subset g(B)$.

Assume f is surjective. Then

$$\begin{aligned} (g \circ f)^{-1}(F^i C) &= f^{-1}(F^i B + \text{Ker}(g)) \\ &= f^{-1}(F^i B) + f^{-1}(\text{Ker}(g)) \\ &= F^i A + \text{Ker}(f) + \text{Ker}(g \circ f) \\ &= F^i A + \text{Ker}(g \circ f) \end{aligned}$$

The first equality because g is strict, the second because f is surjective, the third because f is strict, and the last because $\text{Ker}(f) \subset \text{Ker}(g \circ f)$. $\square$

The following lemma says that subobjects of a filtered object have a well defined filtration independent of a choice of writing the object as a cokernel.

Lemma 19.9. *Let $\mathcal{A}$ be an abelian category. Let (A, F) be a filtered object of $\mathcal{A}$. Let $X \subset Y \subset A$ be subobjects of A . On the object*

$$Y/X = \text{Ker}(A/X \rightarrow A/Y)$$

the quotient filtration coming from the induced filtration on Y and the induced filtration coming from the quotient filtration on A/X agree. Any of the morphisms $X \rightarrow Y$, $X \rightarrow A$, $Y \rightarrow A$, $Y \rightarrow A/X$, $Y \rightarrow Y/X$, $Y/X \rightarrow A/X$ are strict (with induced/quotient filtrations).

Proof. The quotient filtration Y/X is given by $F^p(Y/X) = F^pY/(X \cap F^pY) = F^pY/F^pX$ because $F^pY = Y \cap F^pA$ and $F^pX = X \cap F^pA$. The induced filtration from the injection $Y/X \rightarrow A/X$ is given by

$$\begin{aligned} F^p(Y/X) &= Y/X \cap F^p(A/X) \\ &= Y/X \cap (F^pA + X)/X \\ &= (Y \cap F^pA)/(X \cap F^pA) \\ &= F^pY/F^pX. \end{aligned}$$

Hence the first statement of the lemma. The proof of the other cases is similar. $\square$

Lemma 19.10. *Let $\mathcal{A}$ be an abelian category. Let $A, B, C \in \text{Fil}(\mathcal{A})$. Let $f : A \rightarrow B$ and $g : A \rightarrow C$ be morphisms. Then there exists a pushout*

$$\begin{array}{ccc} A & \xrightarrow{\quad f \quad} & B \\ g \downarrow & & \downarrow g' \\ C & \xrightarrow{\quad f' \quad} & C \amalg_A B \end{array}$$

in $\text{Fil}(\mathcal{A})$. If f is strict, so is f' .

Proof. Set $C \amalg_A B$ equal to $\text{Coker}((g, -f) : A \rightarrow C \oplus B)$ in $\text{Fil}(\mathcal{A})$. This cokernel exists, by Lemma 19.2. It is a pushout, see Example 5.6. Note that $F^p(C \amalg_A B)$ is the image of $F^pC \oplus F^pB$. Hence

$$(f')^{-1}(F^p(C \amalg_A B)) = g(f^{-1}(F^pB)) + F^pC$$

Whence the last statement. $\square$

Lemma 19.11. *Let $\mathcal{A}$ be an abelian category. Let $A, B, C \in \text{Fil}(\mathcal{A})$. Let $f : B \rightarrow A$ and $g : C \rightarrow A$ be morphisms. Then there exists a fibre product*

$$\begin{array}{ccc} B \times_A C & \xrightarrow{\quad g' \quad} & B \\ f' \downarrow & & \downarrow f \\ C & \xrightarrow{\quad g \quad} & A \end{array}$$

in $\text{Fil}(\mathcal{A})$. If f is strict, so is f' .

Proof. This lemma is dual to Lemma 19.10. $\square$

Let $\mathcal{A}$ be an abelian category. Let (A, F) be a filtered object of $\mathcal{A}$. We denote $\text{gr}_F^p(A) = \text{gr}^p(A)$ the object $F^pA/F^{p+1}A$ of $\mathcal{A}$. This defines an additive functor

$$\text{gr}^p : \text{Fil}(\mathcal{A}) \longrightarrow \mathcal{A}, \quad (A, F) \longmapsto \text{gr}^p(A).$$

Recall that we have defined the category $\text{Gr}(\mathcal{A})$ of graded objects of $\mathcal{A}$ in Section 16. For (A, F) in $\text{Fil}(\mathcal{A})$ we may set

$$\text{gr}(A) = \text{the graded object of } \mathcal{A} \text{ whose } p\text{th graded piece is } \text{gr}^p(A)$$

and if $\mathcal{A}$ has countable direct sums, then we simply have

$$\mathrm{gr}(A) = \bigoplus \mathrm{gr}^p(A)$$

This defines an additive functor

$$\mathrm{gr} : \mathrm{Fil}(\mathcal{A}) \longrightarrow \mathrm{Gr}(\mathcal{A}), \quad (A, F) \longmapsto \mathrm{gr}(A).$$

Lemma 19.12. *Let $\mathcal{A}$ be an abelian category.*

- (1) *Let A be a filtered object and $X \subset A$. Then for each p the sequence*

$$0 \rightarrow \mathrm{gr}^p(X) \rightarrow \mathrm{gr}^p(A) \rightarrow \mathrm{gr}^p(A/X) \rightarrow 0$$

is exact (with induced filtration on X and quotient filtration on A/X).

- (2) *Let $f : A \rightarrow B$ be a morphism of filtered objects of $\mathcal{A}$. Then for each p the sequences*

$$0 \rightarrow \mathrm{gr}^p(\mathrm{Ker}(f)) \rightarrow \mathrm{gr}^p(A) \rightarrow \mathrm{gr}^p(\mathrm{Coim}(f)) \rightarrow 0$$

and

$$0 \rightarrow \mathrm{gr}^p(\mathrm{Im}(f)) \rightarrow \mathrm{gr}^p(B) \rightarrow \mathrm{gr}^p(\mathrm{Coker}(f)) \rightarrow 0$$

are exact.

Proof. We have $F^{p+1}X = X \cap F^{p+1}A$, hence the map $\mathrm{gr}^p(X) \rightarrow \mathrm{gr}^p(A)$ is injective. Dually the map $\mathrm{gr}^p(A) \rightarrow \mathrm{gr}^p(A/X)$ is surjective. The kernel of the map $F^p A / F^{p+1} A \rightarrow (F^p A + X) / (F^{p+1} A + X)$ is $(F^{p+1} A + (X \cap F^p A)) / F^{p+1} A \cong F^p X / F^{p+1} X$, hence exactness in the middle. The two short exact sequences of (2) are special cases of the short exact sequence of (1). $\square$

Lemma 19.13. *Let $\mathcal{A}$ be an abelian category. Let $f : A \rightarrow B$ be a morphism of finite filtered objects of $\mathcal{A}$. The following are equivalent*

- (1) *f is strict,*
- (2) *the morphism $\mathrm{Coim}(f) \rightarrow \mathrm{Im}(f)$ is an isomorphism,*
- (3) *$\mathrm{gr}(\mathrm{Coim}(f)) \rightarrow \mathrm{gr}(\mathrm{Im}(f))$ is an isomorphism,*
- (4) *the sequence $\mathrm{gr}(\mathrm{Ker}(f)) \rightarrow \mathrm{gr}(A) \rightarrow \mathrm{gr}(B)$ is exact,*
- (5) *the sequence $\mathrm{gr}(A) \rightarrow \mathrm{gr}(B) \rightarrow \mathrm{gr}(\mathrm{Coker}(f))$ is exact, and*
- (6) *the sequence*

$$0 \rightarrow \mathrm{gr}(\mathrm{Ker}(f)) \rightarrow \mathrm{gr}(A) \rightarrow \mathrm{gr}(B) \rightarrow \mathrm{gr}(\mathrm{Coker}(f)) \rightarrow 0$$

is exact.

Proof. The equivalence of (1) and (2) is Lemma 19.4. By Lemma 19.12 we see that (4), (5), (6) imply (3) and that (3) implies (4), (5), (6). Hence it suffices to show that (3) implies (2). Thus we have to show that if $f : A \rightarrow B$ is an injective and surjective map of finite filtered objects which induces an isomorphism $\mathrm{gr}(A) \rightarrow \mathrm{gr}(B)$, then f induces an isomorphism of filtered objects. In other words, we have to show that $f(F^p A) = F^p B$ for all p . As the filtrations are finite we may prove this by descending induction on p . Suppose that $f(F^{p+1} A) = F^{p+1} B$. Then commutative diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & F^{p+1} A & \longrightarrow & F^p A & \longrightarrow & \mathrm{gr}^p(A) \longrightarrow 0 \\ & & \downarrow f & & \downarrow f & & \downarrow \mathrm{gr}^p(f) \\ 0 & \longrightarrow & F^{p+1} B & \longrightarrow & F^p B & \longrightarrow & \mathrm{gr}^p(B) \longrightarrow 0 \end{array}$$

and the five lemma imply that $f(F^p A) = F^p B$. $\square$

Lemma 19.14. *Let $\mathcal{A}$ be an abelian category. Let $A \rightarrow B \rightarrow C$ be a complex of filtered objects of $\mathcal{A}$. Assume $\alpha : A \rightarrow B$ and $\beta : B \rightarrow C$ are strict morphisms of filtered objects. Then $\text{gr}(\text{Ker}(\beta)/\text{Im}(\alpha)) = \text{Ker}(\text{gr}(\beta))/\text{Im}(\text{gr}(\alpha))$.*

Proof. This follows formally from Lemma 19.12 and the fact that $\text{Coim}(\alpha) \cong \text{Im}(\alpha)$ and $\text{Coim}(\beta) \cong \text{Im}(\beta)$ by Lemma 19.4. $\square$

Lemma 19.15. *Let $\mathcal{A}$ be an abelian category. Let $A \rightarrow B \rightarrow C$ be a complex of filtered objects of $\mathcal{A}$. Assume A, B, C have finite filtrations and that $\text{gr}(A) \rightarrow \text{gr}(B) \rightarrow \text{gr}(C)$ is exact. Then*

- (1) *for each $p \in \mathbf{Z}$ the sequence $\text{gr}^p(A) \rightarrow \text{gr}^p(B) \rightarrow \text{gr}^p(C)$ is exact,*
- (2) *for each $p \in \mathbf{Z}$ the sequence $F^p(A) \rightarrow F^p(B) \rightarrow F^p(C)$ is exact,*
- (3) *for each $p \in \mathbf{Z}$ the sequence $A/F^p(A) \rightarrow B/F^p(B) \rightarrow C/F^p(C)$ is exact,*
- (4) *the maps $A \rightarrow B$ and $B \rightarrow C$ are strict, and*
- (5) *$A \rightarrow B \rightarrow C$ is exact (as a sequence in $\mathcal{A}$).*

Proof. Part (1) is immediate from the definitions. We will prove (3) by induction on the length of the filtrations. If each of A, B, C has only one nonzero graded part, then (3) holds as $\text{gr}(A) = A$, etc. Let n be the largest integer such that at least one of $F^n A, F^n B, F^n C$ is nonzero. Set $A' = A/F^n A$, $B' = B/F^n B$, $C' = C/F^n C$ with induced filtrations. Note that $\text{gr}(A) = F^n A \oplus \text{gr}(A')$ and similarly for B and C . The induction hypothesis applies to $A' \rightarrow B' \rightarrow C'$, which implies that $A/F^p(A) \rightarrow B/F^p(B) \rightarrow C/F^p(C)$ is exact for $p \leq n$. To conclude the same for $p = n+1$, i.e., to prove that $A \rightarrow B \rightarrow C$ is exact we use the commutative diagram

$$\begin{array}{ccccccc}
 0 & \longrightarrow & F^n A & \longrightarrow & A & \longrightarrow & A' \longrightarrow 0 \\
 & & \downarrow & & \downarrow & & \downarrow \\
 0 & \longrightarrow & F^n B & \longrightarrow & B & \longrightarrow & B' \longrightarrow 0 \\
 & & \downarrow & & \downarrow & & \downarrow \\
 0 & \longrightarrow & F^n C & \longrightarrow & C & \longrightarrow & C' \longrightarrow 0
 \end{array}$$

whose rows are short exact sequences of objects of $\mathcal{A}$. The proof of (2) is dual. Of course (5) follows from (2).

To prove (4) denote $f : A \rightarrow B$ and $g : B \rightarrow C$ the given morphisms. We know that $f(F^p(A)) = \text{Ker}(F^p(B) \rightarrow F^p(C))$ by (2) and $f(A) = \text{Ker}(g)$ by (5). Hence $f(F^p(A)) = \text{Ker}(F^p(B) \rightarrow F^p(C)) = \text{Ker}(g) \cap F^p(B) = f(A) \cap F^p(B)$ which proves that f is strict. The proof that g is strict is dual to this. $\square$

20. Spectral sequences

A nice discussion of spectral sequences may be found in [Eis95]. See also [McC01], [Lan02], etc.

Definition 20.1. Let $\mathcal{A}$ be an abelian category.

- (1) A *spectral sequence in $\mathcal{A}$* is given by a system $(E_r, d_r)_{r \geq 1}$ where each E_r is an object of $\mathcal{A}$, each $d_r : E_r \rightarrow E_r$ is a morphism such that $d_r \circ d_r = 0$ and $E_{r+1} = \text{Ker}(d_r)/\text{Im}(d_r)$ for $r \geq 1$.

- (2) A *morphism of spectral sequences* $f : (E_r, d_r)_{r \geq 1} \rightarrow (E'_r, d'_r)_{r \geq 1}$ is given by a family of morphisms $f_r : E_r \rightarrow E'_r$ such that $f_r \circ d_r = d'_r \circ f_r$ and such that f_{r+1} is the morphism induced by f_r via the identifications $E_{r+1} = \text{Ker}(d_r)/\text{Im}(d_r)$ and $E'_{r+1} = \text{Ker}(d'_r)/\text{Im}(d'_r)$.

We will sometimes loosen this definition somewhat and allow E_{r+1} to be an object with a given isomorphism $E_{r+1} \rightarrow \text{Ker}(d_r)/\text{Im}(d_r)$. In addition we sometimes have a system $(E_r, d_r)_{r \geq r_0}$ for some $r_0 \in \mathbf{Z}$ satisfying the properties of the definition above for indices $\geq r_0$. We will also call this a spectral sequence since by a simple renumbering it falls under the definition anyway. In fact, the cases $r_0 = 0$ and $r_0 = -1$ can be found in the literature.

Given a spectral sequence $(E_r, d_r)_{r \geq 1}$ we define

$$0 = B_1 \subset B_2 \subset \dots \subset B_r \subset \dots \subset Z_r \subset \dots \subset Z_2 \subset Z_1 = E_1$$

by the following simple procedure. Set $B_2 = \text{Im}(d_1)$ and $Z_2 = \text{Ker}(d_1)$. Then it is clear that $d_2 : Z_2/B_2 \rightarrow Z_2/B_2$. Hence we can define B_3 as the unique subobject of E_1 containing B_2 such that B_3/B_2 is the image of d_2 . Similarly we can define Z_3 as the unique subobject of E_1 containing B_2 such that Z_3/B_2 is the kernel of d_2 . And so on and so forth. In particular we have

$$E_r = Z_r/B_r$$

for all $r \geq 1$. In case the spectral sequence starts at $r = r_0$ then we can similarly construct B_i, Z_i as subobjects in E_{r_0} . In fact, in the literature one sometimes finds the notation

$$0 = B_r(E_r) \subset B_{r+1}(E_r) \subset B_{r+2}(E_r) \subset \dots \subset Z_{r+2}(E_r) \subset Z_{r+1}(E_r) \subset Z_r(E_r) = E_r$$

to denote the filtration described above but starting with E_r .

Definition 20.2. Let $\mathcal{A}$ be an abelian category. Let $(E_r, d_r)_{r \geq 1}$ be a spectral sequence.

- (1) If the subobjects $Z_\infty = \bigcap Z_r$ and $B_\infty = \bigcup B_r$ of E_1 exist then we define the *limit*³ of the spectral sequence to be the object $E_\infty = Z_\infty/B_\infty$.
- (2) We say that the spectral sequence *degenerates at E_r* if the differentials $d_r, d_{r+1}, \dots$ are all zero.

Note that if the spectral sequence degenerates at E_r , then we have $E_r = E_{r+1} = \dots = E_\infty$ (and the limit exists of course). Also, almost any abelian category we will encounter has countable sums and intersections.

Remark 20.3 (Variant). It is often the case that the terms of a spectral sequence have additional structure, for example a grading or a bigrading. To accommodate this (and to get around certain technical issues) we introduce the following notion. Let $\mathcal{A}$ be an abelian category. Let $(T_r)_{r \geq 1}$ be a sequence of *translation* or *shift* functors, i.e., $T_r : \mathcal{A} \rightarrow \mathcal{A}$ is an isomorphism of categories. In this setting a *spectral sequence* is given by a system $(E_r, d_r)_{r \geq 1}$ where each E_r is an object of $\mathcal{A}$, each $d_r : E_r \rightarrow T_r E_r$ is a morphism such that $T_r d_r \circ d_r = 0$ so that

$$\dots \longrightarrow T_r^{-1} E_r \xrightarrow{T_r^{-1} d_r} E_r \xrightarrow{d_r} T_r E_r \xrightarrow{T_r d_r} T_r^2 E_r \longrightarrow \dots$$

³This notation is not universally accepted. In some references an additional pair of subobjects Z_∞ and B_∞ of E_1 such that $0 = B_1 \subset B_2 \subset \dots \subset B_\infty \subset Z_\infty \subset \dots \subset Z_2 \subset Z_1 = E_1$ is part of the data comprising a spectral sequence!

is a complex and $E_{r+1} = \text{Ker}(d_r)/\text{Im}(T_r^{-1}d_r)$ for $r \geq 1$. It is clear what a *morphism of spectral sequences* means in this setting. In this setting we can still define

$$0 = B_1 \subset B_2 \subset \dots \subset B_r \subset \dots \subset Z_r \subset \dots \subset Z_2 \subset Z_1 = E_1$$

and Z_∞ and B_∞ (if they exist) as above.

21. Spectral sequences: exact couples

Definition 21.1. Let $\mathcal{A}$ be an abelian category.

- (1) An *exact couple* is a datum (A, E, α, f, g) where A, E are objects of $\mathcal{A}$ and α, f, g are morphisms as in the following diagram

$$\begin{array}{ccc} A & \xrightarrow{\alpha} & A \\ & \searrow f & \swarrow g \\ & E & \end{array}$$

with the property that the kernel of each arrow is the image of its predecessor. So $\text{Ker}(\alpha) = \text{Im}(f)$, $\text{Ker}(f) = \text{Im}(g)$, and $\text{Ker}(g) = \text{Im}(\alpha)$.

- (2) A *morphism of exact couples* $t : (A, E, \alpha, f, g) \rightarrow (A', E', \alpha', f', g')$ is given by morphisms $t_A : A \rightarrow A'$ and $t_E : E \rightarrow E'$ such that $\alpha' \circ t_A = t_A \circ \alpha$, $f' \circ t_E = t_A \circ f$, and $g' \circ t_A = t_E \circ g$.

Lemma 21.2. Let (A, E, α, f, g) be an exact couple in an abelian category $\mathcal{A}$. Set

- (1) $d = g \circ f : E \rightarrow E$ so that $d \circ d = 0$,
- (2) $E' = \text{Ker}(d)/\text{Im}(d)$,
- (3) $A' = \text{Im}(\alpha)$,
- (4) $\alpha' : A' \rightarrow A'$ induced by α ,
- (5) $f' : E' \rightarrow A'$ induced by f ,
- (6) $g' : A' \rightarrow E'$ induced by " $g \circ \alpha^{-1}$ ".

Then we have

- (1) $\text{Ker}(d) = f^{-1}(\text{Ker}(g)) = f^{-1}(\text{Im}(\alpha))$,
- (2) $\text{Im}(d) = g(\text{Im}(f)) = g(\text{Ker}(\alpha))$,
- (3) $(A', E', \alpha', f', g')$ is an exact couple.

Proof. Omitted. □

Hence it is clear that given an exact couple (A, E, α, f, g) we get a spectral sequence by setting $E_1 = E$, $d_1 = d$, $E_2 = E'$, $d_2 = d' = g' \circ f'$, $E_3 = E''$, $d_3 = d'' = g'' \circ f''$, and so on.

Definition 21.3. Let $\mathcal{A}$ be an abelian category. Let (A, E, α, f, g) be an exact couple. The *spectral sequence associated to the exact couple* is the spectral sequence $(E_r, d_r)_{r \geq 1}$ with $E_1 = E$, $d_1 = d$, $E_2 = E'$, $d_2 = d' = g' \circ f'$, $E_3 = E''$, $d_3 = d'' = g'' \circ f''$, and so on.

Lemma 21.4. Let $\mathcal{A}$ be an abelian category. Let (A, E, α, f, g) be an exact couple. Let $(E_r, d_r)_{r \geq 1}$ be the spectral sequence associated to the exact couple. In this case we have

$$0 = B_1 \subset \dots \subset B_{r+1} = g(\text{Ker}(\alpha^r)) \subset \dots \subset Z_{r+1} = f^{-1}(\text{Im}(\alpha^r)) \subset \dots \subset Z_1 = E$$

and the map $d_{r+1} : E_{r+1} \rightarrow E_{r+1}$ is described by the following rule: For any (test) object T of $\mathcal{A}$ and any elements $x : T \rightarrow Z_{r+1}$ and $y : T \rightarrow A$ such that $f \circ x = \alpha^r \circ y$ we have

$$d_{r+1} \circ \bar{x} = \overline{g \circ y}$$

where $\bar{x} : T \rightarrow E_{r+1}$ is the induced morphism.

Proof. Omitted. □

Note that in the situation of the lemma we obviously have

$$B_\infty = g \left(\bigcup_r \text{Ker}(\alpha^r) \right) \subset Z_\infty = f^{-1} \left(\bigcap_r \text{Im}(\alpha^r) \right)$$

provided $\bigcup \text{Ker}(\alpha^r)$ and $\bigcap \text{Im}(\alpha^r)$ exist. This produces as limit $E_\infty = Z_\infty / B_\infty$, see Definition 20.2.

Remark 21.5 (Variant). Let $\mathcal{A}$ be an abelian category. Let $S, T : \mathcal{A} \rightarrow \mathcal{A}$ be shift functors, i.e., isomorphisms of categories. We will indicate the n -fold compositions by $S^n A$ and $T^n A$ for $A \in \text{Ob}(\mathcal{A})$ and $n \in \mathbf{Z}$. In this situation an *exact couple* is a datum (A, E, α, f, g) where A, E are objects of $\mathcal{A}$ and $\alpha : A \rightarrow T^{-1}A$, $f : E \rightarrow A$, $g : A \rightarrow SE$ are morphisms such that

$$TE \xrightarrow{Tf} TA \xrightarrow{T\alpha} A \xrightarrow{g} SE \xrightarrow{Sf} SA$$

is an exact complex. Let's visualize this as follows

$$\begin{array}{ccccc} TA & \xrightarrow{\quad T\alpha \quad} & A & \xrightarrow{\quad \alpha \quad} & T^{-1}A \\ & \nwarrow Tf & \nwarrow g & \nwarrow f & \nwarrow T^{-1}g \\ & TE & \cdots & SE & & E & \cdots & T^{-1}SE \end{array}$$

We set $d = g \circ f : E \rightarrow SE$. Then $d \circ S^{-1}d = g \circ f \circ S^{-1}g \circ S^{-1}f = 0$ because $f \circ S^{-1}g = 0$. Set $E' = \text{Ker}(d) / \text{Im}(S^{-1}d)$. Set $A' = \text{Im}(T\alpha)$. Let $\alpha' : A' \rightarrow T^{-1}A'$ induced by α . Let $f' : E' \rightarrow A'$ be induced by f which works because $f(\text{Ker}(d)) \subset \text{Ker}(g) = \text{Im}(T\alpha)$. Finally, let $g' : A' \rightarrow TSE'$ induced by “ $Tg \circ (T\alpha)^{-1}$ ”⁴.

In exactly the same way as above we find

- (1) $\text{Ker}(d) = f^{-1}(\text{Ker}(g)) = f^{-1}(\text{Im}(T\alpha))$,
- (2) $\text{Im}(d) = g(\text{Im}(f)) = g(\text{Ker}(\alpha))$,
- (3) $(A', E', \alpha', f', g')$ is an exact couple for the shift functors TS and T .

We obtain a spectral sequence (as in Remark 20.3) with $E_1 = E$, $E_2 = E'$, etc, with $d_r : E_r \rightarrow T^{r-1}SE_r$ for all $r \geq 1$. Lemma 21.4 tells us that

$$SB_{r+1} = g(\text{Ker}(T^{-r+1}\alpha \circ \dots \circ T^{-1}\alpha \circ \alpha))$$

and

$$Z_{r+1} = f^{-1}(\text{Im}(T\alpha \circ T^2\alpha \circ \dots \circ T^r\alpha))$$

in this situation. The description of the map d_{r+1} is similar to that given in the lemma. (It may be easier to use these explicit descriptions to prove one gets a spectral sequence from such an exact couple.)

⁴This works because $TSE' = \text{Ker}(TSd) / \text{Im}(Td)$ and $Tg(\text{Ker}(T\alpha)) = Tg(\text{Im}(Tf)) = \text{Im}(T(d))$ and $TS(d)(\text{Im}(Tg)) = \text{Im}(TSg \circ TSf \circ Tg) = 0$.

22. Spectral sequences: differential objects

Definition 22.1. Let $\mathcal{A}$ be an abelian category. A *differential object* of $\mathcal{A}$ is a pair (A, d) consisting of an object A of $\mathcal{A}$ endowed with a selfmap d such that $d \circ d = 0$. A *morphism of differential objects* $(A, d) \rightarrow (B, d)$ is given by a morphism $\alpha : A \rightarrow B$ such that $d \circ \alpha = \alpha \circ d$.

Lemma 22.2. Let $\mathcal{A}$ be an abelian category. The category of differential objects of $\mathcal{A}$ is abelian.

Proof. Omitted. □

Definition 22.3. For a differential object (A, d) we denote

$$H(A, d) = \text{Ker}(d) / \text{Im}(d)$$

its *homology*.

Lemma 22.4. Let $\mathcal{A}$ be an abelian category. Let $0 \rightarrow (A, d) \rightarrow (B, d) \rightarrow (C, d) \rightarrow 0$ be a short exact sequence of differential objects. Then we get an exact homology sequence

$$\dots \rightarrow H(C, d) \rightarrow H(A, d) \rightarrow H(B, d) \rightarrow H(C, d) \rightarrow \dots$$

Proof. Apply Lemma 13.12 to the short exact sequence of complexes

$$\begin{array}{ccccccccc} 0 & \rightarrow & A & \rightarrow & B & \rightarrow & C & \rightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \rightarrow & A & \rightarrow & B & \rightarrow & C & \rightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \rightarrow & A & \rightarrow & B & \rightarrow & C & \rightarrow & 0 \end{array}$$

where the vertical arrows are d . □

We come to an important example of a spectral sequence. Let $\mathcal{A}$ be an abelian category. Let (A, d) be a differential object of $\mathcal{A}$. Let $\alpha : (A, d) \rightarrow (A, d)$ be an endomorphism of this differential object. If we assume α injective, then we get a short exact sequence

$$0 \rightarrow (A, d) \rightarrow (A, d) \rightarrow (A/\alpha A, d) \rightarrow 0$$

of differential objects. By Lemma 22.4 we get an exact couple

$$\begin{array}{ccc} H(A, d) & \xrightarrow{\quad \bar{\alpha} \quad} & H(A, d) \\ & \swarrow f \quad \searrow g & \\ & H(A/\alpha A, d) & \end{array}$$

where g is the canonical map and f is the map defined in the snake lemma. By Section 21 we get an associated spectral sequence. Since in this case we have $E_1 = H(A/\alpha A, d)$, it makes sense to define $E_0 = A/\alpha A$ and $d_0 = d$. In other words, we start the spectral sequence with $r = 0$. According to our conventions in Section 20 we define a sequence of subobjects

$$0 = B_0 \subset \dots \subset B_r \subset \dots \subset Z_r \subset \dots \subset Z_0 = E_0$$

with the property that $E_r = Z_r/B_r$. It follows from Lemma 21.4 that for $r \geq 1$ we have

- (1) B_r is the image of $(\alpha^{r-1})^{-1}(dA)$ under the natural map $A \rightarrow A/\alpha A$,

- (2) Z_r is the image of $d^{-1}(\alpha^r A)$ under the natural map $A \rightarrow A/\alpha A$, and
 (3) $d_r : E_r \rightarrow E_r$ is given as follows: given an element $z \in Z_r$ choose an element $y \in A$ such that $d(z) = \alpha^r(y)$. Then $d_r(z + B_r + \alpha A) = y + B_r + \alpha A$.

Warning: It is not necessarily the case that $\alpha A \subset (\alpha^{r-1})^{-1}(dA)$, nor $\alpha A \subset d^{-1}(\alpha^r A)$. It is true that $(\alpha^{r-1})^{-1}(dA) \subset d^{-1}(\alpha^r A)$. We have

$$E_r = \frac{d^{-1}(\alpha^r A) + \alpha A}{(\alpha^{r-1})^{-1}(dA) + \alpha A}.$$

It is not hard to verify directly that (1) – (3) give a spectral sequence.

Definition 22.5. Let $\mathcal{A}$ be an abelian category. Let (A, d) be a differential object of $\mathcal{A}$. Let $\alpha : A \rightarrow A$ be an injective selfmap of A which commutes with d . The *spectral sequence associated to (A, d, α)* is the spectral sequence $(E_r, d_r)_{r \geq 0}$ described above.

Remark 22.6 (Variant). Let $\mathcal{A}$ be an abelian category and let $S, T : \mathcal{A} \rightarrow \mathcal{A}$ be shift functors, i.e., isomorphisms of categories. Assume that $TS = ST$ as functors. Consider pairs (A, d) consisting of an object A of $\mathcal{A}$ and a morphism $d : A \rightarrow SA$ such that $d \circ S^{-1}d = 0$. The category of these objects is abelian. We define $H(A, d) = \text{Ker}(d)/\text{Im}(S^{-1}d)$ and we observe that $H(SA, Sd) = SH(A, d)$ (canonical isomorphism). Given a short exact sequence

$$0 \rightarrow (A, d) \rightarrow (B, d) \rightarrow (C, d) \rightarrow 0$$

we obtain a long exact homology sequence

$$\dots \rightarrow S^{-1}H(C, d) \rightarrow H(A, d) \rightarrow H(B, d) \rightarrow H(C, d) \rightarrow SH(A, d) \rightarrow \dots$$

(note the shifts in the boundary maps). Since $ST = TS$ the functor T defines a shift functor on pairs by setting $T(A, d) = (TA, Td)$. Next, let $\alpha : (A, d) \rightarrow T^{-1}(A, d)$ be injective with cokernel (Q, d) . Then we get an exact couple as in Remark 21.5 with shift functors TS and T given by

$$(H(A, d), S^{-1}H(Q, d), \bar{\alpha}, f, g)$$

where $\bar{\alpha} : H(A, d) \rightarrow T^{-1}H(A, d)$ is induced by α , the map $f : S^{-1}H(Q, d) \rightarrow H(A, d)$ is the boundary map and $g : H(A, d) \rightarrow TH(Q, d) = TS(S^{-1}H(Q, d))$ is induced by the quotient map $A \rightarrow TQ$. Thus we get a spectral sequence as above with $E_1 = S^{-1}H(Q, d)$ and differentials $d_r : E_r \rightarrow T^r SE_r$. As above we set $E_0 = S^{-1}Q$ and $d_0 : E_0 \rightarrow SE_0$ given by $S^{-1}d : S^{-1}Q \rightarrow Q$. If according to our conventions we define $B_r \subset Z_r \subset E_0$, then we have for $r \geq 1$ that

- (1) SB_r is the image of

$$(T^{-r+1}\alpha \circ \dots \circ T^{-1}\alpha)^{-1} \text{Im}(T^{-r}S^{-1}d)$$

under the natural map $T^{-1}A \rightarrow Q$,

- (2) Z_r is the image of

$$(S^{-1}T^{-1}d)^{-1} \text{Im}(\alpha \circ \dots \circ T^{r-1}\alpha)$$

under the natural map $S^{-1}T^{-1}A \rightarrow S^{-1}Q$.

The differentials can be described as follows: if $x \in Z_r$, then pick $x' \in S^{-1}T^{-1}A$ mapping to x . Then $S^{-1}T^{-1}d(x')$ is $(\alpha \circ \dots \circ T^{r-1}\alpha)(y)$ for some $y \in T^{r-1}A$. Then $d_r(x) \in T^r SE_r$ is represented by the class of the image of y in $T^r SE_0 = T^r Q$ modulo $T^r SB_r$.

23. Spectral sequences: filtered differential objects

We can build a spectral sequence starting with a filtered differential object.

Definition 23.1. Let $\mathcal{A}$ be an abelian category. A *filtered differential object* (K, F, d) is a filtered object (K, F) of $\mathcal{A}$ endowed with an endomorphism $d : (K, F) \rightarrow (K, F)$ whose square is zero: $d \circ d = 0$.

To describe the spectral sequence associated to such an object we assume, for the moment, that $\mathcal{A}$ is an abelian category which has countable direct sums and countable direct sums are exact (this is not automatic, see Remark 16.3). Let (K, F, d) be a filtered differential object of $\mathcal{A}$. Note that each $F^n K$ is a differential object by itself. Consider the object $A = \bigoplus F^n K$ and endow it with a differential d by using d on each summand. Then (A, d) is a differential object of $\mathcal{A}$ which comes equipped with a grading. Consider the map

$$\alpha : A \rightarrow A$$

which is given by the inclusions $F^n K \rightarrow F^{n-1} K$. This is clearly an injective morphism of differential objects $\alpha : (A, d) \rightarrow (A, d)$. Hence, by Definition 22.5 we get a spectral sequence. We will call this *the spectral sequence associated to the filtered differential object* (K, F, d) .

Let us figure out the terms of this spectral sequence. First, note that $A/\alpha A = \text{gr}(K)$ endowed with its differential $d = \text{gr}(d)$. Hence we see that

$$E_0 = \text{gr}(K), \quad d_0 = \text{gr}(d).$$

Hence the homology of the graded differential object $\text{gr}(K)$ is the next term:

$$E_1 = H(\text{gr}(K), \text{gr}(d)).$$

In addition we see that E_0 is a graded object of $\mathcal{A}$ and that d_0 is compatible with the grading. Hence clearly E_1 is a graded object as well. But it turns out that the differential d_1 does not preserve this grading; instead it shifts the degree by 1.

To work this out precisely, we define

$$Z_r^p = \frac{F^p K \cap d^{-1}(F^{p+r} K) + F^{p+1} K}{F^{p+1} K}$$

and

$$B_r^p = \frac{F^p K \cap d(F^{p-r+1} K) + F^{p+1} K}{F^{p+1} K}.$$

This notation, although quite natural, seems to be different from the notation in most places in the literature. Perhaps it does not matter, since the literature does not seem to have a consistent choice of notation either. With these choices we see that $B_r \subset E_0$, resp. $Z_r \subset E_0$ (as defined in Section 22) is equal to $\bigoplus_p B_r^p$, resp. $\bigoplus_p Z_r^p$. Hence if we define

$$E_r^p = Z_r^p / B_r^p$$

for $r \geq 0$ and $p \in \mathbf{Z}$, then we have $E_r = \bigoplus_p E_r^p$. We can define a differential $d_r^p : E_r^p \rightarrow E_r^{p+r}$ by the rule

$$z + F^{p+1} K \mapsto dz + F^{p+r+1} K$$

where $z \in F^p K \cap d^{-1}(F^{p+r} K)$.

Lemma 23.2. *Let $\mathcal{A}$ be an abelian category. Let (K, F, d) be a filtered differential object of $\mathcal{A}$. There is a spectral sequence $(E_r, d_r)_{r \geq 0}$ in $\text{Gr}(\mathcal{A})$ associated to (K, F, d) such that $d_r : E_r \rightarrow E_r[r]$ for all r and such that the graded pieces E_r^p and maps $d_r^p : E_r^p \rightarrow E_r^{p+r}$ are as given above. Furthermore, $E_0^p = \text{gr}^p K$, $d_0^p = \text{gr}^p(d)$, and $E_1^p = H(\text{gr}^p K, \text{gr}^p(d))$.*

Proof. If $\mathcal{A}$ has countable direct sums and if countable direct sums are exact, then this follows from the discussion above. In general, we proceed as follows; we strongly suggest the reader skip this proof. Consider the object $A = (F^{p+1}K)$ of $\text{Gr}(\mathcal{A})$, i.e., we put $F^{p+1}K$ in degree p (the funny shift in numbering to get numbering correct later on). We endow it with a differential d by using d on each component. Then (A, d) is a differential object of $\text{Gr}(\mathcal{A})$. Consider the map

$$\alpha : A \rightarrow A[-1]$$

which is given in degree p by the inclusions $F^{p+1}A \rightarrow F^pA$. This is clearly an injective morphism of differential objects $\alpha : (A, d) \rightarrow (A, d)[-1]$. Hence, we can apply Remark 22.6 with $S = \text{id}$ and $T = [1]$. The corresponding spectral sequence $(E_r, d_r)_{r \geq 0}$ in $\text{Gr}(\mathcal{A})$ is the spectral sequence we are looking for. Let us unwind the definitions a bit. First of all we have $E_r = (E_r^p)$ is an object of $\text{Gr}(\mathcal{A})$. Then, since $T^r S = [r]$ we have $d_r : E_r \rightarrow E_r[r]$ which means that $d_r^p : E_r^p \rightarrow E_r^{p+r}$.

To see that the description of the graded pieces hold, we argue as above. Namely, first we have $E_0 = \text{Coker}(\alpha : A \rightarrow A[-1])$ and by our choice of numbering above this gives $E_0^p = \text{gr}^p K$. The first differential is given by $d_0^p = \text{gr}^p d : E_0^p \rightarrow E_0^p$. Next, the description of the boundaries B_r and the cocycles Z_r in Remark 22.6 translates into a straightforward manner into the formulae for Z_r^p and B_r^p given above. $\square$

Lemma 23.3. *Let $\mathcal{A}$ be an abelian category. Let (K, F, d) be a filtered differential object of $\mathcal{A}$. The spectral sequence $(E_r, d_r)_{r \geq 0}$ associated to (K, F, d) has*

$$d_1^p : E_1^p = H(\text{gr}^p K) \longrightarrow H(\text{gr}^{p+1} K) = E_1^{p+1}$$

equal to the boundary map in homology associated to the short exact sequence of differential objects

$$0 \rightarrow \text{gr}^{p+1} K \rightarrow F^p K / F^{p+2} K \rightarrow \text{gr}^p K \rightarrow 0.$$

Proof. This is clear from the formula for the differential d_1^p given just above Lemma 23.2. $\square$

Definition 23.4. Let $\mathcal{A}$ be an abelian category. Let (K, F, d) be a filtered differential object of $\mathcal{A}$. The *induced filtration* on $H(K, d)$ is the filtration defined by $F^p H(K, d) = \text{Im}(H(F^p K, d) \rightarrow H(K, d))$.

Writing out what this means we see that

$$F^p H(K, d) = \frac{\text{Ker}(d) \cap F^p K + \text{Im}(d)}{\text{Im}(d)}$$

and hence we see that

$$\text{gr}^p H(K) = \frac{\text{Ker}(d) \cap F^p K + \text{Im}(d)}{\text{Ker}(d) \cap F^{p+1} K + \text{Im}(d)} = \frac{\text{Ker}(d) \cap F^p K}{\text{Ker}(d) \cap F^{p+1} K + \text{Im}(d) \cap F^p K}$$

Lemma 23.5. *Let $\mathcal{A}$ be an abelian category. Let (K, F, d) be a filtered differential object of $\mathcal{A}$. If Z_∞^p and B_∞^p exist (see proof), then*

- (1) the limit E_∞ exists and is graded having $E_\infty^p = Z_\infty^p/B_\infty^p$ in degree p , and
- (2) the associated graded $\text{gr}(H(K))$ of the cohomology of K is a graded subquotient of the graded limit object E_∞ .

Proof. The objects Z_∞ , B_∞ , and the limit $E_\infty = Z_\infty/B_\infty$ of Definition 20.2 are objects of $\text{Gr}(\mathcal{A})$ by our construction of the spectral sequence in the proof of Lemma 23.2. Since Z_r^p , resp. B_r^p is the p th graded part of Z_r , resp. B_r , if we assume that

$$Z_\infty^p = \bigcap_r Z_r^p = \frac{\bigcap_r (F^p K \cap d^{-1}(F^{p+r} K) + F^{p+1} K)}{F^{p+1} K}$$

and

$$B_\infty^p = \bigcup_r B_r^p = \frac{\bigcup_r (F^p K \cap d(F^{p-r+1} K) + F^{p+1} K)}{F^{p+1} K}.$$

exist, then Z_∞ and B_∞ exist with degree p parts Z_∞^p and B_∞^p (follows from an elementary argument about unions and intersections of graded subobjects). Thus

$$E_\infty^p = \frac{\bigcap_r (F^p K \cap d^{-1}(F^{p+r} K) + F^{p+1} K)}{\bigcup_r (F^p K \cap d(F^{p-r+1} K) + F^{p+1} K)}.$$

where the top and bottom exist. We have

$$(23.5.1) \quad \text{Ker}(d) \cap F^p K + F^{p+1} K \subset \bigcap_r (F^p K \cap d^{-1}(F^{p+r} K) + F^{p+1} K)$$

and

$$(23.5.2) \quad \bigcup_r (F^p K \cap d(F^{p-r+1} K) + F^{p+1} K) \subset \text{Im}(d) \cap F^p K + F^{p+1} K.$$

Thus a subquotient of E_∞^p is

$$\frac{\text{Ker}(d) \cap F^p K + F^{p+1} K}{\text{Im}(d) \cap F^p K + F^{p+1} K} = \frac{\text{Ker}(d) \cap F^p K}{\text{Im}(d) \cap F^p K + \text{Ker}(d) \cap F^{p+1} K}$$

Comparing with the formula given for $\text{gr}^p H(K)$ in the discussion following Definition 23.4 we conclude. $\square$

Definition 23.6. Let $\mathcal{A}$ be an abelian category. Let (K, F, d) be a filtered differential object of $\mathcal{A}$. We say the spectral sequence associated to (K, F, d)

- (1) *weakly converges to $H(K)$* if $\text{gr} H(K) = E_\infty$ via Lemma 23.5,
- (2) *abuts to $H(K)$* if it weakly converges to $H(K)$ and we have $\bigcap F^p H(K) = 0$ and $\bigcup F^p H(K) = H(K)$,

Unfortunately, it seems hard to find a consistent terminology for these notions in the literature.

Lemma 23.7. Let $\mathcal{A}$ be an abelian category. Let (K, F, d) be a filtered differential object of $\mathcal{A}$. The associated spectral sequence

- (1) *weakly converges to $H(K)$* if and only if for every $p \in \mathbf{Z}$ we have equality in equations (23.5.2) and (23.5.1),
- (2) *abuts to $H(K)$* if and only if it weakly converges to $H(K)$ and $\bigcap_p (\text{Ker}(d) \cap F^p K + \text{Im}(d)) = \text{Im}(d)$ and $\bigcup_p (\text{Ker}(d) \cap F^p K + \text{Im}(d)) = \text{Ker}(d)$.

Proof. Immediate from the discussions above. $\square$

24. Spectral sequences: filtered complexes

Definition 24.1. Let $\mathcal{A}$ be an abelian category. A *filtered complex* $K^\bullet$ of $\mathcal{A}$ is a complex of $\text{Fil}(\mathcal{A})$ (see Definition 19.1).

We will denote the filtration on the objects by F . Thus $F^p K^n$ denotes the p th step in the filtration of the n th term of the complex. Note that each $F^p K^\bullet$ is a complex of $\mathcal{A}$. Hence we could also have defined a filtered complex as a filtered object in the (abelian) category of complexes of $\mathcal{A}$. In particular $\text{gr} K^\bullet$ is a graded object of the category of complexes of $\mathcal{A}$.

To describe the spectral sequence associated to such an object we assume, for the moment, that $\mathcal{A}$ is an abelian category which has countable direct sums and countable direct sums are exact (this is not automatic, see Remark 16.3). Let us denote d the differential of K . Forgetting the grading we can think of $\bigoplus K^n$ as a filtered differential object of $\mathcal{A}$. Hence according to Section 23 we obtain a spectral sequence $(E_r, d_r)_{r \geq 0}$. In this section we work out the terms of this spectral sequence, and we endow the terms of this spectral sequence with additional structure coming from the grading of K .

First we point out that $E_0^p = \text{gr}^p K^\bullet$ is a complex and hence is graded. Thus E_0 is bigraded in a natural way. It is customary to use the bigrading

$$E_0 = \bigoplus_{p,q} E_0^{p,q}, \quad E_0^{p,q} = \text{gr}^p K^{p+q}$$

The idea is that $p+q$ should be thought of as the *total degree* of the (co)homology classes. Also, p is called the *filtration degree*, and q is called the *complementary degree*. The differential d_0 is compatible with this bigrading in the following way

$$d_0 = \bigoplus d_0^{p,q}, \quad d_0^{p,q} : E_0^{p,q} \rightarrow E_0^{p,q+1}.$$

Namely, d_0^p is just the differential on the complex $\text{gr}^p K^\bullet$ (which occurs as $\text{gr}^p E_0$ just shifted a bit).

To go further we identify the objects B_r^p and Z_r^p introduced in Section 23 as graded objects and we work out the corresponding decompositions of the differentials. We do this in a completely straightforward manner, but again we warn the reader that our notation is not the same as notation found elsewhere. We define

$$Z_r^{p,q} = \frac{F^p K^{p+q} \cap d^{-1}(F^{p+r} K^{p+q+1}) + F^{p+1} K^{p+q}}{F^{p+1} K^{p+q}}$$

and

$$B_r^{p,q} = \frac{F^p K^{p+q} \cap d(F^{p-r+1} K^{p+q-1}) + F^{p+1} K^{p+q}}{F^{p+1} K^{p+q}}$$

and of course $E_r^{p,q} = Z_r^{p,q}/B_r^{p,q}$. With these definitions it is completely clear that $Z_r^p = \bigoplus_q Z_r^{p,q}$, $B_r^p = \bigoplus_q B_r^{p,q}$, and $E_r^p = \bigoplus_q E_r^{p,q}$. Moreover, we have

$$0 \subset \dots \subset B_r^{p,q} \subset \dots \subset Z_r^{p,q} \subset \dots \subset E_0^{p,q}$$

Also, the map d_r^p decomposes as the direct sum of the maps

$$d_r^{p,q} : E_r^{p,q} \longrightarrow E_r^{p+r,q-r+1}, \quad z + F^{p+1} K^{p+q} \mapsto dz + F^{p+r+1} K^{p+q+1}$$

where $z \in F^p K^{p+q} \cap d^{-1}(F^{p+r} K^{p+q+1})$.

Lemma 24.2. *Let $\mathcal{A}$ be an abelian category. Let $(K^\bullet, F)$ be a filtered complex of $\mathcal{A}$. There is a spectral sequence $(E_r, d_r)_{r \geq 0}$ in the category of bigraded objects of $\mathcal{A}$ associated to $(K^\bullet, F)$ such that d_r has bidegree $(r, -r+1)$ and such that E_r has bigraded pieces $E_r^{p,q}$ and maps $d_r^{p,q} : E_r^{p,q} \rightarrow E_r^{p+r, q-r+1}$ as given above. Furthermore, we have $E_0^{p,q} = \text{gr}^p(K^{p+q})$, $d_0^{p,q} = \text{gr}^p(d^{p+q})$, and $E_1^{p,q} = H^{p+q}(\text{gr}^p(K^\bullet))$.*

Proof. If $\mathcal{A}$ has countable direct sums and if countable direct sums are exact, then this follows from the discussion above. In general, we proceed as follows; we strongly suggest the reader skip this proof. Consider the bigraded object $A = (F^{p+1}K^{p+1+q})$ of $\mathcal{A}$, i.e., we put $F^{p+1}K^{p+1+q}$ in degree (p, q) (the funny shift in numbering to get numbering correct later on). We endow it with a differential $d : A \rightarrow A[0, 1]$ by using d on each component. Then (A, d) is a differential bigraded object. Consider the map

$$\alpha : A \rightarrow A[-1, 1]$$

which is given in degree (p, q) by the inclusion $F^{p+1}K^{p+1+q} \rightarrow F^pK^{p+1+q}$. This is an injective morphism of differential bigraded objects $\alpha : (A, d) \rightarrow (A, d)[-1, 1]$. Hence, we can apply Remark 22.6 with $S = [0, 1]$ and $T = [1, -1]$. The corresponding spectral sequence $(E_r, d_r)_{r \geq 0}$ of bigraded objects is the spectral sequence we are looking for. Let us unwind the definitions a bit. First of all we have $E_r = (E_r^{p,q})$. Then, since $T^r S = [r, -r+1]$ we have $d_r : E_r \rightarrow E_r[r, -r+1]$ which means that $d_r^{p,q} : E_r^{p,q} \rightarrow E_r^{p+r, q-r+1}$.

To see that the description of the graded pieces hold, we argue as above. Namely, first we have

$$E_0 = \text{Coker}(\alpha : A \rightarrow A[-1, 1])[0, -1] = \text{Coker}(\alpha[0, -1] : A[0, -1] \rightarrow A[-1, 0])$$

and by our choice of numbering above this gives

$$E_0^{p,q} = \text{Coker}(F^{p+1}K^{p+q} \rightarrow F^pK^{p+q}) = \text{gr}^p K^{p+q}$$

The first differential is given by $d_0^{p,q} = \text{gr}^p d^{p+q} : E_0^{p,q} \rightarrow E_0^{p,q+1}$. Next, the description of the boundaries B_r and the cocycles Z_r in Remark 22.6 translates into a straightforward manner into the formulae for $Z_r^{p,q}$ and $B_r^{p,q}$ given above. $\square$

Lemma 24.3. *Let $\mathcal{A}$ be an abelian category. Let $(K^\bullet, F)$ be a filtered complex of $\mathcal{A}$. Assume $\mathcal{A}$ has countable direct sums. Let $(E_r, d_r)_{r \geq 0}$ be the spectral sequence associated to $(K^\bullet, F)$.*

(1) *The map*

$$d_1^{p,q} : E_1^{p,q} = H^{p+q}(\text{gr}^p(K^\bullet)) \longrightarrow E_1^{p+1,q} = H^{p+q+1}(\text{gr}^{p+1}(K^\bullet))$$

is equal to the boundary map in cohomology associated to the short exact sequence of complexes

$$0 \rightarrow \text{gr}^{p+1}(K^\bullet) \rightarrow F^p K^\bullet / F^{p+2} K^\bullet \rightarrow \text{gr}^p(K^\bullet) \rightarrow 0.$$

(2) *Assume that $d(F^p K) \subset F^{p+1} K$ for all $p \in \mathbf{Z}$. Then d induces the zero differential on $\text{gr}^p(K^\bullet)$ and hence $E_1^{p,q} = \text{gr}^p(K^\bullet)^{p+q}$. Furthermore, in this case*

$$d_1^{p,q} : E_1^{p,q} = \text{gr}^p(K^\bullet)^{p+q} \longrightarrow E_1^{p+1,q} = \text{gr}^{p+1}(K^\bullet)^{p+q+1}$$

is the morphism induced by d .

Proof. This is clear from the formula given for the differential $d_1^{p,q}$ just above Lemma 24.2. $\square$

Lemma 24.4. *Let $\mathcal{A}$ be an abelian category. Let $\alpha : (K^\bullet, F) \rightarrow (L^\bullet, F)$ be a morphism of filtered complexes of $\mathcal{A}$. Let $(E_r(K), d_r)_{r \geq 0}$, resp. $(E_r(L), d_r)_{r \geq 0}$ be the spectral sequence associated to $(K^\bullet, F)$, resp. $(L^\bullet, F)$. The morphism α induces a canonical morphism of spectral sequences $\{\alpha_r : E_r(K) \rightarrow E_r(L)\}_{r \geq 0}$ compatible with the bigradings.*

Proof. Obvious from the explicit representation of the terms of the spectral sequences. $\square$

Definition 24.5. Let $\mathcal{A}$ be an abelian category. Let $(K^\bullet, F)$ be a filtered complex of $\mathcal{A}$. The *induced filtration* on $H^n(K^\bullet)$ is the filtration defined by $F^p H^n(K^\bullet) = \text{Im}(H^n(F^p K^\bullet) \rightarrow H^n(K^\bullet))$.

Writing out what this means we see that

$$(24.5.1) \quad F^p H^n(K^\bullet, d) = \frac{\text{Ker}(d) \cap F^p K^n + \text{Im}(d) \cap K^n}{\text{Im}(d) \cap K^n}$$

and hence we see that

$$(24.5.2) \quad \text{gr}^p H^n(K^\bullet) = \frac{\text{Ker}(d) \cap F^p K^n}{\text{Ker}(d) \cap F^{p+1} K^n + \text{Im}(d) \cap F^p K^n}$$

(one intermediate step omitted).

Lemma 24.6. *Let $\mathcal{A}$ be an abelian category. Let $(K^\bullet, F)$ be a filtered complex of $\mathcal{A}$. If $Z_\infty^{p,q}$ and $B_\infty^{p,q}$ exist (see proof), then*

- (1) *the limit E_∞ exists and is a bigraded object having $E_\infty^{p,q} = Z_\infty^{p,q} / B_\infty^{p,q}$ in bidegree (p, q) ,*
- (2) *the p th graded part $\text{gr}^p H^n(K^\bullet)$ of the n th cohomology object of $K^\bullet$ is a subquotient of $E_\infty^{p, n-p}$.*

Proof. The objects Z_∞ , B_∞ , and the limit $E_\infty = Z_\infty / B_\infty$ of Definition 20.2 are bigraded objects of $\mathcal{A}$ by our construction of the spectral sequence in Lemma 24.2. Since $Z_r^{p,q}$, resp. $B_r^{p,q}$ is the bidegree (p, q) part of Z_r , resp. B_r , if we assume that

$$Z_\infty^{p,q} = \bigcap_r Z_r^{p,q} = \bigcap_r \frac{F^p K^{p+q} \cap d^{-1}(F^{p+r} K^{p+q+1}) + F^{p+1} K^{p+q}}{F^{p+1} K^{p+q}}$$

and

$$B_\infty^{p,q} = \bigcup_r B_r^{p,q} = \bigcup_r \frac{F^p K^{p+q} \cap d(F^{p-r+1} K^{p+q-1}) + F^{p+1} K^{p+q}}{F^{p+1} K^{p+q}}$$

exist, then Z_∞ and B_∞ exist with bidegree (p, q) parts $Z_\infty^{p,q}$ and $B_\infty^{p,q}$ (follows from an elementary argument about unions and intersections of bigraded objects). Thus

$$E_\infty^{p,q} = \frac{\bigcap_r (F^p K^{p+q} \cap d^{-1}(F^{p+r} K^{p+q+1}) + F^{p+1} K^{p+q})}{\bigcup_r (F^p K^{p+q} \cap d(F^{p-r+1} K^{p+q-1}) + F^{p+1} K^{p+q})}.$$

where the top and the bottom exist. With $n = p + q$ we have

$$(24.6.1) \quad \text{Ker}(d) \cap F^p K^n + F^{p+1} K^n \subset \bigcap_r (F^p K^n \cap d^{-1}(F^{p+r} K^{n+1}) + F^{p+1} K^n)$$

and

$$(24.6.2) \quad \bigcup_r (F^p K^n \cap d(F^{p-r+1} K^{n-1}) + F^{p+1} K^n) \subset \text{Im}(d) \cap F^p K^n + F^{p+1} K^n.$$

Thus a subquotient of $E_\infty^{p,q}$ is

$$\frac{\text{Ker}(d) \cap F^p K^n + F^{p+1} K^n}{\text{Im}(d) \cap F^p K^n + F^{p+1} K^n} = \frac{\text{Ker}(d) \cap F^p K^n}{\text{Im}(d) \cap F^p K^n + \text{Ker}(d) \cap F^{p+1} K^n}$$

Comparing with (24.5.2) we conclude. $\square$

Definition 24.7. Let $\mathcal{A}$ be an abelian category. Let $(E_r, d_r)_{r \geq r_0}$ be a spectral sequence of bigraded objects of $\mathcal{A}$ with d_r of bidegree $(r, -r+1)$. We say such a spectral sequence is

- (1) *regular* if for all $p, q \in \mathbf{Z}$ there is a $b = b(p, q)$ such that the maps $d_r^{p,q} : E_r^{p,q} \rightarrow E_r^{p+r, q-r+1}$ are zero for $r \geq b$,
- (2) *coregular* if for all $p, q \in \mathbf{Z}$ there is a $b = b(p, q)$ such that the maps $d_r^{p-r, q+r-1} : E_r^{p-r, q+r-1} \rightarrow E_r^{p,q}$ are zero for $r \geq b$,
- (3) *bounded* if for all n there are only a finite number of nonzero $E_{r_0}^{p,n-p}$,
- (4) *bounded below* if for all n there is a $b = b(n)$ such that $E_{r_0}^{p,n-p} = 0$ for $p \geq b$.
- (5) *bounded above* if for all n there is a $b = b(n)$ such that $E_{r_0}^{p,n-p} = 0$ for $p \leq b$.

Bounded below means that if we look at $E_r^{p,q}$ on the line $p+q = n$ (whose slope is -1) we obtain zeros as (p, q) moves down and to the right. As mentioned above there is no consistent terminology regarding these notions in the literature.

Lemma 24.8. *In the situation of Definition 24.7. Let $Z_r^{p,q}, B_r^{p,q} \subset E_{r_0}^{p,q}$ be the (p, q) -graded parts of Z_r, B_r defined as in Section 20.*

- (1) *The spectral sequence is regular if and only if for all p, q there exists an $r = r(p, q)$ such that $Z_r^{p,q} = Z_{r+1}^{p,q} = \dots$*
- (2) *The spectral sequence is coregular if and only if for all p, q there exists an $r = r(p, q)$ such that $B_r^{p,q} = B_{r+1}^{p,q} = \dots$*
- (3) *The spectral sequence is bounded if and only if it is both bounded below and bounded above.*
- (4) *If the spectral sequence is bounded below, then it is regular.*
- (5) *If the spectral sequence is bounded above, then it is coregular.*

Proof. Omitted. Hint: If $E_r^{p,q} = 0$, then we have $E_{r'}^{p,q} = 0$ for all $r' \geq r$. $\square$

Definition 24.9. Let $\mathcal{A}$ be an abelian category. Let $(K^\bullet, F)$ be a filtered complex of $\mathcal{A}$. We say the spectral sequence associated to $(K^\bullet, F)$

- (1) *weakly converges to $H^*(K^\bullet)$* if $\text{gr}^p H^n(K^\bullet) = E_\infty^{p,n-p}$ via Lemma 24.6 for all $p, n \in \mathbf{Z}$,
- (2) *abuts to $H^*(K^\bullet)$* if it weakly converges to $H^*(K^\bullet)$ and $\bigcap_p F^p H^n(K^\bullet) = 0$ and $\bigcup_p F^p H^n(K^\bullet) = H^n(K^\bullet)$ for all n ,
- (3) *converges to $H^*(K^\bullet)$* if it is regular, abuts to $H^*(K^\bullet)$, and $H^n(K^\bullet) = \lim_p H^n(K^\bullet)/F^p H^n(K^\bullet)$.

Weak convergence, abutment, or convergence is symbolized by the notation $E_r^{p,q} \Rightarrow H^{p+q}(K^\bullet)$. As mentioned above there is no consistent terminology regarding these notions in the literature.

Lemma 24.10. *Let $\mathcal{A}$ be an abelian category. Let $(K^\bullet, F)$ be a filtered complex of $\mathcal{A}$. The associated spectral sequence*

- (1) *weakly converges to $H^*(K^\bullet)$ if and only if for every $p, q \in \mathbf{Z}$ we have equality in equations (24.6.2) and (24.6.1),*
- (2) *abuts to $H^*(K)$ if and only if it weakly converges to $H^*(K^\bullet)$ and we have $\bigcap_p (\text{Ker}(d) \cap F^p K^n + \text{Im}(d) \cap K^n) = \text{Im}(d) \cap K^n$ and $\bigcup_p (\text{Ker}(d) \cap F^p K^n + \text{Im}(d) \cap K^n) = \text{Ker}(d) \cap K^n$.*

Proof. Immediate from the discussions above. $\square$

Lemma 24.11. *Let $\mathcal{A}$ be an abelian category. Let $(K^\bullet, F)$ be a filtered complex of $\mathcal{A}$. Assume that the filtration on each K^n is finite (see Definition 19.1). Then*

- (1) *the spectral sequence associated to $(K^\bullet, F)$ is bounded,*
- (2) *the filtration on each $H^n(K^\bullet)$ is finite,*
- (3) *the spectral sequence associated to $(K^\bullet, F)$ converges to $H^*(K^\bullet)$,*
- (4) *if $\mathcal{C} \subset \mathcal{A}$ is a weak Serre subcategory and for some r we have $E_r^{p,q} \in \mathcal{C}$ for all $p, q \in \mathbf{Z}$, then $H^n(K^\bullet)$ is in $\mathcal{C}$.*

Proof. Part (1) follows as $E_0^{p,n-p} = \text{gr}^p K^n$. Part (2) is clear from Equation (24.5.1). We will use Lemma 24.10 to prove that the spectral sequence weakly converges. Fix $p, n \in \mathbf{Z}$. The right hand side of (24.6.1) is equal to $F^p K^n \cap \text{Ker}(d) + F^{p+1} K^n$ because $F^{p+r} K^n = 0$ for $r \gg 0$. Thus (24.6.1) is an equality. The left hand side of (24.6.2) is equal to $F^p K^n \cap \text{Im}(d) + F^{p+1} K^n$ because $F^{p-r+1} K^{n-1} = K^{n-1}$ for $r \gg 0$. Thus (24.6.2) is an equality. Since the filtration on $H^n(K^\bullet)$ is finite by (2) we see that we have abutment. To prove we have convergence we have to show the spectral sequence is regular which follows as it is bounded (Lemma 24.8) and we have to show that $H^n(K^\bullet) = \lim_p H^n(K^\bullet)/F^p H^n(K^\bullet)$ which follows from the fact that the filtration on $H^*(K^\bullet)$ is finite proved in part (2).

Proof of (4). Assume that for some $r \geq 0$ we have $E_r^{p,q} \in \mathcal{C}$ for some weak Serre subcategory $\mathcal{C}$ of $\mathcal{A}$. Then $E_{r+1}^{p,q}$ is in $\mathcal{C}$ as well, see Lemma 10.3. By boundedness proved above (which implies that the spectral sequence is both regular and coregular, see Lemma 24.8) we can find an $r' \geq r$ such that $E_\infty^{p,q} = E_{r'}^{p,q}$ for all p, q with $p+q = n$. Thus $H^n(K^\bullet)$ is an object of $\mathcal{A}$ which has a finite filtration whose graded pieces are in $\mathcal{C}$. This implies that $H^n(K^\bullet)$ is in $\mathcal{C}$ by Lemma 10.3. $\square$

Lemma 24.12. *Let $\mathcal{A}$ be an abelian category. Let $(K^\bullet, F)$ be a filtered complex of $\mathcal{A}$. Assume that the filtration on each K^n is finite (see Definition 19.1) and that for some r we have only a finite number of nonzero $E_r^{p,q}$. Then only a finite number of $H^n(K^\bullet)$ are nonzero and we have*

$$\sum (-1)^n [H^n(K^\bullet)] = \sum (-1)^{p+q} [E_r^{p,q}]$$

in $K_0(\mathcal{A}')$ where $\mathcal{A}'$ is the smallest weak Serre subcategory of $\mathcal{A}$ containing the objects $E_r^{p,q}$.

Proof. Denote E_r^{even} and E_r^{odd} the even and odd part of E_r defined as the direct sum of the (p, q) components with $p+q$ even and odd. The differential d_r defines maps $\varphi : E_r^{\text{even}} \rightarrow E_r^{\text{odd}}$ and $\psi : E_r^{\text{odd}} \rightarrow E_r^{\text{even}}$ whose compositions either way give zero. Then we see that

$$\begin{aligned} [E_r^{\text{even}}] - [E_r^{\text{odd}}] &= [\text{Ker}(\varphi)] + [\text{Im}(\varphi)] - [\text{Ker}(\psi)] - [\text{Im}(\psi)] \\ &= [\text{Ker}(\varphi)/\text{Im}(\psi)] - [\text{Ker}(\psi)/\text{Im}(\varphi)] \\ &= [E_{r+1}^{\text{even}}] - [E_{r+1}^{\text{odd}}] \end{aligned}$$

Note that all the intervening objects are in the smallest Serre subcategory containing the objects $E_r^{p,q}$. Continuing in this manner we see that we can increase r at will. Since there are only a finite number of pairs (p, q) for which $E_r^{p,q}$ is nonzero, a property which is inherited by $E_{r+1}, E_{r+2}, \dots$, we see that we may assume that $d_r = 0$. At this stage we see that $H^n(K^\bullet)$ has a finite filtration (Lemma 24.11) whose graded pieces are exactly the $E_r^{p,n-p}$ and the result is clear. $\square$

The following lemma is more a kind of sanity check for our definitions. Surely, if we have a filtered complex such that for every n we have

$$H^n(F^p K^\bullet) = 0 \text{ for } p \gg 0 \quad \text{and} \quad H^n(F^p K^\bullet) = H^n(K^\bullet) \text{ for } p \ll 0,$$

then the corresponding spectral sequence should converge?

Lemma 24.13. *Let $\mathcal{A}$ be an abelian category. Let $(K^\bullet, F)$ be a filtered complex of $\mathcal{A}$. Assume*

- (1) *for every n there exist $p_0(n)$ such that $H^n(F^p K^\bullet) = 0$ for $p \geq p_0(n)$,*
- (2) *for every n there exist $p_1(n)$ such that $H^n(F^p K^\bullet) \rightarrow H^n(K^\bullet)$ is an isomorphism for $p \leq p_1(n)$.*

Then

- (1) *the spectral sequence associated to $(K^\bullet, F)$ is bounded,*
- (2) *the filtration on each $H^n(K^\bullet)$ is finite,*
- (3) *the spectral sequence associated to $(K^\bullet, F)$ converges to $H^*(K^\bullet)$.*

Proof. Fix n . Using the long exact cohomology sequence associated to the short exact sequence of complexes

$$0 \rightarrow F^{p+1} K^\bullet \rightarrow F^p K^\bullet \rightarrow \text{gr}^p K^\bullet \rightarrow 0$$

we find that $E_1^{p, n-p} = 0$ for $p \geq \max(p_0(n), p_0(n+1))$ and $p < \min(p_1(n), p_1(n+1))$. Hence the spectral sequence is bounded (Definition 24.7). This proves (1).

It is clear from the assumptions and Definition 24.5 that the filtration on $H^n(K^\bullet)$ is finite. This proves (2).

Next we prove that the spectral sequence weakly converges to $H^*(K^\bullet)$ using Lemma 24.10. Let us show that we have equality in (24.6.1). Namely, for $p+r > p_0(n+1)$ the map

$$d : F^p K^n \cap d^{-1}(F^{p+r} K^{n+1}) \rightarrow F^{p+r} K^{n+1}$$

ends up in the image of $d : F^{p+r} K^n \rightarrow F^{p+r} K^{n+1}$ because the complex $F^{p+r} K^\bullet$ is exact in degree $n+1$. We conclude that $F^p K^n \cap d^{-1}(F^{p+r} K^{n+1}) = d(F^{p+r} K^n) + \text{Ker}(d) \cap F^p K^n$. Hence for such r we have

$$\text{Ker}(d) \cap F^p K^n + F^{p+1} K^n = F^p K^n \cap d^{-1}(F^{p+r} K^{n+1}) + F^{p+1} K^n$$

which proves the desired equality. To show that we have equality in (24.6.2) we use that for $p-r+1 < p_1(n-1)$ we have

$$d(F^{p-r+1} K^{n-1}) = \text{Im}(d) \cap F^{p-r+1} K^n$$

because the map $F^{p-r+1} K^\bullet \rightarrow K^\bullet$ induces an isomorphism on cohomology in degree $n-1$. This shows that we have

$$F^p K^n \cap d(F^{p-r+1} K^{n-1}) + F^{p+1} K^n = \text{Im}(d) \cap F^p K^n + F^{p+1} K^n$$

for such r which proves the desired equality.

To see that the spectral sequence abuts to $H^*(K^\bullet)$ using Lemma 24.10 we have to show that $\bigcap_p (\text{Ker}(d) \cap F^p K^n + \text{Im}(d) \cap K^n) = \text{Im}(d) \cap K^n$ and $\bigcup_p (\text{Ker}(d) \cap F^p K^n + \text{Im}(d) \cap K^n) = \text{Ker}(d) \cap K^n$. For $p \geq p_0(n)$ we have $\text{Ker}(d) \cap F^p K^n + \text{Im}(d) \cap K^n = \text{Im}(d) \cap K^n$ and for $p \leq p_1(n)$ we have $\text{Ker}(d) \cap F^p K^n + \text{Im}(d) \cap K^n = \text{Ker}(d) \cap K^n$. Combining weak convergence, abutment, and boundedness we see that (2) and (3) are true. $\square$

25. Spectral sequences: double complexes

Let $K^{\bullet,\bullet}$ be a double complex, see Section 18. It is customary to denote $H_I^p(K^{\bullet,\bullet})$ the complex with terms $\text{Ker}(d_1^{p,q})/\text{Im}(d_1^{p-1,q})$ (varying q) and differential induced by d_2 . Then $H_{II}^q(H_I^p(K^{\bullet,\bullet}))$ denotes its cohomology in degree q . It is also customary to denote $H_{II}^q(K^{\bullet,\bullet})$ the complex with terms $\text{Ker}(d_2^{p,q})/\text{Im}(d_2^{p,q-1})$ (varying p) and differential induced by d_1 . Then $H_I^p(H_{II}^q(K^{\bullet,\bullet}))$ denotes its cohomology in degree p . It will turn out that these cohomology groups show up as the terms in the spectral sequence for a filtration on the associated total complex or simple complex, see Definition 18.3.

There are two natural filtrations on the total complex $\text{Tot}(K^{\bullet,\bullet})$ associated to the double complex $K^{\bullet,\bullet}$. Namely, we define

$$F_I^p(\text{Tot}^n(K^{\bullet,\bullet})) = \bigoplus_{i+j=n, i \geq p} K^{i,j} \quad \text{and} \quad F_{II}^p(\text{Tot}^n(K^{\bullet,\bullet})) = \bigoplus_{i+j=n, j \geq p} K^{i,j}.$$

It is immediately verified that $(\text{Tot}(K^{\bullet,\bullet}), F_I)$ and $(\text{Tot}(K^{\bullet,\bullet}), F_{II})$ are filtered complexes. By Section 24 we obtain two spectral sequences. It is customary to denote $({}'E_r, {}'d_r)_{r \geq 0}$ the spectral sequence associated to the filtration F_I and to denote $({}''E_r, {}''d_r)_{r \geq 0}$ the spectral sequence associated to the filtration F_{II} . Here is a description of these spectral sequences.

Lemma 25.1. *Let $\mathcal{A}$ be an abelian category. Let $K^{\bullet,\bullet}$ be a double complex. The spectral sequences associated to $K^{\bullet,\bullet}$ have the following terms:*

- (1) $'E_0^{p,q} = K^{p,q}$ with $'d_0^{p,q} = (-1)^p d_2^{p,q} : K^{p,q} \rightarrow K^{p,q+1}$,
- (2) $''E_0^{p,q} = K^{q,p}$ with $''d_0^{p,q} = d_1^{q,p} : K^{q,p} \rightarrow K^{q+1,p}$,
- (3) $'E_1^{p,q} = H^q(K^{p,\bullet})$ with $'d_1^{p,q} = H^q(d_1^{p,\bullet})$,
- (4) $''E_1^{p,q} = H^q(K^{\bullet,p})$ with $''d_1^{p,q} = (-1)^q H^q(d_2^{\bullet,p})$,
- (5) $'E_2^{p,q} = H_I^p(H_{II}^q(K^{\bullet,\bullet}))$,
- (6) $''E_2^{p,q} = H_{II}^p(H_I^q(K^{\bullet,\bullet}))$.

Proof. Omitted. □

These spectral sequences define two filtrations on $H^n(\text{Tot}(K^{\bullet,\bullet}))$. We will denote these F_I and F_{II} .

Definition 25.2. Let $\mathcal{A}$ be an abelian category. Let $K^{\bullet,\bullet}$ be a double complex. We say the spectral sequence $({}'E_r, {}'d_r)_{r \geq 0}$ *weakly converges to $H^n(\text{Tot}(K^{\bullet,\bullet}))$, abuts to $H^n(\text{Tot}(K^{\bullet,\bullet}))$, or converges to $H^n(\text{Tot}(K^{\bullet,\bullet}))$* if Definition 24.9 applies. Similarly we say the spectral sequence $({}''E_r, {}''d_r)_{r \geq 0}$ *weakly converges to $H^n(\text{Tot}(K^{\bullet,\bullet}))$, abuts to $H^n(\text{Tot}(K^{\bullet,\bullet}))$, or converges to $H^n(\text{Tot}(K^{\bullet,\bullet}))$* if Definition 24.9 applies.

As mentioned above there is no consistent terminology regarding these notions in the literature. In the situation of the definition, we have weak convergence of the first spectral sequence if for all n

$$\text{gr}_{F_I}(H^n(\text{Tot}(K^{\bullet,\bullet}))) = \bigoplus_{p+q=n} {}'E_\infty^{p,q}$$

via the canonical comparison of Lemma 24.6. Similarly the second spectral sequence $({}''E_r, {}''d_r)_{r \geq 0}$ weakly converges if for all n

$$\text{gr}_{F_{II}}(H^n(\text{Tot}(K^{\bullet,\bullet}))) = \bigoplus_{p+q=n} {}''E_\infty^{p,q}$$

via the canonical comparison of Lemma 24.6.

Lemma 25.3. *Let $\mathcal{A}$ be an abelian category. Let $K^{\bullet,\bullet}$ be a double complex. Assume that for every $n \in \mathbf{Z}$ there are only finitely many nonzero $K^{p,q}$ with $p+q = n$. Then*

- (1) *the two spectral sequences associated to $K^{\bullet,\bullet}$ are bounded,*
- (2) *the filtrations F_I, F_{II} on each $H^n(\text{Tot}(K^{\bullet,\bullet}))$ are finite,*
- (3) *the spectral sequences $({}'E_r, {}'d_r)_{r \geq 0}$ and $({}''E_r, {}''d_r)_{r \geq 0}$ converge to $H^*(\text{Tot}(K^{\bullet,\bullet}))$,*
- (4) *if $\mathcal{C} \subset \mathcal{A}$ is a weak Serre subcategory and for some r we have $'E_r^{p,q} \in \mathcal{C}$ for all $p, q \in \mathbf{Z}$, then $H^n(\text{Tot}(K^{\bullet,\bullet}))$ is in $\mathcal{C}$. Similarly for $({}''E_r, {}''d_r)_{r \geq 0}$.*

Proof. Follows immediately from Lemma 24.11. $\square$

Here is our first application of spectral sequences.

Lemma 25.4. *Let $\mathcal{A}$ be an abelian category. Let $K^\bullet$ be a complex. Let $A^{\bullet,\bullet}$ be a double complex. Let $\alpha^p : K^p \rightarrow A^{p,0}$ be morphisms. Assume that*

- (1) *For every $n \in \mathbf{Z}$ there are only finitely many nonzero $A^{p,q}$ with $p+q = n$.*
- (2) *We have $A^{p,q} = 0$ if $q < 0$.*
- (3) *The morphisms α^p give rise to a morphism of complexes $\alpha : K^\bullet \rightarrow A^{\bullet,0}$.*
- (4) *The complex $A^{p,\bullet}$ is exact in all degrees $q \neq 0$ and the morphism $K^p \rightarrow A^{p,0}$ induces an isomorphism $K^p \rightarrow \text{Ker}(d_2^{p,0})$.*

Then α induces a quasi-isomorphism

$$K^\bullet \longrightarrow \text{Tot}(A^{\bullet,\bullet})$$

of complexes. Moreover, there is a variant of this lemma involving the second variable q instead of p .

Proof. The map is simply the map given by the morphisms $K^n \rightarrow A^{n,0} \rightarrow \text{Tot}^n(A^{\bullet,\bullet})$, which are easily seen to define a morphism of complexes. Consider the spectral sequence $({}'E_r, {}'d_r)_{r \geq 0}$ associated to the double complex $A^{\bullet,\bullet}$. By Lemma 25.3 this spectral sequence converges and the induced filtration on $H^n(\text{Tot}(A^{\bullet,\bullet}))$ is finite for each n . By Lemma 25.1 and assumption (4) we have $'E_1^{p,q} = 0$ unless $q = 0$ and $'E_1^{p,0} = K^p$ with differential $'d_1^{p,0}$ identified with d_K^p . Hence $'E_2^{p,0} = H^p(K^\bullet)$ and zero otherwise. This clearly implies $'d_2^{p,q} = {}'d_3^{p,q} = \dots = 0$ for degree reasons. Hence we conclude that $H^n(\text{Tot}(A^{\bullet,\bullet})) = H^n(K^\bullet)$. We omit the verification that this identification is given by the morphism of complexes $K^\bullet \rightarrow \text{Tot}(A^{\bullet,\bullet})$ introduced above. $\square$

Lemma 25.5. *Let $\mathcal{A}$ be an abelian category. Let $M^\bullet$ be a complex of $\mathcal{A}$. Let*

$$a : M^\bullet[0] \longrightarrow (A^{0,\bullet} \rightarrow A^{1,\bullet} \rightarrow A^{2,\bullet} \rightarrow \dots)$$

be a homotopy equivalence in the category of complexes of complexes of $\mathcal{A}$. Then the map $\alpha : M^\bullet \rightarrow \text{Tot}(A^{\bullet,\bullet})$ induced by $M^\bullet \rightarrow A^{0,\bullet}$ is a homotopy equivalence.

Proof. The statement makes sense as a complex of complexes is the same thing as a double complex. The assumption means there is a map

$$b : (A^{0,\bullet} \rightarrow A^{1,\bullet} \rightarrow A^{2,\bullet} \rightarrow \dots) \longrightarrow M^\bullet[0]$$

such that $a \circ b$ and $b \circ a$ are homotopic to the identity in the category of complexes of complexes. This means that $b \circ a$ is the identity of $M^\bullet[0]$ (because there is only one term in degree 0). Also, observe that b is given by a map $b^0 : A^{0,\bullet} \rightarrow M^\bullet$ and zero in all other degrees. Thus b induces a map $\beta : \text{Tot}(A^{\bullet,\bullet}) \rightarrow M^\bullet$ and $\beta \circ \alpha$ is the identity on $M^\bullet$. Finally, we have to show that the map $\alpha \circ \beta$ is homotopic to the identity. For this we choose maps of complexes $h^n : A^{n,\bullet} \rightarrow A^{n-1,\bullet}$ such that

$a \circ b - \text{id} = d_1 \circ h + h \circ d_1$ which exist by assumption. Here $d_1 : A^{n,\bullet} \rightarrow A^{n+1,\bullet}$ are the differentials of the complex of complexes. We will also denote d_2 the differentials of the complexes $A^{n,\bullet}$ for all n . Let $h^{n,m} : A^{n,m} \rightarrow A^{n+1,m}$ be the components of h^n . Then we can consider

$$h' : \text{Tot}(A^{\bullet,\bullet})^k = \bigoplus_{n+m=k} A^{n,m} \rightarrow \bigoplus_{n+m=k-1} A^{n,m} = \text{Tot}(A^{\bullet,\bullet})^{k-1}$$

given by $h^{n,m}$ on the summand $A^{n,m}$. Then we compute that the map

$$d_{\text{Tot}(A^{\bullet,\bullet})} \circ h' + h' \circ d_{\text{Tot}(A^{\bullet,\bullet})}$$

restricted to the summand $A^{n,m}$ is equal to

$$d_1^{n-1,m} \circ h^{n,m} + (-1)^{n-1} d_2^{n-1,m} \circ h^{n,m} + h^{n+1,m} \circ d_1^{n,m} + h^{n,m+1} \circ (-1)^n d_2^{n,m}$$

Since h^n is a map of complexes, the terms $(-1)^{n-1} d_2^{n-1,m} \circ h^{n,m}$ and $h^{n,m+1} \circ (-1)^n d_2^{n,m}$ cancel. The other two terms give $(\alpha \circ \beta)|_{A^{n,m}} - \text{id}_{A^{n,m}}$ because $a \circ b - \text{id} = d_1 \circ h + h \circ d_1$. This finishes the proof. $\square$

26. Double complexes of abelian groups

In this section we put some results on double complexes of abelian groups for which do not (yet) have the analogues results for general abelian categories. Please be careful not to use these lemmas except when the underlying abelian category is the category of abelian groups or some such (e.g., the category of modules over a ring). Some of the arguments will be difficult to follow without drawing “zig-zags” on a napkin – compare with the proof of Algebra, Lemma ??.

Lemma 26.1. *Let $M^\bullet$ be a complex of abelian groups. Let*

$$0 \rightarrow M^\bullet \rightarrow A_0^\bullet \rightarrow A_1^\bullet \rightarrow A_2^\bullet \rightarrow \dots$$

be an exact complex of complexes of abelian groups. Set $A^{p,q} = A_p^q$ to obtain a double complex. Then the map $M^\bullet \rightarrow \text{Tot}(A^{\bullet,\bullet})$ induced by $M^\bullet \rightarrow A_0^\bullet$ is a quasi-isomorphism.

Proof. If there exists a $t \in \mathbf{Z}$ such that $A_0^q = 0$ for $q < t$, then this follows immediately from Lemma 25.4 (with p and q swapped as in the final statement of that lemma). OK, but for every $t \in \mathbf{Z}$ we have a complex

$$0 \rightarrow \sigma_{\geq t} M^\bullet \rightarrow \sigma_{\geq t} A_0^\bullet \rightarrow \sigma_{\geq t} A_1^\bullet \rightarrow \sigma_{\geq t} A_2^\bullet \rightarrow \dots$$

of stupid truncations. Denote $A(t)^{\bullet,\bullet}$ the corresponding double complex. Every element ξ of $H^n(\text{Tot}(A^{\bullet,\bullet}))$ is the image of an element of $H^n(\text{Tot}(A(t)^{\bullet,\bullet}))$ for some t (look at explicit representatives of cohomology classes). Hence ξ is in the image of $H^n(\sigma_{\geq t} M^\bullet)$. Thus the map $H^n(M^\bullet) \rightarrow H^n(\text{Tot}(A^{\bullet,\bullet}))$ is surjective. It is injective because for all t the map $H^n(\sigma_{\geq t} M^\bullet) \rightarrow H^n(\text{Tot}(A(t)^{\bullet,\bullet}))$ is injective and similar arguments. $\square$

Lemma 26.2. *Let $M^\bullet$ be a complex of abelian groups. Let*

$$\dots \rightarrow A_2^\bullet \rightarrow A_1^\bullet \rightarrow A_0^\bullet \rightarrow M^\bullet \rightarrow 0$$

be an exact complex of complexes of abelian groups such that for all $p \in \mathbf{Z}$ the complexes

$$\dots \rightarrow \text{Ker}(d_{A_2^p}^p) \rightarrow \text{Ker}(d_{A_1^p}^p) \rightarrow \text{Ker}(d_{A_0^p}^p) \rightarrow \text{Ker}(d_{M^p}^p) \rightarrow 0$$

are exact as well. Set $A^{p,q} = A_{-p}^q$ to obtain a double complex. Then $\text{Tot}(A^{\bullet,\bullet}) \rightarrow M^\bullet$ induced by $A_0^\bullet \rightarrow M^\bullet$ is a quasi-isomorphism.

Proof. Using the short exact sequences $0 \rightarrow \text{Ker}(d_{A_n^\bullet}^p) \rightarrow A_n^p \rightarrow \text{Im}(d_{A_n^\bullet}^p) \rightarrow 0$ and the assumptions we see that

$$\dots \rightarrow \text{Im}(d_{A_2^\bullet}^p) \rightarrow \text{Im}(d_{A_1^\bullet}^p) \rightarrow \text{Im}(d_{A_0^\bullet}^p) \rightarrow \text{Im}(d_{M^\bullet}^p) \rightarrow 0$$

is exact for all $p \in \mathbf{Z}$. Repeating with the exact sequences $0 \rightarrow \text{Im}(d_{A_n^\bullet}^{p-1}) \rightarrow \text{Ker}(d_{A_n^\bullet}^p) \rightarrow H^p(A_n^\bullet) \rightarrow 0$ we find that

$$\dots \rightarrow H^p(A_2^\bullet) \rightarrow H^p(A_1^\bullet) \rightarrow H^p(A_0^\bullet) \rightarrow H^p(M^\bullet) \rightarrow 0$$

is exact for all $p \in \mathbf{Z}$.

Write $T^\bullet = \text{Tot}(A^{\bullet,\bullet})$. We will show that $H^0(T^\bullet) \rightarrow H^0(M^\bullet)$ is an isomorphism. The same argument works for other degrees. Let $x \in \text{Ker}(d_{T^\bullet}^0)$ represent an element $\xi \in H^0(T^\bullet)$. Write $x = \sum_{i=n,\dots,0} x_i$ with $x_i \in A_i^i$. Assume $n > 0$. Then x_n is in the kernel of $d_{A_n^\bullet}^n$ and maps to zero in $H^n(A_{n-1}^\bullet)$ because it maps to an element which is the boundary of x_{n-1} up to sign. By the first paragraph of the proof, we find that $x_n \bmod \text{Im}(d_{A_n^\bullet}^{n-1})$ is in the image of $H^n(A_{n+1}^\bullet) \rightarrow H^n(A_n^\bullet)$. Thus we can modify x by a boundary and reach the situation where x_n is a boundary. Modifying x once more we see that we may assume $x_n = 0$. By induction we see that every cohomology class ξ is represented by a cocycle $x = x_0$. Finally, the condition on exactness of kernels tells us two such cocycles x_0 and x'_0 are cohomologous if and only if their image in $H^0(M^\bullet)$ are the same. $\square$

Lemma 26.3. *Let $M^\bullet$ be a complex of abelian groups. Let*

$$0 \rightarrow M^\bullet \rightarrow A_0^\bullet \rightarrow A_1^\bullet \rightarrow A_2^\bullet \rightarrow \dots$$

be an exact complex of complexes of abelian groups such that for all $p \in \mathbf{Z}$ the complexes

$$0 \rightarrow \text{Coker}(d_{M^\bullet}^p) \rightarrow \text{Coker}(d_{A_0^\bullet}^p) \rightarrow \text{Coker}(d_{A_1^\bullet}^p) \rightarrow \text{Coker}(d_{A_2^\bullet}^p) \rightarrow \dots$$

are exact as well. Set $A^{p,q} = A_p^q$ to obtain a double complex. Let $\text{Tot}_\pi(A^{\bullet,\bullet})$ be the product total complex associated to the double complex (see proof). Then the map $M^\bullet \rightarrow \text{Tot}_\pi(A^{\bullet,\bullet})$ induced by $M^\bullet \rightarrow A_0^\bullet$ is a quasi-isomorphism.

Proof. Abbreviating $T^\bullet = \text{Tot}_\pi(A^{\bullet,\bullet})$ we define

$$T^n = \prod_{p+q=n} A^{p,q} = \prod_{p+q=n} A_p^q \quad \text{with} \quad d_{T^\bullet}^n = \prod_{n=p+q} (f_p^q + (-1)^p d_{A_p^\bullet}^q)$$

where $f_p^\bullet : A_p^\bullet \rightarrow A_{p+1}^\bullet$ are the maps of complexes in the lemma.

We will show that $H^0(M^\bullet) \rightarrow H^0(T^\bullet)$ is an isomorphism. The same argument works for other degrees. Let $x \in \text{Ker}(d_{T^\bullet}^0)$ represent $\xi \in H^0(T^\bullet)$. Write $x = (x_i)$ with $x_i \in A_i^{-i}$. Note that x_0 maps to zero in $\text{Coker}(A_1^{-1} \rightarrow A_1^0)$. Hence we see that $x_0 = m_0 + d_{A_0^\bullet}^{-1}(y)$ for some $m_0 \in M^0$ and $y \in A_0^{-1}$. Then $d_{M^\bullet}(m_0) = 0$ because $d_{A_0^\bullet}(x_0) = 0$ as $d_{T^\bullet}(x) = 0$. Thus, replacing ξ by something in the image of $H^0(M^\bullet) \rightarrow H^0(T^\bullet)$ we may assume that x_0 is in $\text{Im}(d_{A_0^\bullet}^{-1})$.

Assume $x_0 \in \text{Im}(d_{A_0^\bullet}^{-1})$. We claim that in this case $\xi = 0$. To prove this we find, by induction on n elements $y_0, y_1, \dots, y_n$ with $y_i \in A_i^{-i-1}$ such that $x_0 = d_{A_0^\bullet}^{-1}(y_0)$ and $x_j = f_{j-1}^{-j}(y_{j-1}) + (-1)^j d_{A_j^\bullet}^{-j-1}(y_j)$ for $j = 1, \dots, n$. This is clear for $n = 0$. Proof of induction step: suppose we have found $y_0, \dots, y_{n-1}$. Then $w_n = x_n - f_{n-1}^{-n}(y_{n-1})$ is in the kernel of $d_{A_n^\bullet}^{-n}$ and maps to zero in $H^{-n}(A_{n+1}^\bullet)$ (because it maps to an

element which is the boundary of x_{n+1} up to sign). Exactly as in the proof of Lemma 26.2 the assumptions of the lemma imply that

$$0 \rightarrow H^p(M^\bullet) \rightarrow H^p(A_0^\bullet) \rightarrow H^p(A_1^\bullet) \rightarrow H^p(A_2^\bullet) \rightarrow \dots$$

is exact for all $p \in \mathbf{Z}$. Thus after changing y_{n-1} by an element in $\text{Ker}(d_{A_{n-1}^\bullet}^{n-1})$ we may assume that w_n maps to zero in $H^{-n}(A_n^\bullet)$. This means we can find y_n as desired. Observe that this procedure does not change $y_0, \dots, y_{n-2}$. Hence continuing ad infinitum we find an element $y = (y_i)$ in T^{n-1} with $d_{T^\bullet}(y) = \xi$. This shows that $H^0(M^\bullet) \rightarrow H^0(T^\bullet)$ is surjective.

Suppose that $m_0 \in \text{Ker}(d_{M^\bullet}^0)$ maps to zero in $H^0(T^\bullet)$. Say it maps to the differential applied to $y = (y_i) \in T^{-1}$. Then $y_0 \in A_0^{-1}$ maps to zero in $\text{Coker}(d_{A_1^\bullet}^{-2})$. By assumption this means that $y_0 \bmod \text{Im}(d_{A_0^\bullet}^{-2})$ is the image of some $z \in M^{-1}$. It follows that $m_0 = d_{M^\bullet}^{-1}(z)$. This proves injectivity and the proof is complete. $\square$

Lemma 26.4. *Let $M^\bullet$ be a complex of abelian groups. Let*

$$\dots \rightarrow A_2^\bullet \rightarrow A_1^\bullet \rightarrow A_0^\bullet \rightarrow M^\bullet \rightarrow 0$$

be an exact complex of complexes of abelian groups. Set $A^{p,q} = A_{-p}^q$ to obtain a double complex. Let $\text{Tot}_\pi(A^{\bullet,\bullet})$ be the product total complex associated to the double complex (see proof). Then the map $\text{Tot}_\pi(A^{\bullet,\bullet}) \rightarrow M^\bullet$ induced by $A_0^\bullet \rightarrow M^\bullet$ is a quasi-isomorphism.

Proof. Abbreviating $T^\bullet = \text{Tot}_\pi(A^{\bullet,\bullet})$ we define

$$T^n = \prod_{p+q=n} A^{p,q} = \prod_{p+q=n} A_{-p}^q \quad \text{with} \quad d_{T^\bullet}^n = \prod_{n=p+q} (f_{-p}^q + (-1)^p d_{A_{-p}^\bullet}^q)$$

where $f_p^\bullet : A_p^\bullet \rightarrow A_{p-1}^\bullet$ are the maps of complexes in the lemma. We will show that $T^\bullet$ is acyclic when $M^\bullet$ is the zero complex. This will suffice by the following trick. Set $B_n^\bullet = A_{n+1}^\bullet$ and $B_0^\bullet = M^\bullet$. Then we have an exact sequence

$$\dots \rightarrow B_2^\bullet \rightarrow B_1^\bullet \rightarrow B_0^\bullet \rightarrow 0 \rightarrow 0$$

as in the lemma. Let $S^\bullet = \text{Tot}_\pi(B^{\bullet,\bullet})$. Then there is an obvious short exact sequence of complexes

$$0 \rightarrow M^\bullet \rightarrow S^\bullet \rightarrow T^\bullet[1] \rightarrow 0$$

and we conclude by the long exact cohomology sequence. Some details omitted.

Assume $M^\bullet = 0$. We will show $H^0(T^\bullet) = 0$. The same argument works for other degrees. Let $x = (x_n) \in \text{Ker}(d_{T^\bullet})$ map to $\xi \in H^0(T^\bullet)$ with $x_n \in A^{-n,n} = A_n^n$. Since $M^0 = 0$ we find that $x_0 = f_1^0(y_0)$ for some $y_0 \in A_1^0$. Then $x_1 - d_{A_1^\bullet}^0(y_0) = f_2^1(y_1)$ because it is mapped to zero by f_1^1 as x is a cocycle. for some $y_1 \in A_2^1$. Continuing, using induction, we find $y = (y_i) \in T^{-1}$ with $d_{T^\bullet}(y) = x$ as desired. $\square$

27. Injectives

Definition 27.1. Let $\mathcal{A}$ be an abelian category. An object $J \in \text{Ob}(\mathcal{A})$ is called *injective* if for every injection $A \hookrightarrow B$ and every morphism $A \rightarrow J$ there exists a morphism $B \rightarrow J$ making the following diagram commute

$$\begin{array}{ccc} A & \longrightarrow & B \\ \downarrow & & \nearrow \\ J & & \end{array}$$

Here is the obligatory characterization of injective objects.

Lemma 27.2. *Let $\mathcal{A}$ be an abelian category. Let I be an object of $\mathcal{A}$. The following are equivalent:*

- (1) *The object I is injective.*
- (2) *The functor $B \mapsto \text{Hom}_{\mathcal{A}}(B, I)$ is exact.*
- (3) *Any short exact sequence*

$$0 \rightarrow I \rightarrow A \rightarrow B \rightarrow 0$$

in $\mathcal{A}$ is split.

- (4) *We have $\text{Ext}_{\mathcal{A}}(B, I) = 0$ for all $B \in \text{Ob}(\mathcal{A})$.*

Proof. Omitted. □

Lemma 27.3. *Let $\mathcal{A}$ be an abelian category. Suppose $I_{\omega}, \omega \in \Omega$ is a set of injective objects of $\mathcal{A}$. If $\prod_{\omega \in \Omega} I_{\omega}$ exists then it is injective.*

Proof. Omitted. □

Definition 27.4. Let $\mathcal{A}$ be an abelian category. We say $\mathcal{A}$ has *enough injectives* if every object A has an injective morphism $A \rightarrow J$ into an injective object J .

Definition 27.5. Let $\mathcal{A}$ be an abelian category. We say that $\mathcal{A}$ has *functorial injective embeddings* if there exists a functor

$$J : \mathcal{A} \longrightarrow \text{Arrows}(\mathcal{A})$$

such that

- (1) $s \circ J = \text{id}_{\mathcal{A}}$,
- (2) for any object $A \in \text{Ob}(\mathcal{A})$ the morphism $J(A)$ is injective, and
- (3) for any object $A \in \text{Ob}(\mathcal{A})$ the object $t(J(A))$ is an injective object of $\mathcal{A}$.

We will denote such a functor by $A \mapsto (A \rightarrow J(A))$.

28. Projectives

Definition 28.1. Let $\mathcal{A}$ be an abelian category. An object $P \in \text{Ob}(\mathcal{A})$ is called *projective* if for every surjection $A \rightarrow B$ and every morphism $P \rightarrow B$ there exists a morphism $P \rightarrow A$ making the following diagram commute

$$\begin{array}{ccc} A & \longrightarrow & B \\ \uparrow & \nearrow & \\ P & & \end{array}$$

Here is the obligatory characterization of projective objects.

Lemma 28.2. *Let $\mathcal{A}$ be an abelian category. Let P be an object of $\mathcal{A}$. The following are equivalent:*

- (1) *The object P is projective.*
- (2) *The functor $B \mapsto \text{Hom}_{\mathcal{A}}(P, B)$ is exact.*
- (3) *Any short exact sequence*

$$0 \rightarrow A \rightarrow B \rightarrow P \rightarrow 0$$

in $\mathcal{A}$ is split.

- (4) *We have $\text{Ext}_{\mathcal{A}}(P, A) = 0$ for all $A \in \text{Ob}(\mathcal{A})$.*

Proof. Omitted. $\square$

Lemma 28.3. *Let $\mathcal{A}$ be an abelian category. Suppose P_ω , $\omega \in \Omega$ is a set of projective objects of $\mathcal{A}$. If $\coprod_{\omega \in \Omega} P_\omega$ exists then it is projective.*

Proof. Omitted. $\square$

Definition 28.4. Let $\mathcal{A}$ be an abelian category. We say $\mathcal{A}$ has *enough projectives* if every object A has a surjective morphism $P \rightarrow A$ from a projective object P onto it.

Definition 28.5. Let $\mathcal{A}$ be an abelian category. We say that $\mathcal{A}$ has *functorial projective surjections* if there exists a functor

$$P : \mathcal{A} \longrightarrow \text{Arrows}(\mathcal{A})$$

such that

- (1) $t \circ P = \text{id}_{\mathcal{A}}$,
- (2) for any object $A \in \text{Ob}(\mathcal{A})$ the morphism $P(A)$ is surjective, and
- (3) for any object $A \in \text{Ob}(\mathcal{A})$ the object $s(P(A))$ is a projective object of $\mathcal{A}$.

We will denote such a functor by $A \mapsto (P(A) \rightarrow A)$.

29. Injectives and adjoint functors

Here are some lemmas on adjoint functors and their relationship with injectives. See also Lemma 7.4.

Lemma 29.1. *Let $\mathcal{A}$ and $\mathcal{B}$ be abelian categories. Let $u : \mathcal{A} \rightarrow \mathcal{B}$ and $v : \mathcal{B} \rightarrow \mathcal{A}$ be additive functors with u right adjoint to v . Consider the following conditions:*

- (a) *v transforms injective maps into injective maps,*
- (b) *v is exact, and*
- (c) *u transforms injectives into injectives.*

Then (a) $\Leftrightarrow$ (b) $\Rightarrow$ (c). If $\mathcal{A}$ has enough injectives, then all three conditions are equivalent.

Proof. Observe that v is right exact as a left adjoint (Categories, Lemma ??). Combined with Lemma 7.2 this explains why (a) $\Leftrightarrow$ (b).

Assume (a). Let I be an injective object of $\mathcal{A}$. Let $\varphi : N \rightarrow M$ be an injective map in $\mathcal{B}$ and let $\alpha : N \rightarrow uI$ be a morphism. By adjointness we get a morphism $\alpha : vN \rightarrow I$ and by assumption $v\varphi : vN \rightarrow vM$ is injective. Hence as I is an injective object we get a morphism $\beta : vM \rightarrow I$ extending α . By adjointness again this corresponds to a morphism $\beta : M \rightarrow uI$ extending α . Hence (c) is true.

Assume $\mathcal{A}$ has enough injectives and (c) holds. Let $f : B \rightarrow B'$ be a monomorphism in $\mathcal{B}$, and let $A = \text{Ker}(v(f))$. Choose a monomorphism $g : A \rightarrow I$ with I injective (it exists by assumption). Then g extends to $g' : v(B) \rightarrow I$, whence by adjunction a morphism $B \rightarrow u(I)$. Since $u(I)$ is injective, this morphism extends to $h : B' \rightarrow u(I)$, hence by adjunction a morphism $k : v(B') \rightarrow I$ extending g' . But then k "extends" g , which forces $A = 0$ since g was a monomorphism. Thus (a) is true. $\square$

Remark 29.2. Let $R \rightarrow S$ be a ring map. Let $u : \text{Mod}_S \rightarrow \text{Mod}_R$ be $u(N) = N_R$ and let $v : \text{Mod}_R \rightarrow \text{Mod}_S$ be $v(M) = M \otimes_R S$. Then u is right adjoint to v , and u is exact and v is right exact. But conditions (a), (b), (c) of Lemma 29.1 do not hold in general. For example, if $R = \mathbf{Z}$ and $S = \mathbf{Z}/p\mathbf{Z}$, then the injective S -module $\mathbf{Z}/p\mathbf{Z}$ is not an injective $\mathbf{Z}$ -module. In fact, the lemma shows all injective S -modules are injective as R -modules if and only if $R \rightarrow S$ is a flat ring map.

Lemma 29.3. *Let $\mathcal{A}$ and $\mathcal{B}$ be abelian categories. Let $u : \mathcal{A} \rightarrow \mathcal{B}$ and $v : \mathcal{B} \rightarrow \mathcal{A}$ be additive functors. Assume*

- (1) *u is right adjoint to v ,*
- (2) *v transforms injective maps into injective maps,*
- (3) *$\mathcal{A}$ has enough injectives, and*
- (4) *$vB = 0$ implies $B = 0$ for any $B \in \text{Ob}(\mathcal{B})$.*

Then $\mathcal{B}$ has enough injectives.

Proof. Pick $B \in \text{Ob}(\mathcal{B})$. Pick an injection $vB \rightarrow I$ for I an injective object of $\mathcal{A}$. According to Lemma 29.1 and the assumptions the corresponding map $B \rightarrow uI$ is the injection of B into an injective object. $\square$

Remark 29.4. Let $\mathcal{A}, \mathcal{B}, u : \mathcal{A} \rightarrow \mathcal{B}$ and $v : \mathcal{B} \rightarrow \mathcal{A}$ be as In Lemma 29.3. In the presence of conditions (1) and (2) condition (4) is equivalent to v being faithful. Moreover, condition (4) is needed. An example is to consider the case where the functors u and v are both the zero functor.

Lemma 29.5. *Let $\mathcal{A}$ and $\mathcal{B}$ be abelian categories. Let $u : \mathcal{A} \rightarrow \mathcal{B}$ and $v : \mathcal{B} \rightarrow \mathcal{A}$ be additive functors. Assume*

- (1) *u is right adjoint to v ,*
- (2) *v transforms injective maps into injective maps,*
- (3) *$\mathcal{A}$ has enough injectives,*
- (4) *$vB = 0$ implies $B = 0$ for any $B \in \text{Ob}(\mathcal{B})$, and*
- (5) *$\mathcal{A}$ has functorial injective embeddings.*

Then $\mathcal{B}$ has functorial injective embeddings.

Proof. Let $A \mapsto (A \rightarrow J(A))$ be a functorial injective embedding on $\mathcal{A}$. Then $B \mapsto (B \rightarrow uJ(vB))$ is a functorial injective embedding on $\mathcal{B}$. Compare with the proof of Lemma 29.3. $\square$

Lemma 29.6. *Let $\mathcal{A}$ and $\mathcal{B}$ be abelian categories. Let $u : \mathcal{A} \rightarrow \mathcal{B}$ be a functor. If there exists a subset $\mathcal{P} \subset \text{Ob}(\mathcal{B})$ such that*

- (1) *every object of $\mathcal{B}$ is a quotient of an element of $\mathcal{P}$, and*
- (2) *for every $P \in \mathcal{P}$ there exists an object Q of $\mathcal{A}$ such that $\text{Hom}_{\mathcal{A}}(Q, A) = \text{Hom}_{\mathcal{B}}(P, u(A))$ functorially in A ,*

then there exists a left adjoint v of u .

Proof. By the Yoneda lemma (Categories, Lemma ??) the object Q of $\mathcal{A}$ corresponding to P is defined up to unique isomorphism by the formula $\text{Hom}_{\mathcal{A}}(Q, A) = \text{Hom}_{\mathcal{B}}(P, u(A))$. Let us write $Q = v(P)$. Denote $i_P : P \rightarrow u(v(P))$ the map corresponding to $\text{id}_{v(P)}$ in $\text{Hom}_{\mathcal{A}}(v(P), v(P))$. Functoriality in (2) implies that the bijection is given by

$$\text{Hom}_{\mathcal{A}}(v(P), A) \rightarrow \text{Hom}_{\mathcal{B}}(P, u(A)), \quad \varphi \mapsto u(\varphi) \circ i_P$$

For any pair of elements $P_1, P_2 \in \mathcal{P}$ there is a canonical map

$$\mathrm{Hom}_{\mathcal{B}}(P_2, P_1) \rightarrow \mathrm{Hom}_{\mathcal{A}}(v(P_2), v(P_1)), \quad \varphi \mapsto v(\varphi)$$

which is characterized by the rule $u(v(\varphi)) \circ i_{P_2} = i_{P_1} \circ \varphi$ in $\mathrm{Hom}_{\mathcal{B}}(P_2, u(v(P_1)))$. Note that $\varphi \mapsto v(\varphi)$ is compatible with composition; this can be seen directly from the characterization. Hence $P \mapsto v(P)$ is a functor from the full subcategory of $\mathcal{B}$ whose objects are the elements of $\mathcal{P}$.

Given an arbitrary object B of $\mathcal{B}$ choose an exact sequence

$$P_2 \rightarrow P_1 \rightarrow B \rightarrow 0$$

which is possible by assumption (1). Define $v(B)$ to be the object of $\mathcal{A}$ fitting into the exact sequence

$$v(P_2) \rightarrow v(P_1) \rightarrow v(B) \rightarrow 0$$

Then

$$\begin{aligned} \mathrm{Hom}_{\mathcal{A}}(v(B), A) &= \mathrm{Ker}(\mathrm{Hom}_{\mathcal{A}}(v(P_1), A) \rightarrow \mathrm{Hom}_{\mathcal{A}}(v(P_2), A)) \\ &= \mathrm{Ker}(\mathrm{Hom}_{\mathcal{B}}(P_1, u(A)) \rightarrow \mathrm{Hom}_{\mathcal{B}}(P_2, u(A))) \\ &= \mathrm{Hom}_{\mathcal{B}}(B, u(A)) \end{aligned}$$

Hence we see that we may take $\mathcal{P} = \mathrm{Ob}(\mathcal{B})$, i.e., we see that v is everywhere defined. $\square$

30. Essentially constant systems

In this section we discuss essentially constant systems with values in additive categories.

Lemma 30.1. *Let $\mathcal{I}$ be a category, let $\mathcal{A}$ be a pre-additive Karoubian category, and let $M : \mathcal{I} \rightarrow \mathcal{A}$ be a diagram.*

- (1) *Assume $\mathcal{I}$ is filtered. The following are equivalent*
 - (a) *M is essentially constant,*
 - (b) *$X = \mathrm{colim} M$ exists and there exists a cofinal filtered subcategory $\mathcal{I}' \subset \mathcal{I}$ and for $i' \in \mathrm{Ob}(\mathcal{I}')$ a direct sum decomposition $M_{i'} = X_{i'} \oplus Z_{i'}$ such that $X_{i'}$ maps isomorphically to X and $Z_{i'}$ to zero in $M_{i''}$ for some $i' \rightarrow i''$ in $\mathcal{I}'$.*
- (2) *Assume $\mathcal{I}$ is cofiltered. The following are equivalent*
 - (a) *M is essentially constant,*
 - (b) *$X = \mathrm{lim} M$ exists and there exists an initial cofiltered subcategory $\mathcal{I}' \subset \mathcal{I}$ and for $i' \in \mathrm{Ob}(\mathcal{I}')$ a direct sum decomposition $M_{i'} = X_{i'} \oplus Z_{i'}$ such that X maps isomorphically to $X_{i'}$ and $M_{i''} \rightarrow Z_{i'}$ is zero for some $i'' \rightarrow i'$ in $\mathcal{I}'$.*

Proof. Assume (1)(a), i.e., $\mathcal{I}$ is filtered and M is essentially constant. Let $X = \mathrm{colim} M_i$. Choose i and $X \rightarrow M_i$ as in Categories, Definition ???. Let $\mathcal{I}'$ be the full subcategory consisting of objects which are the target of a morphism with source i . Suppose $i' \in \mathrm{Ob}(\mathcal{I}')$ and choose a morphism $i \rightarrow i'$. Then $X \rightarrow M_i \rightarrow M_{i'}$ composed with $M_{i'} \rightarrow X$ is the identity on X . As $\mathcal{A}$ is Karoubian, we find a direct summand decomposition $M_{i'} = X_{i'} \oplus Z_{i'}$, where $Z_{i'} = \mathrm{Ker}(M_{i'} \rightarrow X)$ and $X_{i'}$ maps isomorphically to X . Pick $i \rightarrow k$ and $i' \rightarrow k$ such that $M_{i'} \rightarrow X \rightarrow M_i \rightarrow M_k$ equals $M_{i'} \rightarrow M_k$ as in Categories, Definition ???. Then we see that $M_{i'} \rightarrow M_k$ annihilates $Z_{i'}$. Thus (1)(b) holds.

Assume (1)(b), i.e., $\mathcal{I}$ is filtered and we have $\mathcal{I}' \subset \mathcal{I}$ and for $i' \in \text{Ob}(\mathcal{I}')$ a direct sum decomposition $M_{i'} = X_{i'} \oplus Z_{i'}$ as stated in the lemma. To see that M is essentially constant we can replace $\mathcal{I}$ by $\mathcal{I}'$, see Categories, Lemma ???. Pick any $i \in \text{Ob}(\mathcal{I})$ and denote $X \rightarrow M_i$ the inverse of the isomorphism $X_i \rightarrow X$ followed by the inclusion map $X_i \rightarrow M_i$. If j is a second object, then choose $j \rightarrow k$ such that $Z_j \rightarrow M_k$ is zero. Since $\mathcal{I}$ is filtered we may also assume there is a morphism $i \rightarrow k$ (after possibly increasing k). Then $M_j \rightarrow X \rightarrow M_i \rightarrow M_k$ and $M_j \rightarrow M_k$ both annihilate Z_j . Thus after postcomposing by a morphism $M_k \rightarrow M_l$ which annihilates the summand Z_k , we find that $M_j \rightarrow X \rightarrow M_i \rightarrow M_l$ and $M_j \rightarrow M_l$ are equal, i.e., M is essentially constant.

The proof of (2) is dual. $\square$

Lemma 30.2. *Let $\mathcal{I}$ be a category. Let $\mathcal{A}$ be an additive, Karoubian category. Let $F : \mathcal{I} \rightarrow \mathcal{A}$ and $G : \mathcal{I} \rightarrow \mathcal{A}$ be functors. The following are equivalent*

- (1) $\text{colim}_{\mathcal{I}} F \oplus G$ exists, and
- (2) $\text{colim}_{\mathcal{I}} F$ and $\text{colim}_{\mathcal{I}} G$ exist.

In this case $\text{colim}_{\mathcal{I}} F \oplus G = \text{colim}_{\mathcal{I}} F \oplus \text{colim}_{\mathcal{I}} G$.

Proof. Assume (1) holds. Set $W = \text{colim}_{\mathcal{I}} F \oplus G$. Note that the projection onto F defines natural transformation $F \oplus G \rightarrow F \oplus G$ which is idempotent. Hence we obtain an idempotent endomorphism $W \rightarrow W$ by Categories, Lemma ???. Since $\mathcal{A}$ is Karoubian we get a corresponding direct sum decomposition $W = X \oplus Y$, see Lemma 4.2. A straightforward argument (omitted) shows that $X = \text{colim}_{\mathcal{I}} F$ and $Y = \text{colim}_{\mathcal{I}} G$. Thus (2) holds. We omit the proof that (2) implies (1). $\square$

Lemma 30.3. *Let $\mathcal{I}$ be a filtered category. Let $\mathcal{A}$ be an additive, Karoubian category. Let $F : \mathcal{I} \rightarrow \mathcal{A}$ and $G : \mathcal{I} \rightarrow \mathcal{A}$ be functors. The following are equivalent*

- (1) $F \oplus G : \mathcal{I} \rightarrow \mathcal{A}$ is essentially constant, and
- (2) F and G are essentially constant.

Proof. Assume (1) holds. In particular $W = \text{colim}_{\mathcal{I}} F \oplus G$ exists and hence by Lemma 30.2 we have $W = X \oplus Y$ with $X = \text{colim}_{\mathcal{I}} F$ and $Y = \text{colim}_{\mathcal{I}} G$. A straightforward argument (omitted) using for example the characterization of Categories, Lemma ?? shows that F is essentially constant with value X and G is essentially constant with value Y . Thus (2) holds. The proof that (2) implies (1) is omitted. $\square$

31. Inverse systems

Let $\mathcal{C}$ be a category. In Categories, Section ?? we defined the notion of an inverse system over a preordered set (with values in the category $\mathcal{C}$). If the preordered set is $\mathbf{N} = \{1, 2, 3, \dots\}$ with the usual ordering such an inverse system over $\mathbf{N}$ is often simply called an *inverse system*. It consists quite simply of a pair $(M_i, f_{ii'})$ where each M_i , $i \in \mathbf{N}$ is an object of $\mathcal{C}$, and for each $i > i'$, $i, i' \in \mathbf{N}$ a morphism $f_{ii'} : M_i \rightarrow M_{i'}$ such that moreover $f_{i'i''} \circ f_{ii'} = f_{ii''}$ whenever this makes sense. It is clear that in fact it suffices to give the morphisms $M_2 \rightarrow M_1$, $M_3 \rightarrow M_2$, and so on. Hence an inverse system is frequently pictured as follows

$$M_1 \xleftarrow{\varphi_2} M_2 \xleftarrow{\varphi_3} M_3 \leftarrow \dots$$

Moreover, we often omit the transition maps φ_i from the notation and we simply say “let (M_i) be an inverse system”.

The collection of all inverse systems with values in $\mathcal{C}$ forms a category with the obvious notion of morphism.

Lemma 31.1. *Let $\mathcal{C}$ be a category.*

- (1) *If $\mathcal{C}$ is an additive category, then the category of inverse systems with values in $\mathcal{C}$ is an additive category.*
- (2) *If $\mathcal{C}$ is an abelian category, then the category of inverse systems with values in $\mathcal{C}$ is an abelian category. A sequence $(K_i) \rightarrow (L_i) \rightarrow (M_i)$ of inverse systems is exact if and only if each $K_i \rightarrow L_i \rightarrow M_i$ is exact.*

Proof. Omitted. □

The limit (see Categories, Section ??) of such an inverse system is denoted $\lim M_i$, or $\lim_i M_i$. If $\mathcal{C}$ is the category of abelian groups (or sets), then the limit always exists and in fact can be described as follows

$$\lim_i M_i = \{(x_i) \in \prod M_i \mid \varphi_i(x_i) = x_{i-1}, i = 2, 3, \dots\}$$

see Categories, Section ?. However, given a short exact sequence

$$0 \rightarrow (A_i) \rightarrow (B_i) \rightarrow (C_i) \rightarrow 0$$

of inverse systems of abelian groups it is not always the case that the associated system of limits is exact. In order to discuss this further we introduce the following notion.

Definition 31.2. Let $\mathcal{C}$ be an abelian category. We say the inverse system (A_i) satisfies the *Mittag-Leffler condition*, or for short is *ML*, if for every i there exists a $c = c(i) \geq i$ such that

$$\operatorname{Im}(A_k \rightarrow A_i) = \operatorname{Im}(A_c \rightarrow A_i)$$

for all $k \geq c$.

It turns out that the Mittag-Leffler condition is good enough to ensure that the $\lim$ -functor is exact, provided one works within the abelian category of abelian groups, modules over a ring, etc. It is shown in a paper by A. Neeman (see [Nee02]) that this condition is not strong enough in an abelian category having AB4* (having exact products).

Lemma 31.3. *Let*

$$0 \rightarrow (A_i) \rightarrow (B_i) \rightarrow (C_i) \rightarrow 0$$

be a short exact sequence of inverse systems of abelian groups.

- (1) *In any case the sequence*

$$0 \rightarrow \lim_i A_i \rightarrow \lim_i B_i \rightarrow \lim_i C_i$$

is exact.

- (2) *If (B_i) is ML, then also (C_i) is ML.*
- (3) *If (A_i) is ML, then*

$$0 \rightarrow \lim_i A_i \rightarrow \lim_i B_i \rightarrow \lim_i C_i \rightarrow 0$$

is exact.

Proof. Nice exercise. See Algebra, Lemma ?? for part (3). □

Lemma 31.4. *Let*

$$(A_i) \rightarrow (B_i) \rightarrow (C_i) \rightarrow (D_i)$$

be an exact sequence of inverse systems of abelian groups. If the system (A_i) is ML, then the sequence

$$\lim_i B_i \rightarrow \lim_i C_i \rightarrow \lim_i D_i$$

is exact.

Proof. Let $Z_i = \text{Ker}(C_i \rightarrow D_i)$ and $I_i = \text{Im}(A_i \rightarrow B_i)$. Then $\lim Z_i = \text{Ker}(\lim C_i \rightarrow \lim D_i)$ and we get a short exact sequence of systems

$$0 \rightarrow (I_i) \rightarrow (B_i) \rightarrow (Z_i) \rightarrow 0$$

Moreover, by Lemma 31.3 we see that (I_i) has (ML), thus another application of Lemma 31.3 shows that $\lim B_i \rightarrow \lim Z_i$ is surjective which proves the lemma. $\square$

The following characterization of essentially constant inverse systems shows in particular that they have ML.

Lemma 31.5. *Let $\mathcal{A}$ be an abelian category. Let (A_i) be an inverse system in $\mathcal{A}$ with limit $A = \lim A_i$. Then (A_i) is essentially constant (see Categories, Definition ??) if and only if there exists an i and for all $j \geq i$ a direct sum decomposition $A_j = A \oplus Z_j$ such that (a) the maps $A_{j'} \rightarrow A_j$ are compatible with the direct sum decompositions, (b) for all j there exists some $j' \geq j$ such that $Z_{j'} \rightarrow Z_j$ is zero.*

Proof. Assume (A_i) is essentially constant. Then there exists an i and a morphism $A_i \rightarrow A$ such that $A \rightarrow A_i \rightarrow A$ is the identity and for all $j \geq i$ there exists a $j' \geq j$ such that $A_{j'} \rightarrow A_j$ factors as $A_{j'} \rightarrow A_i \rightarrow A \rightarrow A_j$ (the last map comes from $A = \lim A_i$). Hence setting $Z_j = \text{Ker}(A_j \rightarrow A)$ for all $j \geq i$ works. Proof of the converse is omitted. $\square$

We will improve on the following lemma in More on Algebra, Lemma ??.

Lemma 31.6. *Let*

$$0 \rightarrow (A_i) \rightarrow (B_i) \rightarrow (C_i) \rightarrow 0$$

be an exact sequence of inverse systems of abelian groups. If (C_i) is essentially constant, then (A_i) has ML if and only if (B_i) has ML.

Proof. After renumbering we may assume that $C_i = C \oplus Z_i$ compatible with transition maps and that for all i there exists an $i' \geq i$ such that $Z_{i'} \rightarrow Z_i$ is zero, see Lemma 31.5.

First, assume $C = 0$, i.e., we have $C_i = Z_i$. In this case choose $1 = n_1 < n_2 < n_3 < \dots$ such that $Z_{n_{i+1}} \rightarrow Z_{n_i}$ is zero. Then $B_{n_{i+1}} \rightarrow B_{n_i}$ factors through $A_{n_i} \subset B_{n_i}$. It follows that for $j \geq i + 1$ we have

$$\text{Im}(A_{n_j} \rightarrow A_{n_i}) \subset \text{Im}(B_{n_j} \rightarrow B_{n_i}) \subset \text{Im}(A_{n_{j-1}} \rightarrow A_{n_i})$$

as subsets of A_{n_i} . Thus the images $\text{Im}(A_{n_j} \rightarrow A_{n_i})$ stabilize for $j \geq i + 1$ if and only if the same is true for the images $\text{Im}(B_{n_j} \rightarrow B_{n_i})$. The equivalence follows from this (small detail omitted).

If $C \neq 0$, denote $B'_i \subset B_i$ the inverse image of C by the map $B_i \rightarrow C \oplus Z_i$. Then by the previous paragraph we see that (B'_i) has ML if and only if (B_i) has ML. Thus we may replace (B_i) by (B'_i) . In this case we have exact sequences $0 \rightarrow A_i \rightarrow B_i \rightarrow C \rightarrow 0$ for all i . It follows that $0 \rightarrow \text{Im}(A_j \rightarrow A_i) \rightarrow \text{Im}(B_j \rightarrow B_i) \rightarrow C \rightarrow 0$

is short exact for all $j \geq i$. Hence the images $\text{Im}(A_j \rightarrow A_i)$ stabilize for $j \geq i$ if and only if the same is true for $\text{Im}(B_j \rightarrow B_i)$ as desired. $\square$

The “correct” version of the following lemma is More on Algebra, Lemma ??.

Lemma 31.7. *Let*

$$(A_i^{-2} \rightarrow A_i^{-1} \rightarrow A_i^0 \rightarrow A_i^1)$$

be an inverse system of complexes of abelian groups and denote $A^{-2} \rightarrow A^{-1} \rightarrow A^0 \rightarrow A^1$ its limit. Denote (H_i^{-1}) , (H_i^0) the inverse systems of cohomologies, and denote H^{-1} , H^0 the cohomologies of $A^{-2} \rightarrow A^{-1} \rightarrow A^0 \rightarrow A^1$. If (A_i^{-2}) and (A_i^{-1}) are ML and (H_i^{-1}) is essentially constant, then $H^0 = \lim H_i^0$.

Proof. Let $Z_i^j = \text{Ker}(A_i^j \rightarrow A_i^{j+1})$ and $I_i^j = \text{Im}(A_i^{j-1} \rightarrow A_i^j)$. Note that $\lim Z_i^0 = \text{Ker}(\lim A_i^0 \rightarrow \lim A_i^1)$ as taking kernels commutes with limits. The systems (I_i^{-1}) and (I_i^0) have ML as quotients of the systems (A_i^{-2}) and (A_i^{-1}) , see Lemma 31.3. Thus an exact sequence

$$0 \rightarrow (I_i^{-1}) \rightarrow (Z_i^{-1}) \rightarrow (H_i^{-1}) \rightarrow 0$$

of inverse systems where (I_i^{-1}) has ML and where (H_i^{-1}) is essentially constant by assumption. Hence (Z_i^{-1}) has ML by Lemma 31.6. The exact sequence

$$0 \rightarrow (Z_i^{-1}) \rightarrow (A_i^{-1}) \rightarrow (I_i^0) \rightarrow 0$$

and an application of Lemma 31.3 shows that $\lim A_i^{-1} \rightarrow \lim I_i^0$ is surjective. Finally, the exact sequence

$$0 \rightarrow (I_i^0) \rightarrow (Z_i^0) \rightarrow (H_i^0) \rightarrow 0$$

and Lemma 31.3 show that $\lim I_i^0 \rightarrow \lim Z_i^0 \rightarrow \lim H_i^0 \rightarrow 0$ is exact. Putting everything together we win. $\square$

Sometimes we need a version of the lemma above where we take limits over big ordinals.

Lemma 31.8. *Let α be an ordinal. Let $K_\beta^\bullet$, $\beta < \alpha$ be an inverse system of complexes of abelian groups over α . If for all $\beta < \alpha$ the complex $K_\beta^\bullet$ is acyclic and the map*

$$K_\beta^n \longrightarrow \lim_{\gamma < \beta} K_\gamma^n$$

is surjective, then the complex $\lim_{\beta < \alpha} K_\beta^\bullet$ is acyclic.

Proof. By transfinite induction we prove this holds for every ordinal α and every system as in the lemma. In particular, whilst proving the result for α we may assume the complexes $\lim_{\gamma < \beta} K_\gamma^n$ are acyclic.

Let $x \in \lim_{\beta < \alpha} K_\beta^0$ with $d(x) = 0$. We will find a $y \in \lim_{\beta < \alpha} K_\beta^{-1}$ with $d(y) = x$. Write $x = (x_\beta)$ where $x_\beta \in K_\beta^0$ is the image of x for $\beta < \alpha$. We will construct $y = (y_\beta)$ by transfinite recursion.

For $\beta = 0$ let $y_0 \in K_0^{-1}$ be any element with $d(y_0) = x_0$.

For $\beta = \gamma + 1$ a successor, we have to find an element y_β which maps both to y_γ by the transition map $f : K_\beta^\bullet \rightarrow K_\gamma^\bullet$ and to x_β under the differential. As a first approximation we choose y'_β with $d(y'_\beta) = x_\beta$. Then the difference $y_\gamma - f(y'_\beta)$ is in the kernel of the differential, hence equal to $d(z_\gamma)$ for some $z_\gamma \in K_\gamma^{-2}$. By

assumption, the map $f^{-2} : K_{\beta}^{-2} \rightarrow K_{\gamma}^{-2}$ is surjective. Hence we write $z_{\gamma} = f(z_{\beta})$ and change y'_{β} into $y_{\beta} = y'_{\beta} + d(z_{\beta})$ which works.

If β is a limit ordinal, then we have the element $(y_{\gamma})_{\gamma < \beta}$ in $\lim_{\gamma < \beta} K_{\gamma}^{-1}$ whose differential is the image of x_{β} . Thus we can argue in exactly the same manner as above using the termwise surjective map of complexes $f : K_{\beta}^{\bullet} \rightarrow \lim_{\gamma < \beta} K_{\gamma}^{\bullet}$ and the fact (see first paragraph of proof) that we may assume $\lim_{\gamma < \beta} K_{\gamma}^{\bullet}$ is acyclic by induction. $\square$

32. Exactness of products

Lemma 32.1. *Let I be a set. For $i \in I$ let $L_i \rightarrow M_i \rightarrow N_i$ be a complex of abelian groups. Let $H_i = \text{Ker}(M_i \rightarrow N_i) / \text{Im}(L_i \rightarrow M_i)$ be the cohomology. Then*

$$\prod L_i \rightarrow \prod M_i \rightarrow \prod N_i$$

is a complex of abelian groups with homology $\prod H_i$.

Proof. Omitted. $\square$

33. Other chapters

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- (4) Categories
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- (6) Sheaves on Spaces
- (7) Sites and Sheaves
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- | | |
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| (88) Algebraization of Formal Spaces | |

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